



COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF PURE AND APPLIED SCIENCES

MATHEMATICS DEPARTMENT

FIRST YEARS (Engineering departments)

ACADEMIC YEAR: 2019-2020

COURSE: MATHEMATICS FOR ENGINEERS II

COURSE CODE: MAT 1266

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COURSE OUTLINE

Title: MATHEMATICS FOR ENGINEERS II

MODULE CODE: MAT 5124

PRE-REQUISITE: MAT 5114 (MATHEMATICS FOR ENGINEERS I)

Description of aims and content

This module is designed in such a way that the knowledge acquired by the students will enable them to perform calculus operations for recording and managing engineering operations such as writing some phenomena in mathematically models. Therefore, it contains powerful tools for handling physics models, engineering models, and analyzing the solutions after resolution the models.

Learning Outcomes:

Knowledge and Understanding

Having successfully completed the module, students should be able to demonstrate knowledge and understanding of:

- i. Use the basic principles of calculus, to apply them in engineering problem resolution.
- ii. To carry out physics operations or engineering operations, calling upon to mathematical skills.

Cognitive/Intellectual skills/Application of knowledge

Having successfully completed the module, students should be able to:

- iii. Present simple arguments and conclusion using the calculus theories.
- iv. Analyze and evaluate problems in mathematics and Engineering.

Communication/ICT/Numeracy/Analytic Techniques/Practical Skills.

Having successfully completed the module, students should be able to:

- v. Apply the mathematical knowledge to solve problems in range of physics and engineering situations.
- vi. Carry out mathematical and numerical manipulation with confidence and accuracy.

General transferable skills

Having successfully completed the module, students should be able to:

- vii. Model real-world problems in the field of physics and engineering by expressing them into mathematical formulation.
- viii. Solve those models and interpret the results.

INDICATIVE CONTENT

Unit I: Ordinary Differential Equations (ODE'S):

1.0. Introduction to ODE'S

1.1. First order linear equations with constant coefficients

1.2. Linear equations of second order with constant and variable coefficients

1.3. Some examples of application in engineering

1.4. System of linear equations

Unit II: Laplace transformations

2.1. Transforms of simple functions

2.2. Basic operational properties

2.3. Transforms of derivatives and integrals

2.4. Initial final value theorems

2.5. Inverse transforms

2.6. Applications of Laplace transforms for solving linear ordinary differential equations up to second order with constant coefficients.

2.7. Examples of engineering applications.

Unit III: Multiple integrals

3.1. Double integration in Cartesian and polar coordinates

3.2. Change of order of integration

3.3. Area as a double integral

3.4. Triple integral in Cartesian coordinates.

Unit IV: Introduction to vector calculus

4.1. Gradient, Divergence, Curl-Solenoidal and irrotational fields

4.2. Vector identities

4.3. Directional derivatives

4.4. Line, surface and volume integrals

REFERENCES

1. Calculus with analytic geometry Howard Anton, second Edition, 1984, by Anton Textbooks, Inc.
2. Differential Equations Richard Bronson, Gabriel Costa, Third Edition, 2006, by The McGraw-Hill Companies
3. Recueil d'exercices et de problèmes d'analyse mathématique.
4. Advanced modern Engineering Mathematics, Second edition, 1999 by Addison-Wesley

ASSESSMENT:

Assignments

Quizes

Examinations: CATS
FINAL

Unit I: ORDINARY DIFFERENTIAL EQUATIONS (ODE'S):

1.0. INTRODUCTION TO ODE'S

1.1.1. Definitions

A differential equation is an equation involving an unknown function and its derivatives.

Example 1.1: The following are differential equations involving the un-known function y .

$$\frac{dy}{dx} = 5x + 2 \quad (1.1)$$

$$e^y \frac{d^2y}{dx^2} + 2 \left(\frac{dy}{dx} \right)^2 = 1 \quad (1.2)$$

$$4 \frac{d^3y}{dx^3} + (\sin x) \frac{d^2y}{dx^2} + 5xy = 0 \quad (1.3)$$

$$\frac{\partial^2 y}{\partial t^2} - 4 \frac{\partial^2 y}{\partial x^2} = 0 \quad (1.4)$$

A differential equation is an **ordinary differential equation** if the unknown function depends on only one independent variable. If the unknown function depends on two or more independent variables, the differential equation is a **partial differential equation**. In this module we will be concerned solely with ordinary differential equations.

Note: The order of a differential equation is the order of the highest derivative appearing in the equation.

Example 1.2: Equations 1.1 through 1.3 are examples of ordinary differential equations, since the unknown function y depends solely on the variable x . Equation 1.4 is a partial differential equation, since y depends on both the independent variables t and x .

1.1.2. Notation

The expressions $y', y'', y''', y^{(4)}, \dots, y^{(n)}$ are often used to represent, respectively, the first, second, third, fourth, ..., n th derivatives of y with respect to the independent variable under consideration. Thus, y'' represents d^2y/dx^2 if the independent variable is x , but represents d^2y/dp^2 if the independent variable is p . Observe that parenthesis is used in $y^{(n)}$ to distinguish it from the n th power, y^n . If the independent variable is time, usually denoted by t , primes are often replaced by dots. Thus, $\dot{y}, \ddot{y},$ and $\dddot{y}$ represent, respectively, $dy/dt, d^2y/dt^2$ and d^3y/dt^3 .

1.1.3. Solutions

A **solution** of a differential equation in the unknown function y and the independent variable x on the interval $\mathfrak{J}$ is a function $y(x)$ that satisfy the differential equation identically for all x in $\mathfrak{J}$

Example 1.3: Is $y(x) = c_1 \sin 2x + c_2 \cos 2x$, where c_1 and c_2 are arbitrary constants, a solution of $y'' + 4y = 0$?

Differentiating y , we find $y' = 2c_1 \cos 2x - 2c_2 \sin 2x$ and $y'' = -4c_1 \sin 2x - 4c_2 \cos 2x$

. Hence,

$$\begin{aligned} y'' + 4y &= -4c_1 \sin 2x - 4c_2 \cos 2x + 4(c_1 \sin 2x + c_2 \cos 2x) \\ &= (-4c_1 + 4c_1) \sin 2x + (-4c_2 + 4c_2) \cos 2x \\ &= 0 \end{aligned}$$

1.2. FIRST ORDER LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

1.2.1. Definitions

- An equation is said to be **linear** if it is linear in the dependent variable.

Example 1.4:

- $y'' \cos x + x^2 y = 0$ Is a linear differential equation.
- $y'' + xy^2 = 0$ Is a nonlinear differential equation.

A differential equation is **homogeneous** if it has no terms that are functions of the independent variable alone. Thus an **inhomogeneous** equation is one in which there are terms that are functions of the independent variables alone.

• **Example 1.5:**

- $y'' + xy + y = 0$ Is a homogeneous equation.
- $y'' + xy + y - 7x = 0$ Is an inhomogeneous equation.

- We call the **first order linear equations with constant coefficients** a linear equation according to the unknown variable y and its derivative y' of the forms:

(E) $y' + ay = q(x)$ **non homogenous equation**

Where a is a constant and $q(x)$ is a function of x or not on the interval $\mathfrak{J}$.

. (E₀) $y' + ay = 0$ **homogenous equation**

1.2.2. Resolution of first order linear equations with constant coefficients

Consider the equation of the form: $y' + ay = q(x)$ (*)

First case: If $q(x) = 0$, then (*) becomes $y' + ay = 0$

$$\Leftrightarrow \frac{dy}{dx} + ay = 0 \Leftrightarrow \frac{dy}{y} = -a dx \rightarrow \int \frac{dy}{y} = - \int a dx + c$$

$$\Leftrightarrow \ln y = -ax + c \rightarrow \boxed{y = \lambda e^{-ax}} \text{ where } \lambda = e^c$$

Second case: If $q(x) \neq 0$, then we suppose that $y = u(x) \cdot v(x)$ and we get:

$$\frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Here we precious that one of the functions is taken arbitrary and the other is definite according to the linear equation.

After substitution into(*), we get:

$$u \frac{dv}{dx} + v \frac{du}{dx} + auv = q(x)$$

$$u \left(\frac{dv}{dx} + av \right) + v \frac{du}{dx} = q(x)$$

Choose the function $v(x) \neq 0$ such that $\frac{dv}{dx} + av = 0$

$$\rightarrow \frac{dv}{dx} = -av \rightarrow \frac{dv}{v} = -a dx \rightarrow v(x) = \lambda e^{-ax} \text{ where } \lambda = e^c$$

With this value of $v(x)$, we have:

$$v \frac{du}{dx} = q(x) \Leftrightarrow du = \frac{q(x)}{v(x)} dx \rightarrow u(x) = \int \frac{q(x)}{v(x)} dx + \lambda_1$$

$$\text{As } y = u(x) \cdot v(x) \rightarrow \boxed{y = \lambda e^{-ax} \cdot \left(\int \frac{q(x)}{v(x)} dx + \lambda_1 \right)}$$

Example 1.6: Solve the following linear ODE'S:

a) $3y' - 5y = x^2 e^{\frac{5}{3}x}$ and find the particular solution with $y(0) = 4$

b) $4y' + 7y = 0$

Solution:

a) $3y' - 5y = x^2 e^{\frac{5}{3}x} \Leftrightarrow y' - \frac{5}{3}y = \frac{x^2}{3} e^{\frac{5}{3}x}$

Suppose that $y = u(x) \cdot v(x) \rightarrow \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$

The equation becomes: $u \frac{dv}{dx} + v \frac{du}{dx} - \frac{5}{3}u \cdot v = \frac{x^2}{3} (2x - 1) \rightarrow u \left(\frac{dv}{dx} - \frac{5}{3}v \right) + v \frac{du}{dx} = \frac{x^2}{3} e^{\frac{5}{3}x}$

Choose $v(x)$ such that $\frac{dv}{dx} - \frac{5}{3}v = 0 \rightarrow \frac{dv}{v} = \frac{5}{3}dx \rightarrow v(x) = \lambda e^{\frac{5}{3}x}$

If $v(x) = \lambda e^{\frac{5}{3}x}$, then $v \frac{du}{dx} = \frac{x^2}{3} e^{\frac{5}{3}x} \rightarrow du = \frac{\frac{x^2}{3} e^{\frac{5}{3}x}}{v} dx \leftrightarrow du = \frac{\frac{x^2}{3} e^{\frac{5}{3}x}}{\lambda e^{\frac{5}{3}x}} dx \leftrightarrow du = \frac{x^2}{3\lambda} dx \rightarrow u(x) = \frac{x^3}{9\lambda} + \lambda_1 \quad \forall \lambda \neq 0$

As $y = u(x) \cdot v(x) = \lambda e^{\frac{5}{3}x} \cdot \left(\frac{x^3}{9\lambda} + \lambda_1 \right) = e^{\frac{5}{3}x} \left(\frac{x^3}{9} + \lambda_2 \right)$ where $\lambda_2 = \lambda_1 \cdot \lambda$

A particular solution is finding as: $y(0) = 4 \leftrightarrow 4 = \lambda_2$

Then a particular solution is $y(x) = e^{\frac{5}{3}x} \left(\frac{x^3}{9} + 4 \right)$

b) $4y' + 7y = 0 \leftrightarrow y' = -\frac{7}{4}y \leftrightarrow \frac{dy}{dx} = -\frac{7}{4}y \rightarrow \frac{dy}{y} = -\frac{7}{4}dx \rightarrow \ln y = -\frac{7}{4}x \rightarrow y = \lambda e^{-\frac{7}{4}x}$

1.1.3. Linear equations none solvable according to the derivatives

a) Equations of first order of degree n according to y'

Consider an ODE'S $(y')^n + a_1(y')^{n-1} + a_2(y')^{n-2} + \dots + a_{n-1}y' + a_n = 0$

To solve this equation according to y' is to obtained as solution of equation $y' = f_1(x, y), y' = f_2(x, y), \dots, y' = f_k(x, y)$ where $k \leq n$.

The general integral of considered function will be written as the set of integrals $\phi_1(x, y, c) = 0, \phi_2(x, y, c) = 0, \phi_3(x, y, c) = 0, \dots, \phi_k(x, y, c) = 0$ where $\phi_i(x, y, c) = 0$ the general integral of the equation is $y' = f_i(x, y, c), i = 1, 2, \dots, k$.

In each point of domain of plan where y' takes the real values it passes through k integrals curves

Example 1.7: Solve the equation $3(y')^2 + 2y' - x = 0$

Solution: Suppose that $y' = t$

$$\rightarrow dy = t dx \leftrightarrow dx = \frac{dy}{t} \quad (*)$$

The equation becomes: $3t^2 + 2t - x = 0 \rightarrow x = 3t^2 + 2t$

$$\rightarrow dx = (6t + 2)dt \quad (**)$$

$$(*) = (**) \leftrightarrow \frac{dy}{t} = (6t + 2)dt \rightarrow dy = (6t^2 + 2t)dt \rightarrow \int dy = \int (6t^2 + 2t)dt$$

$$\leftrightarrow y = 2t^3 + t^2 + c$$

In parametric form we have: $\begin{cases} x = 3t^2 + 2t \\ y = 2t^3 + t^2 + c \end{cases}$

b) Equations of the form $f(y, y') = 0$ or $f(x, y') = 0$

Those equations can be solved easily according to y' and we get in general the equations with separable variables.

Solve the equation $f(y, y') = 0$ is to write y as function of y' .

Consider $y = \varphi(y')$, suppose $y' = p \rightarrow dy = p dx$ (*)

$$y = \varphi(y') \rightarrow y = \varphi(p) \rightarrow dy = \varphi'(p) dp \quad (**)$$

$$(*) = (**) \leftrightarrow p dx = \varphi'(p) dp \leftrightarrow dx = \frac{\varphi'(p) dp}{p}$$

$$\rightarrow x = \int \frac{\varphi'(p)}{p} dp + c \text{ and } y = \varphi(p)$$

Example 1.8: Solve the equation: $y = (y')^2 - 4(y')^3$

Answer: Suppose $y' = p \rightarrow dy = p dx$ (*)

The equation becomes: $y = p^2 - 4p^3 \rightarrow dy = (2p - 12p^2) dp$ (**)

$$(*) = (**) \leftrightarrow p dx = (2p - 12p^2) dp \rightarrow dx = (2 - 12p) dp$$

$$\rightarrow x = 2p - 6p^2 + c$$

The general solution in the parametric form is $\begin{cases} x = 2p - 6p^2 + c \\ y = p^2 - 4p^3 \end{cases}$

1.2. LINEAR EQUATIONS OF SECOND ORDER WITH CONSTANT AND VARIABLE COEFFICIENTS.

1.2.1. Definition

We call linear equation of second order all equations of the form:

$$y'' + a(x)y' + b(x)y = f(x) \quad (***)$$

Where $a(x)$, $b(x)$ and $f(x)$ are the functions definite and continuous on the interval $\mathfrak{I}$.

Remarks 1.1:

➤ If $f(x) = 0$, then the equation (***) is called **homogeneous**;

- If $a(x) = c_1$ and $b(x) = c_2$ where $c_1, c_2 \in \mathbb{R}$, then the equation (***) is **equation with constants coefficients**.

1.2.2. General Properties

Two functions $y_1(x)$ and $y_2(x)$ are:

- ✓ linearly independent on the interval $[a, b]$ if

$$c_1 y_1(x) + c_2 y_2(x) = 0 \rightarrow c_1 = c_2 = 0$$
- ✓ linearly dependent on the interval $[a, b]$ if there exist a real $\beta \in \mathbb{R}$ such that

$$y_2(x) = \beta y_1(x)$$

Consider a homogeneous equation as: $y'' + ay' + by = 0$ (1)

Theorem 1.1: If functions $y_1(x)$ and $y_2(x)$ are the solution of the linear homogeneous equation $y'' + ay' + by = 0$, then $y(x) = c_1 y_1(x) + c_2 y_2(x)$ is also a solution of that equation.

Proof: Consider $y = c_1 y_1 + c_2 y_2 \rightarrow y' = c_1 y_1' + c_2 y_2' \rightarrow y'' = c_1 y_1'' + c_2 y_2''$

The equation $y'' + a(x)y' + b(x)y = 0$ becomes:

$$c_1 y_1'' + c_2 y_2'' + a(x)[c_1 y_1' + c_2 y_2'] + b(x)[c_1 y_1 + c_2 y_2] = 0$$

If we group the terms we get:

$$c_1 \left[\underbrace{y_1'' + a(x)y_1' + b(x)y_1}_{=0} \right] + c_2 \left[\underbrace{y_2'' + a(x)y_2' + b(x)y_2}_{=0} \right] = 0$$

$\rightarrow y(x) = c_1 y_1(x) + c_2 y_2(x)$ is also a solution of the equation (1)

THE WRONSKIAN

We call Wronskian determinant or Wronskian of a set of functions $\{y_1(x), y_2(x), y_3(x), \dots, y_n(x)\}$ on the interval $a \leq x \leq b$, having the property that each function possesses $n - 1$ derivatives on this interval, is the determinant

$$W(y_1, y_2, y_3, \dots, y_n) = \begin{vmatrix} y_1 & y_2 & y_3 & \cdots & y_n \\ y_1' & y_2' & y_3' & \cdots & y_n' \\ y_1'' & y_2'' & y_3'' & \cdots & y_n'' \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ y_1^{(n-1)} & y_2^{(n-1)} & y_3^{(n-1)} & \cdots & y_n^{(n-1)} \end{vmatrix}$$

Theorem 1.2: If the functions $y_1(x)$ and $y_2(x)$ are linearly dependent on interval $[a, b]$, then its Wronskian is identically zero on $[a, b]$. Indeed if the functions $y_1(x)$ and $y_2(x)$ are linearly dependent on interval $[a, b]$, then there exist a real $\beta \in \mathbb{R}$ such that

$$y_2(x) = \beta y_1(x), \text{ the Wronskian}$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} y_1 & \beta y_1 \\ y_1' & \beta y_1' \end{vmatrix} = \beta \begin{vmatrix} y_1 & y_1 \\ y_1' & y_1' \end{vmatrix} = 0$$

Example 1.9: Find the Wronskian of the set $\{\sin 3x, \cos 3x\}$

Answer:
$$W(\sin 3x, \cos 3x) = \begin{vmatrix} \sin 3x & \cos 3x \\ \frac{d(\sin 3x)}{dx} & \frac{d(\cos 3x)}{dx} \end{vmatrix}$$

$$= \begin{vmatrix} \sin 3x & \cos 3x \\ 3\cos 3x & -3\sin 3x \end{vmatrix} = -3(\sin^2 3x + \cos^2 3x) = -3$$

1.2.3. The second-order linear Homogeneous Differential Equation with Constant Coefficients

a) General form

The general form of that ODE'S is:

$$ay'' + by' + cy = 0 \quad (2)$$

Where **a**, **b**, and **c** are constants.

Example 1.10: $y'' + 3y' - 4y = 0$ is a second-order linear Homogeneous Differential Equation with Constant Coefficients

b) Resolution of this ODE'S

For solving this equation we use the characteristic equation as:

If we have the equation of the form (2), then its **Characteristic equation** is:

$$a\lambda^2 + b\lambda + c = 0 \quad (3)$$

From (3), we get: $\Delta = b^2 - 4ac$

Here we have three cases for the value of λ :

First case: If $\Delta > 0$, then $\lambda = \frac{-b \pm \sqrt{\Delta}}{2a} \leftrightarrow \begin{cases} \lambda_1 = \frac{-b + \sqrt{\Delta}}{2a} \\ \lambda_2 = \frac{-b - \sqrt{\Delta}}{2a} \end{cases}$

In this case the partial solutions of (2) are: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

Hence, the general solution is $y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$

Example 1.11: Solve the equation $y'' + 3y' - 4y = 0$

Answer: The characteristic equation is $\lambda^2 + 3\lambda - 4 = 0 \rightarrow \lambda = 1 \text{ or } -4$

i.e $\lambda_1 = 1$ and $\lambda_2 = -4$

The general solution is $y = c_1 e^x + c_2 e^{-4x}, \forall c_1, c_2 \in \mathbb{R}$

Second case: If $\Delta = 0$, then $\lambda = \frac{b}{2a} \leftrightarrow \lambda_1 = \lambda_2 = \frac{-b}{2a}$

In this case the partial solutions of (2) are: $y_1 = e^{\lambda x}$ and $y_2 = x e^{\lambda x}$

Hence, the general solution is $y = c_1 y_1 + c_2 y_2 = e^{\lambda x} (c_1 + c_2 x)$

Example 1.12: Solve the equation $y'' + 4y' + 4y = 0$

Answer: The characteristic equation is $\lambda^2 + 4\lambda + 4 = 0 \rightarrow \lambda = -2$

i.e $\lambda_1 = -2$ and $\lambda_2 = -2$

The general solution is $y = e^{-2x} (c_1 + c_2 x), \forall c_1, c_2 \in \mathbb{R}$

Third case: If $\Delta < 0$, then $\lambda = \alpha \pm i\beta \leftrightarrow \lambda_1 = \alpha + i\beta; \lambda_2 = \alpha - i\beta$

In this case the partial solutions of (2) are: $y_1 = e^{\lambda_1 x}$ and $y_2 = e^{\lambda_2 x}$

i.e $y_1 = e^{x(\alpha+i\beta)} = e^{\alpha x} \cdot e^{i\beta x} = e^{\alpha x} (\cos \beta x + i \sin \beta x)$

and

$$y_2 = e^{x(\alpha-i\beta)} = e^{\alpha x} \cdot e^{-i\beta x} = e^{\alpha x} (\cos \beta x - i \sin \beta x)$$

Remarks 1.2:

- If y_1 and y_2 are solution of (2), then $z_1 = \frac{y_1 + y_2}{2}$ is also a solution of (2)
- If y_1 and y_2 are solution of (2), then $z_1 = \frac{y_1 - y_2}{2i}$ is also a solution of (2)
- If z_1 and z_2 are solution of (2), then $y = c_1 z_1 + c_2 z_2$ is also a solution of (2)
- When the Wronskian is identically zero and the functions are not known to be solutions of the same linear differential equation.

Hence, the general solution is $y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Example 1.13: Solve the equation $y'' + 10y' + 26y = 0$

Answer: The characteristic equation is $\lambda^2 + 10\lambda + 26 = 0 \rightarrow \lambda = -5 \pm i \leftrightarrow \begin{cases} \lambda_1 = -5 + i \\ \lambda_2 = -5 - i \end{cases}$

i.e $y_1 = e^{x(-5+i)} = e^{-5x} \cdot e^{ix} = e^{-5x} (\cos x + i \sin x)$

and

$$y_2 = e^{x(\alpha-i\beta)} = e^{\alpha x} \cdot e^{-i\beta x} = e^{\alpha x} (\cos x - i \sin x)$$

The general solution is $y = e^{-5x}(c_1 \cos x + c_2 \sin x), \forall c_1, c_2 \in \mathbb{R}$

1.2.4. The second-order linear non-Homogeneous Differential Equation with constants Coefficients

a) General form

The general form of a second-order linear non-Homogeneous Differential Equation with constants Coefficients is:

$$y'' + py' + qy = f(x) \quad (3)$$

Where $p, q \in \mathbb{R}$

Let y_p denote any *particular solution* of (3) and let y_h (henceforth called the *homogeneous or complementary solution*) represent the general solution of the associated homogeneous equation (2).

Theorem 1.3: The general solution of $y'' + py' + qy = f(x)$ is $y = y_h + y_p$

b) Method of resolution

b.1) Method of undetermined coefficients

We use this method if the function $f(x)$ has one of the following forms:

- i) $f(x) = P_n(x)$ Where $P_n(x)$ an n^{th} -degree polynomial in x .
- ii) $f(x) = P_n(x)e^{\alpha x}$
- iii) $f(x) = P_n(x)e^{\alpha x} \cos \beta x$ or $P_n(x)e^{\alpha x} \sin \beta x$

First case: If $f(x) = P_n(x)$, then we find the particular solution when we suppose:

$$1^0) y_p = Q_n(x) \text{ if } q \neq 0$$

$$2^0) y_p = Q_{n+1}(x) \text{ if } p \neq 0 \text{ and } q = 0$$

$$3^0) y_p = Q_{n+2}(x) \text{ if } p = q = 0$$

Where $Q_n(x)$ an n^{th} -degree polynomial in x to be determined.

Example 1.14: Find the general solution of the equation $y'' - y = x^2 - x$

Answer: The associated homogeneous equation is $y'' - y = 0$

Its characteristic equation is $\lambda^2 - 1 = 0 \rightarrow \lambda_1 = 1$ or $\lambda_2 = -1$

The general solution of homogeneous equation is $y_h = c_1 e^x + c_2 e^{-x}$

The particular solution of non-homogeneous equation is in the form:

$$y_p = ax^2 + bx + c \rightarrow y'_p = 2ax + b \text{ and } y''_p = 2a.$$

The differential equation $y'' - y = x^2 - x$ becomes:

$$2a - (ax^2 + bx + c) = x^2 - x \leftrightarrow -ax^2 - bx + 2a - c = x^2 - x$$

By identification, we get:
$$\begin{cases} -a = 1 \\ -b = -1 \\ 2a - c = 0 \end{cases} \leftrightarrow \begin{cases} a = -1 \\ b = 1 \\ c = -2 \end{cases}$$

The particular solution of non-homogeneous equation is: $y_p = -x^2 + x - 2$

The general solution of non-homogeneous equation is $y = y_h + y_p$

$$y = c_1 e^x + c_2 e^{-x} - x^2 + x - 2$$

Example 1.15: Find the general solution of the equation $y'' + y' = x^3 - x^2$

Answer: The associated homogeneous equation is $y'' + y' = 0$

Its characteristic equation is $\lambda^2 + \lambda = 0 \rightarrow \lambda_1 = 0$ or $\lambda_2 = -1$

The general solution of homogeneous equation is $y_h = c_1 + c_2 e^{-x}$

The particular solution of non-homogeneous equation is in the form:

$$y_p = x(ax^3 + bx^2 + cx + d) = ax^4 + bx^3 + cx^2 + dx \rightarrow y'_p = 4ax^3 + 3bx^2 + 2cx + d \text{ and } y''_p = 12ax^2 + 6bx + 2c.$$

The differential equation $y'' + y' = x^3 - x^2$ becomes:

$$12ax^2 + 6bx + 2c + (4ax^3 + 3bx^2 + 2cx + d) = x^3 - x^2$$

$$\leftrightarrow 4ax^3 + (12a + 3b)x^2 + (6b + 2c)x + (2c + d) = x^3 - x^2$$

By identification, we get:
$$\begin{cases} 4a = 1 \\ 12a + 3b = -1 \\ 6b + 2c = 0 \\ 2c + d = 0 \end{cases} \leftrightarrow \begin{cases} a = \frac{1}{4} \\ b = \frac{-4}{3} \\ c = 4 \\ d = -8 \end{cases}$$

The particular solution of non-homogeneous equation is: $y_p = \frac{1}{4}x^4 + \frac{-4}{3}x^3 + 4x^2 - 8x$

The general solution of non-homogeneous equation is $y = y_h + y_p$

$$y = c_1 + c_2 e^{-x} + \frac{1}{4}x^4 + \frac{-4}{3}x^3 + 4x^2 - 8x$$

Second case: If $f(x) = P_n(x)e^{ax}$, then we find the particular solution when we suppose:

1⁰) $y_p = Q_n(x)e^{\alpha x}$, if α is not a root of correspondent characteristic equation;

2⁰) $y_p = Q_{n+1}(x)e^{\alpha x}$, if α is a simple root of correspondent characteristic equation;

3⁰) $y_p = Q_{n+2}(x)e^{\alpha x}$, if α is a double root of correspondent characteristic equation.

Where $Q_n(x)$ an n^{th} -degree polynomial in x to be determined.

Example 1.16: Find the general solution of the equation $y'' - 4y' = 4x^2e^x$

Answer: The associated homogeneous equation is $y'' - 4y' = 0$

Its characteristic equation is $\lambda^2 - 4\lambda = 0 \rightarrow \lambda_1 = 0$ or $\lambda_2 = 4$

The general solution of homogeneous equation is $y_h = c_1 + c_2e^{4x}$

As $\alpha = 1$ is not a root of a correspondent characteristic equation, then

The particular solution of non-homogeneous equation is in the form:

$$y_p = (ax^2 + bx + c)e^x \rightarrow y'_p = e^x[ax^2 + (4a + b)x + (a + 2b)]$$

$$\text{and } y''_p = e^x[ax^2 + (4a + b)x + (a + 2b)]$$

The differential equation $y'' - 4y' = 4x^2e^x$ becomes:

$$e^x[ax^2 + (4a + b)x + (a + 2b)] - 4e^x[ax^2 + (4a + b)x + (a + 2b)] = 4x^2e^x$$

$$\Leftrightarrow [-3ax^2 + (-4a - 3b)x + (2a - 2b - 3c)]e^x = 4x^2e^x$$

By identification, we get:
$$\begin{cases} -3a = 4 \\ -4a - 3b = 0 \\ 2a - 2b - 3c \end{cases} \Leftrightarrow \begin{cases} a = \frac{-4}{3} \\ b = \frac{16}{9} \\ c = \frac{-56}{27} \end{cases}$$

The particular solution of non-homogeneous equation is: $y_p = \left(\frac{-4}{3}x^2 + \frac{16}{9}x - \frac{56}{27}\right)e^x$

The general solution of non-homogeneous equation is $y = y_h + y_p$

$$y = c_1 + c_2e^{4x} + \left(\frac{-4}{3}x^2 + \frac{16}{9}x - \frac{56}{27}\right)e^x$$

Example 1.17: Find the general solution of the equation $y'' + 10y' + 25y = 4e^{-5x}$

Answer: The associated homogeneous equation is $y'' + 10y' + 25y = 0$

Its characteristic equation is $\lambda^2 + 10\lambda + 25 = (\lambda + 5)^2 = 0 \rightarrow \lambda_1 = \lambda_2 = -5$

The general solution of homogeneous equation is $y_h = c_1e^{-5x} + c_2xe^{-5x}$

As $\alpha = -5$ is a double root of a correspondent characteristic equation, then

The particular solution of non-homogeneous equation is in the form:

$$y_p = (ax^2 + bx + c)e^{-5x} \rightarrow y_p' = e^{-5x}[-5ax^2 + (2a - 5b)x + (b - 5c)]$$

$$\text{and } y_p'' = e^{-5x}[25ax^2 + (-20a + 25b)x + (2a - 10b + 25c)]$$

The differential equation $y'' + 10y' + 25y = 4e^{-5x}$ becomes:

$$\{[25ax^2 + (-20a + 25b)x + (2a - 10b + 25c)] + 10 \cdot [-5ax^2 + (2a - 5b)x + (b - 5c)] + 25 \cdot (ax^2 + bx + c)\} \cdot e^{-5x} = 4e^{-5x}$$

$$\Leftrightarrow 2ae^{-5x} = 4e^{-5x} \rightarrow 2a = 4 \Leftrightarrow a = 2, b = c = 0$$

The particular solution of non-homogeneous equation is: $y_p = 2x^2e^{-5x}$

The general solution of non-homogeneous equation is $y = y_h + y_p$

$$y = (c_1 + c_2x + 2x^2)e^{-5x}$$

Third case: If $f(x) = P_n(x)e^{\alpha x} \cos \beta x$ or $P_n(x)e^{\alpha x} \sin \beta x$, then we find the particular solution when we suppose:

1⁰) $y_p = e^{\alpha x}[Q_n(x) \cos \beta x + R_n(x) \sin \beta x]$, if $\alpha + i\beta$ is not a root of correspondent characteristic equation;

2⁰) $y_p = xe^{\alpha x}[Q_n(x) \cos \beta x + R_n(x) \sin \beta x]$, if $\alpha + i\beta$ is a root of correspondent characteristic equation.

Where $Q_n(x)$ and $R_n(x)$ the n^{th} -degree polynomial in x to be determined.

Example 1.18: Find the general solution of the equation $y'' + 2y' + 5y = e^{-x} \cos 2x$

Answer: The associated homogeneous equation is $y'' + 2y' + 5y = 0$

Its characteristic equation is $\lambda^2 + 2\lambda + 5 = 0 \rightarrow \lambda_1 = -1 + 2i; \lambda_2 = -1 - 2i$

The general solution of homogeneous equation is $y_h = e^{-x}(c_1 \cos 2x + c_2 \sin 2x)$

As $-1 + 2i$ is a root of a correspondent characteristic equation, then

The particular solution of non-homogeneous equation is in the form:

$$y_p = xe^{-x}[a \cos \beta x + b \sin \beta x], \rightarrow y_p' = e^{-x}[(-a + 2b) \cos x + (-2ab - b) \sin 2x]$$

$$\text{and } y_p'' = e^{-x}\{[(-a + 2b)x + a] \cos x + [(-2ab - b)x + b] \sin 2x\}$$

The differential equation $y'' + 2y' + 5y = e^{-x} \cos 2x$ becomes:

$$\begin{aligned} & \{[(-a + 2b)x + a] \cos x + [(-2ab - b)x + b] \sin 2x + 2 \\ & \quad \cdot [(-a + 2b) \cos x + (-2ab - b) \sin 2x] + 5x[a \cos \beta x + b \sin \beta x]\} \cdot e^{-x} \\ & = e^{-x} \cos 2x \end{aligned}$$

By identification, we get: $\begin{cases} 4b = 1 \\ -4a = 0 \end{cases} \leftrightarrow \begin{cases} b = \frac{1}{4} \\ a = 0 \end{cases}$

The particular solution of non-homogeneous equation is: $y_p = \frac{x}{4} e^{-x} \sin 2x$

The general solution of non-homogeneous equation is $y = y_h + y_p$

$$y = \left(c_1 \cos 2x + c_2 \sin 2x + \frac{x}{4} \sin 2x \right) e^{-x}$$

Remarks 1.3:

- If we can write $f(x)$ in the form of the sum of two functions $f(x) = f_1(x) + f_2(x)$, then we separate the equations as:

$$\begin{cases} y'' + py' + qy = f_1(x) \\ y'' + py' + qy = f_2(x) \end{cases}$$

And the particular solution is $y_p = y_{p_1} + y_{p_2}$ where y_{p_1} and y_{p_2} are respectively the particular solutions of those two above equations.

- We can also generalize the third case if $f(x) = P(x)e^x \cos \beta x + S(x)e^x \sin \beta x$
- If $\alpha + i\beta$ is not a root of correspondent characteristic equation, then we suppose $y = Q(x)e^x \cos \beta x + R(x)e^x \sin \beta x$ where $Q(x)$ or $R(x)$ are the degree equal to the highest degree of $S(x)$
 - If $\alpha + i\beta$ is a root of correspondent characteristic equation, then we suppose
- $$y = x[Q(x)e^x \cos \beta x + R(x)e^x \sin \beta x]$$

b.2) Method of variation of parameters or constants

For finding a particular solution of the equation (3), once the solution of the associated homogeneous equation $y'' + py' + qy = 0$ is known. Recall that if y_1 and y_2 are independent solutions of $y'' + py' + qy = 0$, then the general solution of $y'' + py' + qy = 0$ is

$$y_h = y_1 + y_2$$

For applying this method, then the particular solution of (3) is

$$y_p = c_1(x)y_1 + c_2(x)y_2 \quad (4)$$

Where $c_1(x)$ and $c_2(x)$ are functions of x to be determined, and y_1, y_2 are the solution of $y'' + py' + qy = 0$

Derivate (4) respect to x , we get:

$$y'_p = c'_1 y_1 + c_1 y'_1 + c'_2 y_2 + c_2 y'_2 = (c'_1 y_1 + c'_2 y_2) + (c_1 y'_1 + c_2 y'_2)$$

Choose the functions $c_1(x)$ and $c_2(x)$ such that $c'_1 y_1 + c'_2 y_2 = 0$ (5)

After

(5),

then

$$y'_p = c_1 y'_1 + c_2 y'_2$$

If we derive that expression, then we get:

$$y''_p = c'_1 y'_1 + c_1 y''_1 + c'_2 y'_2 + c_2 y''_2$$

Substitute y_p, y'_p , and y''_p in the equation (3) we get:

$$\begin{aligned} c'_1 y'_1 + c_1 y''_1 + c'_2 y'_2 + c_2 y''_2 + p(c_1 y'_1 + c_2 y'_2) + q(c_1 y_1 + c_2 y_2) &= f(x) \\ \Leftrightarrow c_1 \underbrace{(y''_1 + p y'_1 + q y_1)}_{=0} + c_2 \underbrace{(y''_2 + p y'_2 + q y_2)}_{=0} + c'_1 y'_1 + c'_2 y'_2 &= f(x) \end{aligned}$$

The equation (3), becomes:

$$c'_1 y'_1 + c'_2 y'_2 = f(x) \quad (6)$$

The function $y = c_1 y_1 + c_2 y_2$ is the solution of the equation (3) if

$$\begin{cases} c'_1 y_1 + c'_2 y_2 = 0 \\ c'_1 y'_1 + c'_2 y'_2 = f(x) \end{cases} \rightarrow W = \begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix} \neq 0$$

We can then find $c_1(x)$ and $c_2(x)$ for solving the above system, we find that:

$$c'_1(x) = \varphi_1(x) \text{ and } c'_2(x) = \varphi_2(x)$$

We take $c_1(x) = \int \varphi_1(x) dx + c_3$ and $c_2(x) = \int \varphi_2(x) dx + c_4$

The general solution of (3), is obtained when we replace the functions $c_1(x)$ and $c_2(x)$ by their values in the expression $y = c_1 y_1 + c_2 y_2$

$$\rightarrow y = y_1(x) \left[\int \varphi_1(x) dx + c_3 \right] + y_2(x) \left[\int \varphi_2(x) dx + c_4 \right]$$

Example 1.19: Find the general solution of the equation $y'' + 4y' + 4y = \frac{e^{-2x}}{x^2}$

Answer: The associated homogeneous equation is $y'' + 4y' + 4y = 0$

Its characteristic equation is $\lambda^2 + 4\lambda + 4 = 0 \rightarrow \lambda_1 = \lambda_2 = -2$

The general solution of homogeneous equation is $y_h = c_1 e^{-2x} + c_2 x e^{-2x}$

$\rightarrow y_1 = e^{-2x}$ and $y_2 = x e^{-2x}$

The general solution of non-homogeneous equation is $y_p = c_1 y_1 + c_2 y_2$

Where $y_1 = e^{-2x}$ and $y_2 = x e^{-2x}$

$$W(y_1, y_2) = \begin{vmatrix} e^{-2x} & x e^{-2x} \\ -2e^{-2x} & (1-2x)e^{-2x} \end{vmatrix} = e^{-4x} \neq 0$$

For calculating $c_1(x)$ and $c_2(x)$, we solve the system:

$$\begin{cases} c_1' y_1 + c_2' y_2 = 0 \\ c_1' y_1' + c_2' y_2' = f(x) \end{cases} \Leftrightarrow \begin{cases} c_1' e^{-2x} + c_2' x e^{-2x} = 0 \\ -2c_1' e^{-2x} + c_2' (1-2x) e^{-2x} = \frac{e^{-2x}}{x^2} \end{cases}$$

$$\Leftrightarrow c_1' = \frac{1}{e^{-4x}} \begin{vmatrix} 0 & x e^{-2x} \\ \frac{e^{-2x}}{x^2} & (1-2x) e^{-2x} \end{vmatrix} = \frac{-1}{x}$$

$$\rightarrow c_1 = -\ln|x| + c_3$$

$$c_2' = \frac{1}{e^{-4x}} \begin{vmatrix} e^{-2x} & 0 \\ -2e^{-2x} & \frac{e^{-2x}}{x^2} \end{vmatrix} = \frac{1}{x^2}$$

$$\rightarrow c_2 = -\frac{1}{x} + c_4$$

The general solution of non-homogeneous equation is $y_p = c_1 y_1 + c_2 y_2$

$$\rightarrow y_p = -e^{-2x} \ln|x| - e^{-2x} = (-1 - \ln|x|) e^{-2x}$$

The general solution of a given problem is:

$$y = y_h + y_p = c_1 e^{-2x} + c_2 x e^{-2x} + (-1 - \ln|x|) e^{-2x} = e^{-2x} (c_1 + c_2 x - 1 - \ln|x|)$$

1.2.4. The second-order linear Homogeneous Differential Equation with variable Coefficients

Problem: Knowing one particular solution $y_1(x)$ of the homogeneous equation

$$y'' + a(x)y' + b(x)y = 0 \quad (7)$$

How can we find the second solution $y_2(x)$?

Where $a(x)$ and $b(x)$ are functions of independent variable x

Answer: Consider the solutions $y_1(x)$ and $y_2(x)$ as linearly independent solutions

i.e. $\frac{y_2(x)}{y_1(x)} \neq \text{constant}$

In this case, we consider a solution $v(x)$ such that $\frac{y_2(x)}{y_1(x)} = v(x) \leftrightarrow y_2(x) = v(x) \cdot y_1(x)$

The solution $y_2(x)$ is a solution of equation (7) if and only if verifies that equation

$$\text{i.e: } y_2'' + ay_2' + by_2 = 0 \quad (*)$$

If we replace y_2 in (*) by $v \cdot y_1$, then we get :

$$(v''y_1 + 2v'y_1' + vy_1'') + a(v'y_1 + vy_1') + bvy_1 = 0 \quad (**)$$

If we group the terms in v , v' and v'' , then (**) becomes:

$$\begin{aligned} v''y_1 + v'(2y_1' + ay_1) + v \underbrace{(y_1'' + ay_1' + by_1)}_{=0} &= 0 \\ \Leftrightarrow v''y_1 + v'(2y_1' + ay_1) &= 0 \Leftrightarrow \frac{v''}{v'} = -2\frac{y_1'}{y_1} - a \\ \rightarrow \ln|v'| &= -2\ln|y_1| - \int a(x)dx \rightarrow v' = \frac{1}{y_1^2} e^{-\int a(x)dx} \\ \rightarrow v(x) &= \int \frac{e^{-\int a(x)dx}}{y_1^2} dx + c \end{aligned}$$

Example 1.20: Find the general solution of the equation:

$$x^2y'' - xy' + y = 0, \text{ such that } y_1(x) = x, \text{ where } x > 0$$

$$\text{Answer: } x^2y'' - xy' + y = 0 \Leftrightarrow y'' - \frac{1}{x}y' + \frac{1}{x^2}y = 0$$

$$\begin{aligned} v(x) &= \int \frac{e^{-\int a(x)dx}}{y_1^2} dx + c \rightarrow v(x) = \int \frac{e^{\int \frac{1}{x}dx}}{x^2} dx + c = \int \frac{e^{\ln x}}{x^2} dx + c = \int \frac{x}{x^2} dx + c \\ &= \int \frac{1}{x} dx + c = \ln x + c \end{aligned}$$

$$v(x) = \ln x + c$$

$$\text{As } y_2(x) = v(x) \cdot y_1(x) \rightarrow y_2(x) = x(\ln x + c) \rightarrow y_2(x) = x \ln x, \text{ when } c = 0$$

The general solution is $y = c_1y_1 + c_2y_2 = c_1x + c_2x \ln x, \forall c_1, c_2 \in \mathbb{R}$

1.3. SOME EXAMPLES OF APPLICATION IN ENGINEERING

1.3.1. Application of first- order linear differential equations in engineering

a) Temperature problems

Newton's law of cooling, which is equally applicable to heating, states that '*The time rate of change of the temperature of a body is proportional to the temperature difference between the body and its surrounding medium*'. Let T denote the temperature of the body and let T_m denote the temperature of surrounding medium. Then the time rate of change of the temperature of the body is $\frac{dT}{dt}$, and Newton's law of cooling can be formulated as $\frac{dT}{dt} = -k(T - T_m)$, or as:

$$\frac{dT}{dt} + kT = kT_m \quad (8)$$

Where k is a positive constant of proportionality. Once k is chosen positive, the minus sign is required in Newton's law to make $\frac{dT}{dt}$ negative in a cooling process, when T is greater than T_m , and positive in a heating process, when T is less than T_m .

Example 1.21: A body at a temperature of 50° F is placed outdoors where the temperature is 100° F. If after 5 minutes the temperature of a body is 60° F. Find:

- How long it will take the body to reach a temperature of 75° F?
- The temperature of the body after 20 minutes?

Answer: Here $T_m = 100^\circ$ F, the surrounding medium is the outside air.

We have $\frac{dT}{dt} + kT = 100k \Leftrightarrow \dot{T} + kT = 100k$

This differential equation is linear.

As $100k \neq 0$, then we suppose that $T(t) = u(t) \cdot v(t)$

$$\dot{T} = u\dot{v} + v\dot{u}$$

$$\dot{T} + kT = 100k \Leftrightarrow u\dot{v} + v\dot{u} + kuv = 100k \Leftrightarrow u(\dot{v} + kv) + v\dot{u} = 100k$$

Choose the function $v(t) \neq 0$ such that $\dot{v} + kv = 0 \rightarrow v = ce^{-\int k dt} = ce^{-kt}$

$$v\dot{u} = 100k \Leftrightarrow du = 100ke^{kt} dt, \text{ when } c = 1$$

$$\Leftrightarrow u = \int 100ke^{kt} dt + c_1 = 100e^{kt} + c_1$$

$$T(t) = u(t) \cdot v(t) = e^{-kt} \cdot (100e^{kt} + c_1) = c_1 e^{-kt} + 100$$

$$T(t) = c_1 e^{-kt} + 100 \quad (*)$$

Since $T = 50$ when $t = 0$, it follows from (*) that $50 = c_1 e^{-k(0)} + 100$, or $c_1 = -50$.

Substituting this value into (*), we obtain:

$$T(t) = -50e^{-kt} + 100 \quad (**)$$

At $t = 5$, we are given that $T = 60$; Hence from (**), $60 = -50e^{-5k} + 100$. Solving for k , we obtain: $k = -\frac{1}{5} \ln \frac{40}{50} = 0.045$

Substituting this value into (**), we obtain:

$$T(t) = -50e^{-0.045t} + 100 \quad (***)$$

a) We require t when $T = 75$. Substituting $T = 75$ into (***), we have

$$75 = -50e^{-0.045t} + 100 \text{ or } e^{-0.045t} = \frac{1}{2} \rightarrow -0.045t = \ln \frac{1}{2}$$

$$t = 15.4 \text{ min}$$

b) We require T when $t = 20$. Substituting $t = 20$ into (***), we have

$$T = -50e^{-0.045(20)} + 100 = -50(0.41) + 100 = 79.5^\circ F$$

$$T = 79.5^\circ F$$

b) Falling body problems

Consider a vertically falling body of mass m that is being influenced only by gravity g and an air resistance that is proportional to the velocity of the body. Assume that both gravity and mass remain constant and, for convenience, choose the downward direction as the positive direction.

Newton's second law of motion: The net force acting on a body is equal to the time rate of change of the momentum of the body; or, for constant mass,

$$F = m \frac{dv}{dt} \quad (9)$$

Where F is the net force on the body and v is the velocity of the body, both at time t

For the problem at hand, there are two forces acting on the body:

- ✓ The force due to gravity given by the weight w of the body, which equals mg ;
- ✓ The force due to air resistance given by $-kv$, where $k > 0$ is a constant of proportionality. The minus sign required because this force opposes the velocity; that is, it acts in the upward, or negative, direction.

The net force F on the body is, therefore, $F = mg - kv$. Substituting this result into (9), we get:

$$mg - kv = m \frac{dv}{dt}$$

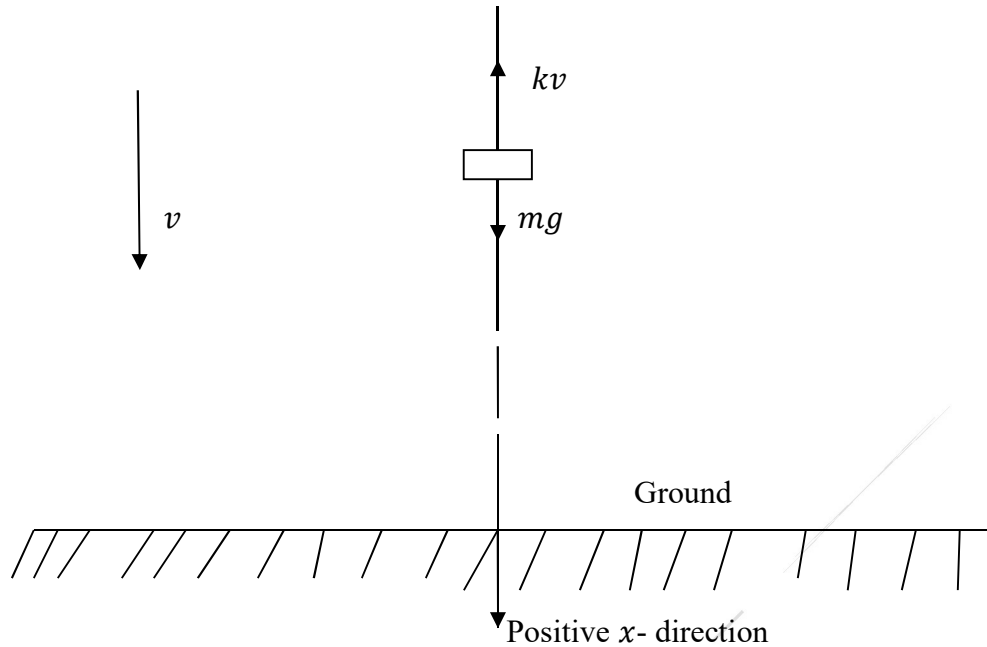
Or

$$\frac{dv}{dt} + \frac{k}{m}v = g \quad (10)$$

As equation of motion for the body.

If air resistance is negligible or nonexistent, then $k = 0$ and (10) simplifies to

$$\frac{dv}{dt} = g \quad (11)$$



When $k > 0$, the limiting velocity v_l is defined by

$$v_l = \frac{mg}{k} \quad (11)$$

Remarks 1.4: Equations (9), (10), and (11), are valid only if the given conditions are satisfied. These equations are not valid if, for example, air resistance is not proportional to but to the velocity square, or if the upward direction is taken to be the positive direction.

Example 1.22: A body weighing 64N is dropped from a height of 100m with an initial velocity of 10 m/sec . Assume that the air resistance is proportional to the velocity of the body. If the limiting velocity is known to be 128 m/sec . Find:

- An expression for the velocity of the body at any time t ;
- An expression for the position of the body at any time t

Answer:

a) Here $w = 64 \text{ kg}$, Since $w = mg \rightarrow 64 = mg \rightarrow m = \frac{64}{9.8} = 6.5$

Given that $v_l = 128 \text{ m/sec}$, it follows from (11) that $128 = \frac{64}{k} \rightarrow k = \frac{1}{2}$. Substituting these values into (10), we obtain the linear differential equation:

$$\frac{dv}{dt} + \frac{1}{13}v = 9.8 \leftrightarrow \dot{v} + \frac{1}{13}v = 9.8$$

This differential equation is linear.

As $9.8 \neq 0$, then we suppose that $v(t) = u(t) \cdot w(t)$

$$\dot{v} = u\dot{w} + w\dot{u}$$

$$\dot{v} + \frac{1}{13}v = 9.8 \leftrightarrow u\dot{w} + w\dot{u} + \frac{1}{13}uw = 9.8 \leftrightarrow u\left(\dot{w} + \frac{1}{13}w\right) + w\dot{u} = 9.8$$

Choose the function $w(t) \neq 0$ such that $\dot{w} + \frac{1}{13}w = 0 \rightarrow w = ce^{-\int \frac{1}{13}dt} = ce^{-\frac{1}{13}t}$

$$w\dot{u} = 9.8 \leftrightarrow du = 9.8e^{\frac{1}{13}t}dt, \text{ when } c = 1$$

$$\leftrightarrow u = \int 9.8e^{\frac{1}{13}t}dt + c_1 = 127.4e^{\frac{1}{13}t} + c_1$$

$$v(t) = u(t) \cdot w(t) = e^{-\frac{1}{13}t} \cdot \left(127.4e^{\frac{1}{13}t} + c_1\right) = c_1e^{-\frac{1}{13}t} + 127.4$$

$$v(t) = c_1e^{-\frac{1}{13}t} + 127.4 \quad (*)$$

At $t = 0$, we are given that $v = 10 \text{ m/sec}$. Substituting these values into (*), we get:

$$10 = c_1e^{-\frac{1}{13}(0)} + 127.4 \rightarrow c_1 = -117.4$$

The velocity at any time t is given by:

$$v(t) = -117.4e^{-\frac{1}{13}t} + 127.4 \quad (**)$$

b) Since $v = \frac{dx}{dt}$, where x is displacement, (**) can be rewritten as:

$$\frac{dx}{dt} = -117.4e^{-\frac{1}{13}t} + 127.4 \rightarrow dx = \left(-117.4e^{-\frac{1}{13}t} + 127.4\right)dt$$

$$\rightarrow x = \int \left(-117.4e^{-\frac{1}{13}t} + 127.4\right)dt$$

$$\leftrightarrow x = 1526.2e^{-\frac{1}{13}t} + 127.4t + c_2 \quad (***)$$

At $t = 0$, we have $x = 0$. Thus, (***) gives:

$$0 = 1526.2e^{-\frac{1}{13}(0)} + 127.4(0) + c_2 \rightarrow c_2 = -1526.2$$

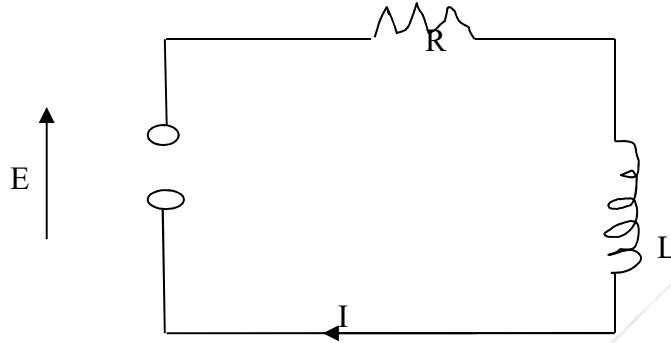
The displacement at any time t , is then given by:

$$x(t) = 1526.2e^{-\frac{1}{13}t} + 127.4t - 1526.2$$

c) Electrical circuits

The basic equation governing the amount of current I (in amperes) in a simple RL circuit consisting of resistance R (in ohms), an inductor L (in henries), and an electromotive force (abbreviated emf) E (in volts) is:

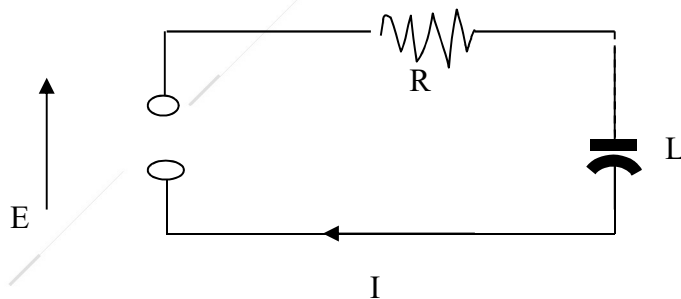
$$\frac{dI}{dt} + \frac{R}{L}I = \frac{E}{L} \quad (12)$$



For an RC circuit consisting of a resistance, a capacitance C (in farads), an emf, and no inductance, the equation governing the amount of electrical charge q (in coulombs) on the capacitor is:

$$\frac{dq}{dt} + \frac{1}{RC}q = \frac{E}{R} \quad (13)$$

The relationship between q and I is $I = \frac{dq}{dt} \quad (14)$



Example 1.23:

a) An RL circuit has an emf of 5volts, a resistance of 50 ohms, an inductance of 1henry, and no initial current. Find the current in the circuit at any time t ?

b) An RC circuit has an emf given (in volts) $400 \cos 2t$, a resistance of 100ohms, and a capacitance of 10^{-2} farad. Initially there is no charge on the capacitor. Find the current in the circuit at any time t ?

Answers:

a) Here $E = 5$, $R = 50$, and $L = 1$, hence (12) becomes:

$$\frac{dI}{dt} + 50I = 5$$

Here the solution of this linear differential equation is: $I = ce^{-50t} + \frac{1}{10}$

At $t = 0, I = 0$; thus $0 = ce^{-50(0)} + \frac{1}{10} \rightarrow c = -\frac{1}{10}$

The current at any time t is: $I(t) = -\frac{1}{10}e^{-50t} + \frac{1}{10}$

b) We first find the charge q and then use (14) to obtain the current.

Here $E = 400 \cos 2t$, $R = 100$, and $C = 10^{-2}$, hence (13) becomes:

$$\frac{dq}{dt} + q = 4 \cos 2t$$

This equation is linear, and its solution is

$$q(t) = ce^{-t} + \frac{8}{5} \sin 2t + \frac{4}{5} \cos 2t$$

At $t = 0, q = 0$, hence,

$$0 = ce^0 + \frac{8}{5} \sin 0 + \frac{4}{5} \cos 0 \rightarrow c = -\frac{4}{5}$$

$$q(t) = -\frac{4}{5}e^{-t} + \frac{8}{5} \sin 2t + \frac{4}{5} \cos 2t$$

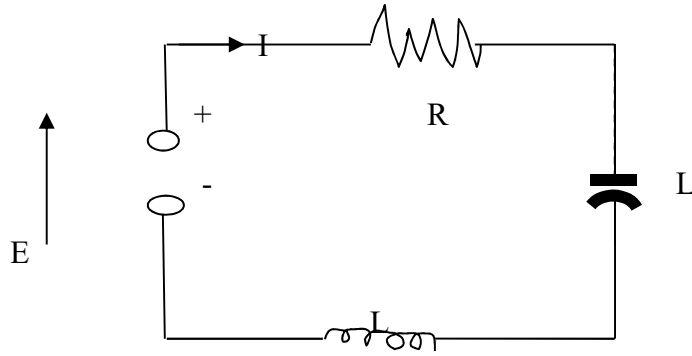
And using (14), we get:

$$I(t) = \frac{dq}{dt} = \frac{4}{5}e^{-t} + \frac{16}{5} \cos 2t + \frac{4}{5} \sin 2t$$

1.3.2. Application of second- order linear differential equations in engineering

Electrical circuit problems

The simple electrical circuit consists of a resistor R in ohms, a capacitor C in farads, an inductor L in henries, and an electromotive force (emf) $E(t)$ in volts, usually a battery or a generator, all connected in series. The current I flowing through the circuit is measured in amperes and the charge q on the capacitor is measured in coulombs.



Kirchhoff's loop law: The algebraic sum of the voltage drops in a simple closed circuits zero.

It is known that the voltage drops across a resistor, and an inductor are respectively RI , $(1/C)q$, and $L(dI/dt)$ where q is the charge on the capacitor. The voltage drop across an emf is $-E(t)$. Thus, from Kirchhoff's loop law, we have:

$$RI + L \frac{dI}{dt} + \frac{1}{C}q - E(t) = 0 \quad (15)$$

The relationship between q and I is:

$$I = \frac{dq}{dt} \rightarrow \frac{dI}{dt} = \frac{d^2q}{dt^2} \quad (16)$$

Substituting these values into (15), we get:

$$\begin{aligned} R \frac{dq}{dt} + L \frac{d^2q}{dt^2} + \frac{1}{C}q - E(t) &= 0 \\ \rightarrow \frac{d^2q}{dt^2} + \frac{R}{L} \frac{dq}{dt} + \frac{1}{LC}q &= \frac{1}{L}E(t) \end{aligned} \quad (17)$$

The initial conditions for q are:

$$q(0) = q_0 \text{ and } \left. \frac{dq}{dt} \right|_{t=0} = I(0) = I_0 \quad (18)$$

To obtain a differential equation for the current, we differentiate equation (15) with respect t and then substitute equation (16) directly into the resulting equation to obtain:

$$\frac{d^2I}{dt^2} + \frac{R}{L} \frac{dI}{dt} + \frac{1}{LC}I = \frac{1}{L} \frac{dE(t)}{dt} \quad (19)$$

The first initial condition is $I(0) = I_0$. The second initial condition is obtained from equation (15) by solving for $\frac{dI}{dt}$ and then setting $t = 0$. Thus,

$$\left. \frac{dI}{dt} \right|_{t=0} = \frac{1}{L}E(0) - \frac{R}{L}I_0 - \frac{1}{LC}q_0 \quad (20)$$

Note: The expression for the current can be gotten either by solving equation (19) directly or by solving equation (17) for the charge and then differentiating that expression.

Example 1.24:

An RCL circuit connected in series has $R=10\text{ohms}$, $C=10^{-2}\text{ farad}$, $L= \frac{1}{2}\text{ henry}$, and an applied voltage $E=12\text{ volts}$. Assuming no initial current at $t=0$ when the voltage is first applied, find the subsequent current in the system?

Answer: Substituting the given values into equation (19), we get the homogeneous equation:

$$\frac{d^2I}{dt^2} + 20\frac{dI}{dt} + 200I = 0$$

Its characteristic equation is: $\lambda^2 + 20\lambda + 200 = 0$,

The roots are:

$$\lambda_1 = -10 + 10i, \text{ and } \lambda_2 = -10 - 10i$$

$$\text{So, } I(t) = e^{-10t}(c_1 \cos 10t + c_2 \sin 10t) \quad (*)$$

The initial conditions are: $I_0 = I(0)$, and the equation (20),

$$\left. \frac{dI}{dt} \right|_{t=0} = \frac{12}{1/2} - \left(\frac{10}{1/2} \right)(0) - \frac{1}{(1/2)(10^{-2})}(0) = 24$$

Applying these conditions to (*), we get $c_1 = 0$ and $c_2 = \frac{12}{5}$, thus:

$$I(t) = \frac{12}{5}e^{-10t} \sin 10t$$

EXERCISES

1. Find the particular solution of the following equation:

$$x(x-1)y' + y = x^2(2x-1) \text{ with } y(2) = 4$$

2. Solve the following equations:

$$\text{a) } y' - 7y = 14x \quad \text{b) } y' + \frac{2}{x}y = x, y(1) = 0 \quad \text{c) } \frac{dN}{dt} + \frac{1}{t}N = t, N(2) = 8$$

$$\text{d) } 25\frac{dT}{dt} + T = 80e^{-0.04t} \quad \text{e) } \frac{dy}{dx} + 5y = 0 \quad \text{f) }$$

3. Solve the following linear ODE'S

a) $y' - \frac{4}{x}y = \frac{x+1}{x}$ b) $y' + y \cos x = \frac{1}{2} \sin 2x$ c) $y' + \frac{1-2x}{x^2}y = 1$

4. a) What can one say about the general solution of $y'' - 8y' = 0$ if two particular solutions are known to be $y_1 = e^{8x}$, and $y_2 = 1$?

b) What can one say about the general solution of $y'' + y' = 0$ if two particular solutions are known to be $y_1 = 8$, and $y_2 = 1$?

5. Solve the following differential equations:

a) $\ddot{x} - 20\dot{x} + 64x = 0$ b) $\frac{d^2R}{d\theta^2} + 5\frac{dR}{d\theta} = 0$ c) $y'' + y' + \frac{1}{4}y = 0$

d) $y'' - y' - 2y = 4x^2$, e) $y'' - y' - 2y = e^{3x}$ f) $\ddot{y} - 6\dot{y} + 25y = 50t^3 - 36t^2 - 63t + 18$

6. Solve the following initial-value problems:

a) $y'' - y' - 2y = e^{3x}; y(0) = 1, y'(0) = 2$ b) $y'' - y' - 2y = e^{3x}; y(0) = 2, y'(0) = 1$

c) $y'' + 4y = \sin^2 2x; y(\pi) = 0, y'(\pi) = 0$ d) $y'' + y = 0; y(2) = 0, y'(2) = 0$

7. A body at unknown temperature is placed in a room which is held at a constant temperature of 30°F . If after 10 minutes the temperature of the body is 0°F and after 20 minutes the temperature of the body is 15°F . Find the unknown initial temperature?

8. A steel ball weighing 2 N is dropped from a height of 3000m with no velocity. As it falls, the ball encounters air resistance numerically equal to $v/8$ (in pounds), where v denotes the velocity of the ball (in meter per second). Find:

a) The limiting velocity for the ball?

b) The time required for the ball to hit the ground

9. An RL circuit has an emf given (in volts) by $3 \sin 2t$, a resistance of 10ohms, an inductance of 0.5 henry, and an initial current of 6 amperes. Find the current in the circuit at any time t ?

10. An RCL circuit connected in series has $R=180$ ohms, $C= 1/280$ farad, $L=20$ henries, and an applied voltage $E(t)= 10 \sin t$. Assuming no initial charge on the capacitor, but an initial current of 1 ampere at $t=0$ when the voltage is first applied. Find the subsequent charge on the capacitor?

1.4. SYSTEMS OF ODES WITH CONSTANT COEFFICIENTS

1.4.1. INTRODUCTION

In the introduction to this section we briefly discussed how a system of differential equations can arise from a population problem in which we keep track of the population of both the prey and the predator. It makes sense that the number of prey present will affect the number of the predator

present. Likewise, the number of predator present will affect the number of prey present. Therefore the differential equation that governs the population of either the prey or the predator should in some way depend on the population of the other. This will lead to two differential equations that must be solved simultaneously in order to determine the population of the prey and the predator.

The whole point of this is to notice that systems of differential equations can arise quite easily from naturally occurring situations. Developing an effective predator-prey system of differential equations is not the subject of this chapter. However, systems can arise from n^{th} order linear differential equations as well. Before we get into this however, let's write down a system and get some terminology out of the way.

We are going to be looking at first order, linear systems of differential equations. These terms mean the same thing that they have meant up to this point. The largest derivative anywhere in the system will be a first derivative and all unknown functions and their derivatives will only occur to the first power and will not be multiplied by other unknown functions.

Example 1.25: This is an example of a system of first order, linear differential equations:

$$\begin{cases} x' = x + 2y \\ y' = 3x + 2y \end{cases}$$

We call this kind of system a ***coupled*** system since knowledge of y is required in order to find x and likewise knowledge of x is required to find y . We will worry about how to go about solving these later. At this point we are only interested in becoming familiar with some of the basics of systems.

Now, as mentioned earlier, we can write an n^{th} order linear differential equation as a system. Let's see how that can be done.

Example 1.26: Write the following 2^{nd} order differential equation as a system of first order, linear differential equations.

$$2y'' - 5y' + y = 0; \quad y(3) = 6, \quad y'(3) = -1$$

Solution

We can write higher order differential equations as a system with a very simple change of variable. We'll start by defining the following two new functions.

$$\begin{cases} x_1(t) = y(t) \\ x_2(t) = y'(t) \end{cases}$$

Now notice that if we differentiate both sides of these we get,

$$\begin{cases} x_1' = y' = x_2 \\ x_2' = y'' = -\frac{1}{2}y + \frac{5}{2}y' = -\frac{1}{2}x_1 + \frac{5}{2}x_2 \end{cases}$$

Note the use of the differential equation in the second equation. We can also convert the initial conditions over to the new functions.

$$\begin{cases} x_1(3) = y(3) = 6 \\ x_2(3) = y'(3) = -1 \end{cases}$$

Putting all of this together gives the following system of differential equations.

$$\begin{cases} x_1' = x_2 & x_1(3) = 6 \\ x_2' = -\frac{1}{2}x_1 + \frac{5}{2}x_2 & x_2(3) = -1 \end{cases}$$

We will call the system in the above example an **Initial Value Problem** just as we did for differential equations with initial conditions.

1.4.2 DIFFERENTIAL OPERATORS AND AN OPERATOR METHOD

1.4.2.1 Types of linear systems

The general linear system of two first-order differential equation in two unknown functions x and y is of the form:

$$\begin{cases} a_1(t)x' + a_2(t)y' + a_3(t)x + a_4(t)y = f_1(t) \\ b_1(t)x' + b_2(t)y' + b_3(t)x + b_4(t)y = f_2(t) \end{cases} \quad (1.1)$$

Where $x' = \frac{dx}{dt}$ and $y' = \frac{dy}{dt}$

Example 1.27: The following example is two first- order differential equation

$$\begin{cases} 2x' + 3y' - 2x + y = t^2 \\ x' - 2y' + 3x + 4y = e^t \end{cases}$$

The solution of (1.1) is an ordered pair of real functions (f, g) such that $x = f(t)$ and $y = g(t)$ simultaneously satisfy both equations of the system (1.1) on some real interval

$a \leq t \leq b$.

The general linear system of n first-order differential equations in n unknown functions $y_1, y_2, y_3, \dots, y_n$ is of the form:

$$\begin{cases} a_{11}(t)y'_1 + a_{12}(t)y'_2 + \cdots + a_{1n}(t)y'_n + \beta_{11}(t)y_1 + \beta_{12}(t)y_2 + \beta_{13}(t)y_3 + \cdots + \beta_{1n}(t)y_n = f_1(t) \\ a_{21}(t)y'_1 + a_{22}(t)y'_2 + \cdots + a_{2n}(t)y'_n + \beta_{21}(t)y_1 + \beta_{22}(t)y_2 + \beta_{23}(t)y_3 + \cdots + \beta_{2n}(t)y_n = f_2(t) \\ \vdots \\ a_{n1}(t)y'_1 + a_{n2}(t)y'_2 + \cdots + a_{nn}(t)y'_n + \beta_{n1}(t)y_1 + \beta_{n2}(t)y_2 + \beta_{n3}(t)y_3 + \cdots + \beta_{nn}(t)y_n = f_n(t) \end{cases} \quad (1.2)$$

If we use the summation sign, then (1.2) becomes:

$$\sum_{i=1}^n \sum_{j=1}^n (a_{ij}(t)y'_j + \beta_{ij}(t)y_j) \quad (1.3)$$

Where i is index indicating us the number of equations that we have, and j is index indicate us the number of unknown functions that we have in our system.

Example 1.28: The following example is three first- order differential equation

$$\begin{cases} x' + y' - 2z' + 2x - 3y + z = t \\ 2x' - y' + 3z' + x + 4y - 5z = \sin t \\ x' + 2y' + z' - 3x + 2y - z = \cos t \end{cases}$$

The solution of this example is an ordered triple of real functions (f, g, h) such that $x = f(t)$, $y = g(t)$ and $z = h(t)$ simultaneously satisfy both equations of the system on some real interval $a \leq t \leq b$.

Systems of the form (1.1) to (1.3) contained only first derivatives, and we now consider the basic linear system involving higher derivatives.

The following given system of linear equation is a linear system of two second-order differential equations in two unknown functions x and y :

$$\begin{cases} a_1(t)x'' + a_2(t)y'' + a_3(t)x' + a_4(t)y' + a_5(t)x + a_6(t)y = f_1(t) \\ b_1(t)x'' + b_2(t)y'' + b_3(t)x' + b_4(t)y' + b_5(t)x + b_6(t)y = f_2(t) \end{cases} \quad (1.4)$$

In this course we shall be concerned with systems having constant coefficients.

Example 1.29: The following example is two first- order differential equation

$$\begin{cases} 2x'' + 5y'' + 7x' + 3y' + 2y = 3t + 1 \\ 3x'' + 2y'' - 2y' + 4x + y = 0 \end{cases}$$

We consider the special type of linear system (1.1), which is of the form:

$$\begin{cases} x' = a_{11}(t)x + a_{12}(t)y + f_1(t) \\ y' = b_{11}(t)x + b_{12}(t)y + f_2(t) \end{cases} \quad (1.5)$$

This is the **normal form** in the case of two linear differential equations in two unknown functions.

Example 1.30:

$$(a) \begin{cases} x' = t^2x + (t+1)y + t^3 \\ y' = te^tx + t^3y - e^t \end{cases} \text{ This is a system with variable coefficients}$$

$$(b) \begin{cases} x' = 5x + 7y + t^2 \\ y' = 2x - 3y + 2t \end{cases} \text{ This is a system with constant coefficients}$$

The normal form in the general case of linear system of n differential equations in n unknown functions $x_1, x_2, x_3, \dots, x_n$ is

$$\begin{cases} x'_1 = a_{11}(t)x_1 + a_{12}(t)x_2 + a_{13}(t)x_3 + \dots + a_{1n}(t)x_n + f_1(t) \\ x'_2 = a_{21}(t)x_1 + a_{22}(t)x_2 + a_{23}(t)x_3 + \dots + a_{2n}(t)x_n + f_2(t) \\ \vdots \\ x'_n = a_{n1}(t)x_1 + a_{n2}(t)x_2 + a_{n3}(t)x_3 + \dots + a_{nn}(t)x_n + f_n(t) \end{cases} \quad (1.6)$$

An important fundamental property of a normal linear system (1.6) is its relationship to a single n th-order linear differential equation in one unknown function. Specifically, consider the **normalized** (meaning, the coefficient of the highest derivative is one) n th-order linear differential equation:

$$x^{(n)} + a_1(t)x^{(n-1)} + a_2(t)x^{(n-2)} + \dots + a_{n-1}(t)x' + a_n(t)x = f(t) \quad (1.7)$$

In the one unknown function x .

$$\text{Let } x_1 = x, \quad x_2 = x', \quad x_3 = x'', \dots, x_{n-1} = x^{(n-2)}, x_n = x^{(n-1)} \quad (1.8)$$

From (1.8), we have :

$$x' = x'_1, \quad x'' = x'_2, \dots, x^{(n-1)} = x'_{n-1}, x^{(n)} = x'_n \quad (4.9)$$

Then using both (1.8) and (1.9), the single n th-order equation (1.7) can be transformed into

$$\begin{cases} x'_1 = x_2 \\ x'_2 = x_3 \\ \vdots \\ x'_{n-1} = x_n \\ x'_n = -a_n(t)x_1 - a_{n-1}(t)x_2 - \dots - a_1(t)x_n + f(t) \end{cases} \quad (1.10)$$

Which is a special case of the normal linear system (1.6) of n equations in n unknown functions. Thus, we see that a single n th-order linear differential equation of form (1.7) in one unknown function is indeed intimately related to a normal linear system (1.6) of n first-order differential equation in n unknown functions.

1.4.2.2. Differential Operators

1.4.2.2.1. Definition

In this section we shall present a symbolic operator (*differential operators D*) method for solving linear systems with constant coefficients.

Let x be an n -times differentiable function of the independent variable t .

$$\frac{dx}{dt} = Dx$$

$$Dx = x', D^2x = x'', \dots, D^n x = x^{(n)} = \frac{d^n x}{dt^n}$$

1.4.2.3. Properties of differential operators

- i. $(D + c)x = Dx + cx \leftrightarrow x' + cx$ where c is a constant
 - ii. $(\alpha D^n + \beta D^m)x = \alpha D^n x + \beta D^m x \leftrightarrow \alpha x^{(n)} + \beta x^{(m)}$ where α , and β are the constants
 - iii. $D[\alpha f(t) + \beta g(t)] = \alpha Df(t) + \beta Dg(t)$ means that **differential operators is linear operator**
 - iv. $(a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n)x = a_0 x^{(n)} + a_1 x^{(n-1)} + \dots + a_{n-1} x' + a_n x$
- The expression $a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n$ by itself, where $a_0, a_1, \dots, a_{n-1}, a_n$ are constants, is called **linear differential operator** with constant coefficients.

Example 1.31: Consider the linear differential operator

$$3D^2 + 5D - 2.$$

If x is a twice differentiable function of t , then $(3D^2 + 5D - 2)x$ denotes $3x'' + 5x' - 2x$

Consider $x = e^{2t}t^3$, then we have

$$\begin{aligned} (3D^2 + 5D - 2)(e^{2t}t^3) &= 3 \frac{d^2}{dt^2}(e^{2t}t^3) + 5 \frac{d}{dt}(e^{2t}t^3) - 2(e^{2t}t^3) \\ &= 3 \frac{d}{dt}(3e^{2t}t^2 + 2e^{2t}t^3) + 5(3e^{2t}t^2 + 2e^{2t}t^3) - 2e^{2t}t^3 \\ &= 3(6e^{2t}t^2 + 6e^{2t}t + 4e^{2t}t^3 + 6e^{2t}t^2) + 5(3e^{2t}t^2 + 2e^{2t}t^3) - 2e^{2t}t^3 \\ &= (22t^3 + 51t^2 + 16t)e^{2t} \end{aligned}$$

We shall now discuss certain useful properties of linear differential operator with constant coefficients. In order to facilitate our discussion, we shall let L denote this operator. We have:

$$L \equiv a_0 D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n,$$

where $a_0, a_1, \dots, a_{n-1}, a_n$ are constants.

Properties of operator L

- i. $L[\alpha f(t) + \beta g(t)] = \alpha Lf(t) + \beta Lg(t)$
- ii. $L_1 L_2 f = L_2 L_1 f = Lf$

Example 1.32:

Let $L_1 = D^2 + 1$, $L_2 = 3D + 2$, $f(t) = t^3$

$$\begin{aligned} L_1 L_2 f &= (D^2 + 1)(3D + 2)t^3 = (D^2 + 1)(9t^2 + 2t^3) \\ &= 9(D^2 + 1)t^2 + 2(D^2 + 1)t^3 = 9(2 + t^2) + 2(6t + t^3) \\ &= 2t^3 + 9t^2 + 12t + 18 \end{aligned}$$

Or

$$L_1 L_2 f = Lf, \text{ where } L \equiv (D^2 + 1)(3D + 2) = 3D^3 + 2D^2 + 3D + 2$$

$$Lf = (3D^3 + 2D^2 + 3D + 2)t^3 = 3D^3 t^3 + 2D^2 t^3 + 3D t^3 + 2t^3 = 2t^3 + 9t^2 + 12t + 18$$

1.4.3. Resolution of linear system using operator method

We now proceed to explain a symbolic operator method for solving linear systems with constant coefficients.

We consider a linear system of the form:

$$\begin{cases} L_1 x + L_2 y = f_1(t) \\ L_3 x + L_4 y = f_2(t) \end{cases} \quad (1.11)$$

Example 1.32: Solve the following linear system using differential operator

$$\begin{cases} 2x' - 2y' - 3x = t \\ 2x' + 2y' + 3x + 8y = 2 \end{cases}$$

Solution: Introducing operator notation this system takes the following form:

$$\begin{aligned} &\begin{cases} (2D - 3)x - 2Dy = t \\ (2D + 3)x + (2D + 8)y = 2 \end{cases} \quad \begin{vmatrix} (2D + 8) \\ 2D \end{vmatrix} \\ \rightarrow (2D + 8)(2D - 3)x + 2D(2D + 3)x &= (2D + 8)t + (2D)2 \\ \Leftrightarrow (8D^2 + 16D - 24)x &= 2 + 8t + 0 \\ \Leftrightarrow (D^2 + 2D - 3)x = t + \frac{1}{4} &\Leftrightarrow x'' + 2x' - 3x = t + \frac{1}{4} \\ \rightarrow x(t) = c_1 e^t + c_2 e^{-3t} - \frac{1}{3}t - \frac{11}{36} \end{aligned}$$

$$(2D - 3)x - 2Dy = t \rightarrow 2Dy = -t + (2D - 3)x \rightarrow Dy = \frac{1}{2}[-t + (2D - 3)x]$$

$$Dy = \frac{1}{2} \left[-t + (2D - 3)(c_1 e^t + c_2 e^{-3t} - \frac{1}{3}t - \frac{11}{36}) \right]$$

$$Dy = \frac{1}{2} \left[-t + (2c_1 e^t - 6c_2 e^{-3t} - \frac{2}{3} - 3(c_1 e^t + c_2 e^{-3t} - \frac{1}{3}t - \frac{11}{36})) \right]$$

$$Dy = \frac{1}{2} \left[-c_1 e^t - 9c_2 e^{-3t} + t + \frac{1}{6} \right]$$

$$y(t) = \frac{1}{2} \left[-c_1 e^t + 3c_2 e^{-3t} + \frac{t}{6} + \frac{t^2}{2} \right]$$

EXERCISES

- I. Use the operator method described to find the general solution in the following linear systems:

$$\begin{array}{ll} 1. \begin{cases} x' + y' - 2x - 4y = e^t \\ x' + y' - y = e^{4t} \end{cases} & 5. \begin{cases} x' + y' + 2y = \sin t \\ x' + y' - x - y = 0 \end{cases} \\ 2. \begin{cases} 5x' + y' - 3x + y = 0 \\ 4x' + y' - 3x = -3t \end{cases} & 6. \begin{cases} 2x' + y' + x + y = t^2 + 4t \\ x' + y' + 2x + 2y = 2t^2 - 2t \end{cases} \\ 3. \begin{cases} 2x' + 4y' + xy = 3e^t \\ x' + y' + 2x + 2y = e^t \end{cases} & 7. \begin{cases} x'' + y' - x + y = 1 \\ y'' + x' - x + y = 0 \end{cases} \\ 4. \begin{cases} x' + y' - x = -2t \\ x' + y' - 3x - y = t^2 \end{cases} & 8. \begin{cases} x'' + 4y' + x - 4y = 0 \\ x' + y' - x + 9y = e^{2t} \end{cases} \end{array}$$

- II. Transform the single linear differential equation into a system of first-order differential equations.

$$\begin{array}{ll} 1. x'' - 3x' + 2x = t^2 & 3. x''' + 2x'' - x' - 2x = e^{3t} \\ 2. x''' + tx'' + 2t^2x' - 5t^4 = 0 & 4. x^{iv} - t^2x'' + 2tx = \cos t \end{array}$$

1.4.4. Homogeneous linear systems with constant coefficients: Two equations in two unknown functions.

1.4.4.1. Introduction

In this case, we consider the following general form of homogeneous linear system:

$$\begin{cases} x' = a_1x + b_1y \\ y' = a_2x + b_2y \end{cases} \quad (1.12)$$

Where a_1, b_1, a_2 , and b_2 are real constants.

Let us assume that the solutions of (4.12) have the following form:

$$\begin{cases} x = Ae^{\lambda t} \\ y = Be^{\lambda t} \end{cases} \quad (1.13)$$

Where A, B , and λ are constants. If we substitute (1.13) into (1.12), then we obtain

$$\begin{cases} A\lambda e^{\lambda t} = a_1Ae^{\lambda t} + b_1Be^{\lambda t} \\ B\lambda e^{\lambda t} = a_2Ae^{\lambda t} + b_2Be^{\lambda t} \end{cases} \leftrightarrow \begin{cases} A\lambda e^{\lambda t} - a_1Ae^{\lambda t} - b_1Be^{\lambda t} = 0 \\ B\lambda e^{\lambda t} - a_2Ae^{\lambda t} - b_2Be^{\lambda t} = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} (\lambda e^{\lambda t} - a_1 A e^{\lambda t})A - b_1 B e^{\lambda t} = 0 \\ -a_2 A e^{\lambda t} + (\lambda e^{\lambda t} - b_2 B e^{\lambda t})B = 0 \end{cases} \Leftrightarrow \begin{cases} [(\lambda - a_1)A - b_1 B]e^{\lambda t} = 0 \\ [-a_2 A + (\lambda - b_2)B]e^{\lambda t} = 0 \end{cases}$$

As $e^{\lambda t} \neq 0$, then we have:

$$\begin{cases} (a_1 - \lambda)A + b_1 B = 0 \\ a_2 A + (b_2 - \lambda)B = 0 \end{cases}$$

Here unknowns are A and B .

This system has the trivial solution $A = B = 0$.

Theorem 1.4.1: *A system of n homogeneous linear algebraic equations in n unknown variables has a nontrivial solution if and only if the determinant of coefficients of the system is equal to zero.*

The above system has nontrivial solution iff:

$$\begin{vmatrix} a_1 - \lambda & b_1 \\ a_2 & b_2 - \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} a_1 - \lambda & b_1 \\ a_2 & b_2 - \lambda \end{vmatrix} = 0$$

$$\Leftrightarrow \lambda^2 - (a_1 + b_2)\lambda + (a_1 b_2 - a_2 b_1) = 0 \quad (1.14)$$

in the unknown λ .

This equation is called **the characteristic equation** associated with the system (1.12). Its roots λ_1 and λ_2 are called the **Characteristic roots**. According to the values of λ_1 and λ_2 we consider three cases:

First case: The roots λ_1 and λ_2 are real and distinct.

Second case: The roots λ_1 and λ_2 are conjugate complex.

Third case: The roots λ_1 and λ_2 are real and equal.

1.4.4.2. Analysis of those three cases

First case: The roots of the characteristic equation are real and distinct

If the roots λ_1 and λ_2 of (1.14) are real and distinct, then summarize this case in the following theorem

Theorem 1.4.2: *If the roots λ_1 and λ_2 of (1.14) associated with the system (1.12) are real and distinct, then the system (1.12) has two nontrivial linearly independent solutions of the form*

$$\begin{cases} x = A_1 e^{\lambda_1 t} \\ y = B_1 e^{\lambda_1 t} \end{cases} \text{ and } \begin{cases} x = A_2 e^{\lambda_2 t} \\ y = B_2 e^{\lambda_2 t} \end{cases}$$

Where $A_1, B_1, A_2,$ and B_2 are definite constants.

The general solution of the system (1.12) may thus be written as:

$$\begin{cases} x = c_1 A_1 e^{\lambda_1 t} + c_2 A_2 e^{\lambda_2 t} \\ y = c_1 B_1 e^{\lambda_1 t} + c_2 B_2 e^{\lambda_2 t} \end{cases}$$

Where $c_1,$ and c_2 are arbitrary constants.

Example 1.33: Solve the following system of differential equations:

$$\begin{cases} x' = 6x - 3y \\ y' = 2x + y \end{cases}$$

Answer: We assume that a solution has this form:

$$\begin{cases} x = Ae^{\lambda t} \\ y = Be^{\lambda t} \end{cases} \rightarrow \begin{cases} x' = A\lambda e^{\lambda t} \\ y' = B\lambda e^{\lambda t} \end{cases}$$

The given equation becomes:

$$\begin{aligned} \begin{cases} x' = 6x - 3y \\ y' = 2x + y \end{cases} &\leftrightarrow \begin{cases} A\lambda e^{\lambda t} = 6Ae^{\lambda t} - 3Be^{\lambda t} \\ B\lambda e^{\lambda t} = 2Ae^{\lambda t} + Be^{\lambda t} \end{cases} \\ \begin{cases} [(6 - \lambda)A - 3B]e^{\lambda t} = 0 \\ [2A + (1 - \lambda)B]e^{\lambda t} = 0 \end{cases} &\leftrightarrow \begin{cases} (6 - \lambda)A - 3B = 0 \\ 2A + (1 - \lambda)B = 0 \end{cases} \quad (*) \end{aligned}$$

Where λ is unknown variable.

For nontrivial solutions of this system we must have:

$$\begin{vmatrix} 6 - \lambda & -3 \\ 2 & 1 - \lambda \end{vmatrix} = 0 \leftrightarrow \lambda^2 - 7\lambda + 12 = 0 \rightarrow \lambda_1 = 3 \text{ and } \lambda_2 = 4.$$

Setting $\lambda = \lambda_1 = 3$ in (*), we obtain: $\begin{cases} 3A - 3B = 0 \\ 2A - 2B = 0 \end{cases} \rightarrow A = B = 1$

We find the nontrivial solution as: $\begin{cases} x = e^{3t} \\ y = e^{3t} \end{cases}$

Now setting $\lambda = \lambda_2 = 4$ in (*), we obtain:

$$\begin{cases} 2A - 3B = 0 \\ 2A - 3B = 0 \end{cases} \rightarrow A = 3, B = 2$$

We find the nontrivial solution as: $\begin{cases} x = 3e^{4t} \\ y = 2e^{4t} \end{cases}$

The general solution is: $\begin{cases} x = c_1 e^{3t} + 3c_2 e^{4t} \\ y = c_1 e^{3t} + 2c_2 e^{4t} \end{cases}$

Where c_1 , and c_2 are arbitrary constants.

Second case: The roots of the characteristic equation are conjugate complex

If the roots λ_1 and λ_2 of (4.14) are conjugate complex numbers $\alpha + i\beta$ and $\alpha - i\beta$, then we still

obtain two distinct solutions as:
$$\begin{cases} x = K_1 e^{(\alpha+i\beta)t} \\ y = H_1 e^{(\alpha+i\beta)t} \end{cases} \text{ and } \begin{cases} x = K_2 e^{(\alpha-i\beta)t} \\ y = H_2 e^{(\alpha-i\beta)t} \end{cases}$$

Where K_1, H_1, K_2 , and H_2 are complex constants and we have:

$$K_1 = A_1 + iA_2, H_1 = B_1 + iB_2$$

$$\begin{aligned} & \begin{cases} x = (A_1 + iA_2)e^{(\alpha+i\beta)t} \\ y = (B_1 + iB_2)e^{(\alpha+i\beta)t} \end{cases} \leftrightarrow \begin{cases} x = (A_1 + iA_2)e^{\alpha t} e^{i\beta t} \\ y = (B_1 + iB_2)e^{\alpha t} e^{i\beta t} \end{cases} \\ & \leftrightarrow \begin{cases} x = (A_1 + iA_2)e^{\alpha t}(\cos \beta t + i \sin \beta t) \\ y = (B_1 + iB_2)e^{\alpha t}(\cos \beta t + i \sin \beta t) \end{cases} \\ & \leftrightarrow \begin{cases} x = e^{\alpha t}[(A_1 \cos \beta t - A_2 \sin \beta t) + i(A_2 \cos \beta t + A_1 \sin \beta t)] \\ y = e^{\alpha t}[(B_1 \cos \beta t - B_2 \sin \beta t) + i(B_2 \cos \beta t + B_1 \sin \beta t)] \end{cases} \quad (1.15) \end{aligned}$$

It can be shown that a pair $[f_1(t) + if_2(t), g_1(t) + ig_2(t)]$ of complex functions is a solution of the system (1.12) iff both the pair $[f_1(t), g_1(t)]$ consisting of their real parts and the pair $[f_2(t), g_2(t)]$ consisting of their imaginary parts are solution of (1.12).

$$\text{Thus both the real part } \begin{cases} x = e^{\alpha t}(A_1 \cos \beta t - A_2 \sin \beta t) \\ y = e^{\alpha t}(B_1 \cos \beta t - B_2 \sin \beta t) \end{cases} \quad (1.16)$$

$$\text{And the imaginary part } \begin{cases} x = e^{\alpha t}(A_2 \cos \beta t + A_1 \sin \beta t) \\ y = e^{\alpha t}(B_2 \cos \beta t + B_1 \sin \beta t) \end{cases} \quad (1.17)$$

Of the solution (1.15) of the system (1.12) are also solutions of (1.12). Furthermore, the solutions (1.16) and (1.17) are linearly independent.

Now, we summarize the above results in the following theorem:

Theorem 1.4.3: *If the roots λ_1 and λ_2 of (1.14) associated with the system (1.12) are the conjugate complex numbers $\alpha \pm i\beta$, then the system (1.12) has two real linearly independent solutions of the form*

$$\begin{cases} x = e^{\alpha t}(A_1 \cos \beta t - A_2 \sin \beta t) \\ y = e^{\alpha t}(B_1 \cos \beta t - B_2 \sin \beta t) \end{cases} \text{ and } \begin{cases} x = e^{\alpha t}(A_2 \cos \beta t + A_1 \sin \beta t) \\ y = e^{\alpha t}(B_2 \cos \beta t + B_1 \sin \beta t) \end{cases}$$

Where A_1, B_1, A_2 , and B_2 are definite constants.

The general solution of the system (1.12) may thus be written as:

$$\begin{cases} x = e^{\alpha t} [c_1(A_1 \cos \beta t - A_2 \sin \beta t) + c_2(A_2 \cos \beta t + A_1 \sin \beta t)] \\ y = e^{\alpha t} [c_1(B_1 \cos \beta t - B_2 \sin \beta t) + c_2(B_2 \cos \beta t + B_1 \sin \beta t)] \end{cases}$$

Where c_1 , and c_2 are arbitrary constants.

Example 1.4.11: Solve the following system of differential equations:

$$\begin{cases} x' = 3x + 2y \\ y' = -5x + y \end{cases} \quad (**)$$

Answer: We assume that a solution has this form:

$$\begin{cases} x = Ae^{\lambda t} \\ y = Be^{\lambda t} \end{cases} \rightarrow \begin{cases} x' = A\lambda e^{\lambda t} \\ y' = B\lambda e^{\lambda t} \end{cases}$$

The given equation becomes:

$$\begin{aligned} \begin{cases} x' = 6x - 3y \\ y' = 2x + y \end{cases} &\leftrightarrow \begin{cases} A\lambda e^{\lambda t} = 3Ae^{\lambda t} + 2Be^{\lambda t} \\ B\lambda e^{\lambda t} = -5Ae^{\lambda t} + Be^{\lambda t} \end{cases} \\ \begin{cases} [(3 - \lambda)A + 2B]e^{\lambda t} = 0 \\ [-5A + (1 - \lambda)B]e^{\lambda t} = 0 \end{cases} &\leftrightarrow \begin{cases} (3 - \lambda)A + 2B = 0 \\ -5A + (1 - \lambda)B = 0 \end{cases} \quad (***) \end{aligned}$$

Where λ is unknown variable.

For nontrivial solutions of this system we must have:

$$\begin{vmatrix} 3 - \lambda & 2 \\ -5 & 1 - \lambda \end{vmatrix} = 0 \leftrightarrow \lambda^2 - 4\lambda + 13 = 0 \rightarrow \lambda_1 = 2 + 3i \text{ and } \lambda_2 = 2 - 3i.$$

Setting $\lambda = \lambda_1 = 2 + 3i$ in (***), we obtain:

$$\begin{cases} (1 - 3i)A + 2B = 0 \\ -5A + (-1 - 3i)B = 0 \end{cases} \rightarrow A = 2, B = -1 + 3i$$

We find the nontrivial solution as:

$$\begin{cases} x = 2e^{(2+3i)t} \\ y = (-1 + 3i)e^{(2+3i)t} \end{cases} \leftrightarrow \begin{cases} x = e^{2t}[(2 \cos 3t + i(2 \sin 3t))] \\ y = e^{2t}[(-\cos 3t - 3 \sin 3t) + i(3 \cos 3t - \sin 3t)] \end{cases}$$

Since both the real and imaginary parts of this solution of system (**) are themselves solutions of (**), we thus get the two real solutions as:

$$\begin{cases} x = 2e^{2t} \cos 3t \\ y = -e^{2t}(\cos 3t + 3 \sin 3t) \end{cases} \text{ and } \begin{cases} x = 2e^{2t} \sin 3t \\ y = e^{2t}(3 \cos 3t - \sin 3t) \end{cases}$$

As those both solutions are linearly independent, then the general solution is written as:

$$\begin{aligned}x &= 2e^{2t}(c_1 \cos 3t + c_2 \sin 3t) \\y &= e^{2t}[c_1(-\cos 3t - 3\sin 3t) + c_2(3\cos 3t - \sin 3t)]\end{aligned}$$

Where c_1 , and c_2 are arbitrary constants.

Third case: The roots of the characteristic equation are real and equal

If the roots λ_1 and λ_2 of (1.14) are real and equal, it would appear that we could find only one solution of the form (1.13). Except in the special sub-case in which $a_1 = b_2 \neq 0, a_2 = b_1 = 0$, this is indeed true. In general, how shall we then proceed to find a second, linearly independent solution? This would lead us to expect a second solution of the form:

$$\begin{cases} x = Ate^{\lambda t} \\ y = Bte^{\lambda t} \end{cases}$$

However, the situation here is not quite so simple. We must actually seek a second solution of the form:

$$\begin{cases} x = (A_1t + A_2)e^{\lambda t} \\ y = (B_1t + B_2)e^{\lambda t} \end{cases} \quad (1.18)$$

We summarize this case in the following theorem:

Theorem 1.4.4: *If the roots λ_1 and λ_2 of (1.14) associated with the system (1.12) are real and equal. Let λ denote their common value. Further assume that system (4.12) is not such that $a_1 = b_2 \neq 0, a_2 = b_1 = 0$; then the system (1.12) has two linearly independent solutions of the form*

$$\begin{cases} x = Ate^{\lambda t} \\ y = Bte^{\lambda t} \end{cases} \text{ and } \begin{cases} x = (A_1t + A_2)e^{\lambda t} \\ y = (B_1t + B_2)e^{\lambda t} \end{cases}$$

Where A, B, A_1, B_1, A_2 , and B_2 are definite constant, A_1 and B_1 are not both zero, and $B_1/A_1 = B/A$.

The general solution of the system (1.12) may thus be written as:

$$\begin{cases} x = c_1 Ate^{\lambda t} + c_2 (A_1t + A_2)e^{\lambda t} \\ y = c_1 Bte^{\lambda t} + c_2 (B_1t + B_2)e^{\lambda t} \end{cases}$$

Where c_1 , and c_2 are arbitrary constants.

Example 1.34: Solve the following system of differential equations: $\begin{cases} x' = 4x - y \\ y' = x + 2y \end{cases}$

Answer: We assume that a solution has this form: $\begin{cases} x = Ae^{\lambda t} \\ y = Be^{\lambda t} \end{cases} \rightarrow \begin{cases} x' = A\lambda e^{\lambda t} \\ y' = B\lambda e^{\lambda t} \end{cases}$

The given equation becomes: $\begin{cases} x' = 4x - y \\ y' = x + 2y \end{cases} \leftrightarrow \begin{cases} A\lambda e^{\lambda t} = 4Ae^{\lambda t} - Be^{\lambda t} \\ B\lambda e^{\lambda t} = Ae^{\lambda t} + 2Be^{\lambda t} \end{cases}$

$$\begin{cases} [(4 - \lambda)A - B]e^{\lambda t} = 0 \\ [A + (2 - \lambda)B]e^{\lambda t} = 0 \end{cases} \leftrightarrow \begin{cases} (4 - \lambda)A - B = 0 \\ A + (2 - \lambda)B = 0 \end{cases} \quad (****)$$

Where λ is unknown variable.

For nontrivial solutions of this system we must have:

$$\begin{vmatrix} 4 - \lambda & -1 \\ 1 & 2 - \lambda \end{vmatrix} = 0 \leftrightarrow \lambda^2 - 6\lambda + 9 = 0 \rightarrow \lambda_1 = \lambda_2 = 3$$

Setting $\lambda = 3$ in (****), we obtain: $\begin{cases} A - B = 0 \\ A - B = 0 \end{cases} \rightarrow A = B = 1$

We find the nontrivial solution as: $\begin{cases} x = e^{3t} \\ y = e^{3t} \end{cases}$

Since the roots of the characteristic equation are both equal to 3, we must seek a second solution of the form (1.18), with $\lambda = 3$. That is, we must determine A_1, B_1, A_2 , and B_2 (with A_1 , and B_1

not both zero) such that: $\begin{cases} x = (A_1 t + A_2)e^{3t} \\ y = (B_1 t + B_2)e^{3t} \end{cases} \quad (&)$

It is a solution of a given system in our example.

If we substitute this system in a given example, then we get:

$$\begin{cases} (3A_1 t + 3A_2 + A_1)e^{3t} = 4(A_1 t + A_2)e^{3t} - (B_1 t + B_2)e^{3t} \\ (3B_1 t + 3B_2 + B_1)e^{3t} = (A_1 t + A_2)e^{3t} + 2(B_1 t + B_2)e^{3t} \end{cases} \leftrightarrow \begin{cases} (A_1 - B_1)t + (A_2 - A_1 - B_2) = 0 \\ (A_1 - B_1)t + (A_2 - B_1 - B_2) = 0 \end{cases}$$

In order for those equations to be identities, we must have:

$$\begin{cases} A_1 - B_1 = 0, & A_2 - A_1 - B_2 = 0 \\ A_1 - B_1 = 0, & A_2 - B_1 - B_2 = 0 \end{cases} \quad (&&)$$

Thus in order for (&) to be a solution of a given system, the constants A_1, B_1, A_2 , and B_2 must be chosen to satisfy the equations (&&). From the equations $A_1 - B_1 = 0$, we see that $A_1 = B_1$. The other two equations of (&&) show that A_2 , and B_2 must satisfy the following relation:

$$A_2 - B_2 = A_1 = B_1 \quad (&&&)$$

We may choose any convenient nonzero values for A_1 , and B_1 . We choose $A_1 = B_1 = 1$. Then (&&&) reduces to $A_2 - B_2 = 1$, and we can choose any convenient values for A_2 , and B_2 that will satisfy this equation. We choose $A_2 = 1, B_2 = 0$. Thus the relation (&) becomes:

$$\begin{cases} x = (t + 1)e^{3t} \\ y = te^{3t} \end{cases}$$

The general solution of a given system is:

$$\begin{cases} x = c_1 e^{3t} + c_2 (t + 1)e^{3t} \\ y = c_1 e^{3t} + c_2 t e^{3t} \end{cases}$$

Where c_1 , and c_2 are arbitrary constants

EXERCISES

I. Find the general solution of each of the linear systems

1. $\begin{cases} x' = 5x + 2y \\ y' = 4x - y \end{cases}$

2. $\begin{cases} x' = 5x - y \\ y' = 3x + y \end{cases}$

3. $\begin{cases} x' = x + 2y \\ y' = 3x + 2y \end{cases}$

4. $\begin{cases} x' = 2x + 3y \\ y' = -x - 2y \end{cases}$

5. $\begin{cases} x' = 3x + y \\ y' = 4x + 3y \end{cases}$

6. $\begin{cases} x' = 6x - y \\ y' = 3x + 2y \end{cases}$

7. $\begin{cases} x' = 3x + 4y \\ y' = 2x - 3y \end{cases}$

8. $\begin{cases} x' = x - 4y \\ y' = x + y \end{cases}$

9. $\begin{cases} x' = x - 3y \\ y' = 3x - y \end{cases}$

10. $\begin{cases} x' = 3x - 2y \\ y' = 2x + 3y \end{cases}$

11. $\begin{cases} x' = 5x + 4y \\ y' = -x + y \end{cases}$

12. $\begin{cases} x' = x - 2y \\ y' = 2x - 4y \end{cases}$

II. Find the particular solution of the linear system that satisfies the stated initial conditions:

1. $\begin{cases} x' = -2x + 7y \\ y' = 3x + 2y \\ x(0) = 9, y(0) = -1 \end{cases}$

2. $\begin{cases} x' = -2x + y \\ y' = 7x + 4y \\ x(0) = 6, y(0) = 2 \end{cases}$

3. $\begin{cases} x' = 2x - 8y \\ y' = x + 6y \\ x(0) = 4, y(0) = 1 \end{cases}$

4. $\begin{cases} x' = 7x - 4y \\ y' = 2x + 3y \\ x(0) = 2, y(0) = -1 \end{cases}$

5. $\begin{cases} x' = x - 2y \\ y' = 8x - 7y \\ x(0) = 6, y(0) = 8 \end{cases}$

6. $\begin{cases} x' = 3x + 5y \\ y' = -2x + 5y \\ x(0) = 5, y(0) = -1 \end{cases}$

III. a) Consider the linear system

$$\begin{cases} tx' = a_1 x + b_1 y \\ ty' = a_2 x + b_2 y \end{cases}$$

Where a_1, a_2, b_1 and b_2 are real constants. Show that the transformation $t = e^w$ transforms this system into a linear system with constant coefficients.

b) Use the result of a) to solve the system:

$$\begin{cases} tx' = x + y \\ ty' = -3x + 5y \end{cases}$$

c) Use the result of a) to solve the system:

$$\begin{cases} tx' = 2x + 3y \\ ty' = 2x + y \end{cases}$$

1.5. MATRIX METHOD FOR SOLVING LINEAR SYTEMS WITH CONSTANT COEFFICIENTS CASE OF TWO EQUATIONS IN TWO UNKNOWN FUNCTIONS

1.5.1 Matrix method for solving homogeneous linear

1.5.1.1.Introduction

Consider the following homogeneous linear system:

$$\begin{cases} x'_1 = a_{11}x_1 + a_{12}x_2 \\ x'_2 = a_{21}x_1 + a_{22}x_2 \end{cases} \quad (1.19)$$

Where the coefficients $a_{11}, a_{12}, a_{21}, a_{22}$ are real constants.

In matrix notation (1.19) is written as:

$$\begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ \Leftrightarrow X' = AX \quad (1.20)$$

Where A is called *coefficient matrix* and X is a *vector of unknown functions*.

The nontrivial solutions of (4.20) are the form: $\begin{cases} x_1 = \beta_1 e^{\lambda t} \\ x_2 = \beta_2 e^{\lambda t} \end{cases} \quad (1.21)$

Where β_1, β_2 , and λ are constant; and $\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \rightarrow X = \beta e^{\lambda t} \quad (1.21)$

Where β , is a constant vector and are constant λ is a number.

Now, (1.20) becomes: $\beta \lambda e^{\lambda t} = A \beta e^{\lambda t} \Leftrightarrow \beta \lambda = A \beta \quad (1.22)$

$$(A - \lambda I) \beta = 0$$

Where I , is the 2×2 identity matrix.

$$(A - \lambda I) \beta = 0 \Leftrightarrow \left[\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right] \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Now, we have the following system of two homogeneous linear equations in the two unknowns β_1 , and β_2 .

$$\begin{cases} (a_{11} - \lambda) \beta_1 + a_{12} \beta_2 = 0 \\ a_{12} \beta_1 + (a_{22} - \lambda) \beta_2 = 0 \end{cases} \quad (1.23)$$

This system has nontrivial solution iff

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0$$

This is called *characteristic equation* of the coefficient matrix $A = (a_{ij})$.

$$\begin{vmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{vmatrix} = 0$$

$$\Leftrightarrow \lambda^2 - (a_{11} + a_{22})\lambda + (a_{11}a_{22} - a_{12}a_{21}) = 0$$

The roots λ_1 and λ_2 of this equation are *the characteristic values (Eigen values)* of A .

If we substituting each characteristic value $\lambda_i, (i = 1, 2)$, into system(1.23), we obtain the corresponding nontrivial solution:

$$\beta^i = \begin{pmatrix} \beta_{1i} \\ \beta_{2i} \end{pmatrix}$$

This is called *characteristic vector (Eigen vector)* corresponding to the characteristic values $\lambda_i, (i = 1, 2)$.

Example 1.34: Convert the following system to matrix form $\begin{cases} x_1' = 4x_1 + 7x_2 \\ x_2' = -2x_1 - 5x_2 \end{cases}$

Solution: First write the system so that each side is a vector. $\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 4x_1 + 7x_2 \\ -2x_1 - 5x_2 \end{pmatrix}$

Now the right side can be written as a matrix multiplication, $\begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 4 & 7 \\ -2 & -5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

1.5.1.2. Analysis of different values of characteristic values

First case: The characteristic values are distinct

Theorem 1.5.1: Consider the homogeneous linear system:

$$\begin{cases} x_1' = a_{11}x_1 + a_{12}x_2 \\ x_2' = a_{21}x_1 + a_{22}x_2 \end{cases}$$

That is, the vector differential equation: $\mathbf{X}' = \mathbf{A}\mathbf{X}$

Where $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

and coefficients $a_{11}, a_{12}, a_{21}, a_{22}$ are real constants.

Suppose the two characteristic values λ_1 and λ_2 of A are distinct real or complex; and let $\beta^{(1)}$ and $\beta^{(2)}$ be a pair of respective corresponding characteristic vector of A .

Then on every real interval, the vector functions defined by $\beta^{(1)}e^{\lambda_1 t}$ and $\beta^{(2)}e^{\lambda_2 t}$ form a linearly independent set of solutions of (1.19); and

$$X = c_1\beta^{(1)}e^{\lambda_1 t} + c_2\beta^{(2)}e^{\lambda_2 t}$$

Where c_1 , and c_2 are arbitrary constants.

Example 1.35: Solve the following linear system of differential equation using the matrix method?

$$\begin{cases} x_1' = 6x_1 - 3x_2 \\ x_2' = 2x_1 + x_2 \end{cases}$$

Answer:

$$\begin{cases} x_1' = 6x_1 - 3x_2 \\ x_2' = 2x_1 + x_2 \end{cases} \leftrightarrow \begin{pmatrix} x_1' \\ x_2' \end{pmatrix} = \begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Characteristic equation is:

$$|A - \lambda I| = \begin{vmatrix} 6 - \lambda & -3 \\ 2 & 1 - \lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 6 - \lambda & -3 \\ 2 & 1 - \lambda \end{vmatrix} = 0 \leftrightarrow \lambda^2 - 7\lambda + 12 = 0 \rightarrow \lambda_1 = 3, \lambda_2 = 4$$

We use (1.22) for finding the characteristic vectors as:

Let $\lambda = \lambda_1 = 3$ and $\beta = \beta^{(1)} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$, then (1.22) becomes:

$$\begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = 3 \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} \leftrightarrow \begin{cases} 6\beta_1 - 3\beta_2 = 3\beta_1 \\ 2\beta_1 + \beta_2 = 3\beta_1 \end{cases} \rightarrow \beta_1 = \beta_2$$

Let $\beta_1 = \beta_2 = 1$, then $\beta^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Then by above theorem we get $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} \leftrightarrow x = \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix}$

Let $\lambda = \lambda_1 = 4$ and $\beta = \beta^{(2)} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$, then (1.22) becomes:

$$\begin{pmatrix} 6 & -3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = 4 \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} \leftrightarrow \begin{cases} 6\beta_1 - 3\beta_2 = 4\beta_1 \\ 2\beta_1 + \beta_2 = 4\beta_1 \end{cases} \rightarrow 2\beta_1 = 3\beta_2$$

Let $\beta_1 = 3, \beta_2 = 2$, then $\beta^{(2)} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$.

Then by above theorem we get $x = \begin{pmatrix} 3 \\ 2 \end{pmatrix} e^{3t} \leftrightarrow x = \begin{pmatrix} 3e^{4t} \\ 2e^{4t} \end{pmatrix}$

The general solution is $X = c_1 \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix} + c_2 \begin{pmatrix} 3e^{4t} \\ 2e^{4t} \end{pmatrix}$

Where c_1 and c_2 are arbitrary constants. A general solution of a given homogeneous linear system

is $\begin{cases} x_1 = c_1 e^{3t} + 3c_2 e^{4t} \\ x_2 = c_1 e^{3t} + 2c_2 e^{4t} \end{cases}$

Second case: The repeated characteristic value

Theorem 1.5.2: Consider the homogeneous linear system: $\begin{cases} x'_1 = a_{11}x_1 + a_{12}x_2 \\ x'_2 = a_{21}x_1 + a_{22}x_2 \end{cases}$

That is, the vector differential equation: $\mathbf{X}' = \mathbf{A}\mathbf{X}$

Where $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$

and coefficients $a_{11}, a_{12}, a_{21}, a_{22}$ are real constants which are not such that $a_{11} = a_{12} \neq 0, a_{21} = a_{22} = 0$.

Suppose the two characteristic values λ_1 and λ_2 of A are real and equal and let λ denote their common value. Let β be a corresponding characteristic vector of A and let γ be a vector satisfying the equation:

$$(A - \lambda I)\gamma = \beta$$

Then on every real interval, the vector functions defined by $\beta e^{\lambda t}$ and $(\beta t + \gamma) e^{\lambda t}$

Form a linearly independent set of solutions of (4.20), and $X = c_1 \beta e^{\lambda t} + c_2 (\beta t + \gamma) e^{\lambda t}$

Where c_1 , and c_2 are arbitrary constants.

Example 1.35: Solve the following linear system of differential equation using the matrix method?

$$\begin{cases} x'_1 = 4x_1 - x_2 \\ x'_2 = x_1 + 2x_2 \end{cases}$$

Answer:

$$\begin{cases} x'_1 = 4x_1 - x_2 \\ x'_2 = x_1 + 2x_2 \end{cases} \leftrightarrow \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

Characteristic equation is: $|A - \lambda I| = \begin{vmatrix} 4 - \lambda & -1 \\ 1 & 2 - \lambda \end{vmatrix} = 0$

$$\begin{vmatrix} 4 - \lambda & -1 \\ 1 & 2 - \lambda \end{vmatrix} = 0 \leftrightarrow \lambda^2 - 6\lambda + 9 = 0 \rightarrow \lambda_1 = 3 = \lambda_2$$

We apply the above theorem for finding the characteristic vectors as:

Let $\lambda = 3$. With $\lambda = 3$ and $\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$, then (4.22) becomes:

$$\begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} = 3 \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix} \leftrightarrow \begin{cases} 4\beta_1 - \beta_2 = 3\beta_1 \\ \beta_1 + 2\beta_2 = 3\beta_1 \end{cases} \rightarrow \beta_1 = \beta_2$$

Let $\beta_1 = \beta_2 = 1$, then $\beta = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Then by above theorem we get $x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{3t} \leftrightarrow x = \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix}$

By the above theorem, a linearly independent solution is of the form: $(\beta t + \gamma) e^{\lambda t}$

Where $\beta = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $\lambda = 3$, and γ satisfies $(A - \lambda I)\gamma = \beta$. Thus $\gamma = \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix}$ satisfies

$$\left[\begin{pmatrix} 4 & -1 \\ 1 & 2 \end{pmatrix} - 3 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right] \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \leftrightarrow \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \gamma_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rightarrow \begin{cases} \gamma_1 - \gamma_2 = 1 \\ \gamma_1 - \gamma_2 = 1 \end{cases} \rightarrow \gamma_1 = \gamma_2 + 1$$

If $\gamma_2 = 0 \rightarrow \gamma_1 = 1$, then we find that $\gamma = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

Then by above theorem we get $X = (\beta t + \gamma) e^{\lambda t} \leftrightarrow X = \left[\begin{pmatrix} 1 \\ 1 \end{pmatrix} t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] e^{3t}$

$$X = \begin{pmatrix} (t+1)e^{3t} \\ te^{3t} \end{pmatrix}$$

The general solution is: $X = c_1 \begin{pmatrix} e^{3t} \\ e^{3t} \end{pmatrix} + c_2 \begin{pmatrix} (t+1)e^{3t} \\ te^{3t} \end{pmatrix}$

Where c_1 and c_2 are arbitrary constants. A general solution of a given homogeneous linear system

is
$$\begin{cases} x_1 = c_1 e^{3t} + c_2 (t+1) e^{3t} \\ x_2 = c_1 e^{3t} + c_2 t e^{3t} \end{cases}$$

EXERCISES

Find the general solution of each of the homogeneous linear systems, using the vector- matrix methods, where in each exercise $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$.

$$1. x' = \begin{pmatrix} 5 & -2 \\ 4 & -1 \end{pmatrix} x \quad 2. x' = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix} x \quad 3. x' = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} x \quad 4. x' = \begin{pmatrix} 2 & 3 \\ -1 & -2 \end{pmatrix} x$$

$$5. x' = \begin{pmatrix} 3 & 1 \\ 4 & 3 \end{pmatrix} x \quad 6. x' = \begin{pmatrix} 6 & -1 \\ 3 & 2 \end{pmatrix} x \quad 7. x' = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} x \quad 8. x' = \begin{pmatrix} 3 & 2 \\ -6 & -1 \end{pmatrix} x$$

$$\begin{array}{llll}
9. x' = \begin{pmatrix} 1 & -4 \\ 1 & 1 \end{pmatrix} x & 10. x' = \begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} x & 11. x' = \begin{pmatrix} 1 & -3 \\ 3 & 1 \end{pmatrix} x & 12. x' = \begin{pmatrix} 5 & -4 \\ 2 & 1 \end{pmatrix} x \\
13. x' = \begin{pmatrix} 1 & -2 \\ 2 & -3 \end{pmatrix} x & 14. x' = \begin{pmatrix} 6 & -4 \\ 1 & 2 \end{pmatrix} x & 15. x' = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} x & 16. x' = \begin{pmatrix} 7 & 4 \\ -1 & 3 \end{pmatrix} x
\end{array}$$

1.6. MATRIX METHOD FOR SOLVING LINEAR SYSTEMS WITH CONSTANT COEFFICIENTS CASE OF n EQUATIONS IN n UNKNOWN FUNCTIONS

1.6.1. Introduction

In this section we extend the methods of the previous section to a homogeneous linear system of n first-order differential equations in n unknown functions and having real constant coefficients.

Consider a homogeneous linear system of the form:

$$\begin{cases} x'_1 = a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ x'_2 = a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ x'_n = a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{cases} \quad (1.24)$$

Where the coefficients a_{ij} , ($1 \leq i \leq n$, and $1 \leq j \leq n$) are constants; and A is $n \times n$ constant matrix.

In matrix notation the relation (1.24) is written as:

$$\begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

In the *homogeneous linear vector differential equation* the relation (1.24) is expressed as:

$$x' = Ax \quad (1.25)$$

Definition 1.6:

Consider an $n \times 1$ column-vector following functions as a solution of (1.24):

$$\Phi = \begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_n \end{pmatrix}$$

whose components $\phi_1, \phi_2, \phi_3, \dots, \phi_n$ each have a continuous derivative on the real interval $a < t < b$, and which is such that :

$$\begin{cases} \phi_1' = a_{11}\phi_1 + a_{12}\phi_2 + \cdots + a_{1n}\phi_n \\ \phi_2' = a_{21}\phi_1 + a_{22}\phi_2 + \cdots + a_{2n}\phi_n \\ \vdots \\ \phi_n' = a_{n1}\phi_1 + a_{n2}\phi_2 + \cdots + a_{nn}\phi_n \end{cases}$$

For all $t \in [a, b]$. In the other words, the components $\phi_1, \phi_2, \phi_3, \dots, \phi_n$ of ϕ are such that

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \phi_1(t) \\ \phi_2(t) \\ \vdots \\ \phi_n(t) \end{pmatrix}$$

Simultaneously satisfy all n equations of the system (4.24) identically on $t \in [a, b]$.

Theorem 1.6.1: Any linear combination of n solutions of homogeneous linear system (1.24) is itself a solution of the system (1.24).

Theorem 1.6.2: There exist sets of n linearly independent solutions of homogeneous linear system (1.24). Every solution of the system (1.24) can be written as a linear combination of any n linearly independent solutions of the system (1.24).

Definition 1.6.2:

$$\text{Let } \phi_1 = \begin{pmatrix} \phi_{11} \\ \phi_{21} \\ \vdots \\ \phi_{n1} \end{pmatrix}, \phi_2 = \begin{pmatrix} \phi_{12} \\ \phi_{22} \\ \vdots \\ \phi_{n2} \end{pmatrix}, \dots, \phi_n = \begin{pmatrix} \phi_{1n} \\ \phi_{2n} \\ \vdots \\ \phi_{nn} \end{pmatrix}$$

Be n linearly independent solutions of the homogeneous linearly system (1.24), let $c_1, c_2, c_3, \dots, c_n$ be n arbitrary constants. Then the solution

$$x = c_1\phi_1(t) + c_2\phi_2(t) + \cdots + c_n\phi_n(t)$$

$$\text{that is: } \begin{cases} x_1 = c_1\phi_{11}(t) + c_2\phi_{12}(t) + \cdots + c_n\phi_{1n}(t) \\ x_2 = c_1\phi_{21}(t) + c_2\phi_{22}(t) + \cdots + c_n\phi_{2n}(t) \\ \vdots \\ x_n = c_1\phi_{n1}(t) + c_2\phi_{n2}(t) + \cdots + c_n\phi_{nn}(t) \end{cases}$$

Is called a general solution of the system (1.24).

Definition 1.6.3:

Consider the n vector functions $\phi_1(t) = \begin{pmatrix} \phi_{11}(t) \\ \phi_{21}(t) \\ \vdots \\ \phi_{n1}(t) \end{pmatrix}, \phi_2(t) = \begin{pmatrix} \phi_{12}(t) \\ \phi_{22}(t) \\ \vdots \\ \phi_{n2}(t) \end{pmatrix}, \dots, \phi_n(t) = \begin{pmatrix} \phi_{1n}(t) \\ \phi_{2n}(t) \\ \vdots \\ \phi_{nn}(t) \end{pmatrix}$, then the determinant of $n \times n$ matrix is: $\begin{vmatrix} \phi_{11} & \phi_{12} & \cdots & \phi_{1n} \\ \phi_{21} & \phi_{22} & \cdots & \phi_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \phi_{n1} & \phi_{n2} & \cdots & \phi_{nn} \end{vmatrix}$ is called the

Wronskian of n vector functions $\phi_1, \phi_2, \phi_3, \dots, \phi_n$ and is denoted as: $W(\phi_1, \phi_2, \phi_3, \dots, \phi_n)(t)$.

Theorem 1.6.2: Let $\phi_1, \phi_2, \phi_3, \dots, \phi_n$ be n solutions of the homogeneous linear system (1.24) on interval $a \leq t \leq b$. Then $W(\phi_1, \phi_2, \phi_3, \dots, \phi_n)(t) \neq 0$ or $W(\phi_1, \phi_2, \phi_3, \dots, \phi_n)(t) = 0, \forall t \in [a, b]$

The characteristic equation of the coefficient matrix is:

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix} = 0 \quad (1.26)$$

Where $\lambda = \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \vdots \\ \lambda_n \end{pmatrix}$ are the characteristic values of A .

Let $x = \beta e^{\lambda t}$ be a solution of (4.25), then $\beta \lambda e^{\lambda t} = A \beta e^{\lambda t} \rightarrow (A - \lambda I) \beta = 0$

Where $\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_n \end{pmatrix}$ is the characteristic vectors of A .

Each $\lambda^i, i = 1, 2, 3, \dots, n$ characteristic values, correspond to each $\beta^i, i = 1, 2, 3, \dots, n$.

1.6.2. Analysis of different cases

First case: n distinct characteristic values

Theorem 1.6.4: Consider the homogeneous linear equation of the form (1.24) and (1.25).

Suppose each of the n characteristic values $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_n$ of A is distinct; and let $\beta^{(i)} = \begin{pmatrix} \beta_{1i} \\ \beta_{2i} \\ \vdots \\ \beta_{ni} \end{pmatrix}, i = 1, 2, \dots, n$, be a set of n vector functions defined by: $\beta^{(1)}e^{\lambda_1 t}, \beta^{(2)}e^{\lambda_2 t}, \dots, \beta^{(n)}e^{\lambda_n t}$ form a linearly independent set of solutions of (4.24), that is (4.25); and

$$x = c_1 \beta^{(1)} e^{\lambda_1 t} + c_2 \beta^{(2)} e^{\lambda_2 t} + \dots + c_n \beta^{(n)} e^{\lambda_n t}$$

Where $c_1, c_2, \dots, c_n$ are n arbitrary constants, is a general solution of (1.24).

Example 1.37: Solve the following linear system of differential equation using the matrix method?

$$\begin{cases} x'_1 = 7x_1 - x_2 + 6x_3 \\ x'_2 = -10x_1 + 4x_2 - 12x_3 \\ x'_3 = -2x_1 + x_2 - x_3 \end{cases}$$

Answer:

$$\begin{cases} x'_1 = 7x_1 - x_2 + 6x_3 \\ x'_2 = -10x_1 + 4x_2 - 12x_3 \\ x'_3 = -2x_1 + x_2 - x_3 \end{cases} \leftrightarrow \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix} = \begin{pmatrix} 7 & -1 & 6 \\ -10 & 4 & -12 \\ -2 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Assuming a solution of this is of the form: $x = \beta e^{\lambda t}$ that is:

$$x_1 = \beta_1 e^{\lambda_1 t}, x_2 = \beta_2 e^{\lambda_2 t}, x_3 = \beta_3 e^{\lambda_3 t}$$

$$A = \begin{pmatrix} 7 & -1 & 6 \\ -10 & 4 & -12 \\ -2 & 1 & -1 \end{pmatrix} \rightarrow |A - \lambda I| = \begin{vmatrix} 7 - \lambda & -1 & 6 \\ -10 & 4 - \lambda & -12 \\ -2 & 1 & -1 - \lambda \end{vmatrix} = 0$$

$$\rightarrow \lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 5$$

$$A\beta = \lambda\beta$$

Let consider $\lambda = \lambda_1 = 2$, then we get:

$$\begin{pmatrix} 7 & -1 & 6 \\ -10 & 4 & -12 \\ -2 & 1 & -1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = 2 \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \rightarrow \beta_2 = -\beta_1 \text{ and } \beta_3 = -\beta_1$$

Let $\beta_1 = 1$, then $\beta_2 = -1$ and $\beta_3 = -1$

$$\beta^{(1)} = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} e^{2t} \\ -e^{2t} \\ -e^{2t} \end{pmatrix}$$

Let consider $\lambda = \lambda_2 = 3$, then we get:

$$\begin{pmatrix} 7 & -1 & 6 \\ -10 & 4 & -12 \\ -2 & 1 & -1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{3} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \rightarrow \beta_2 = -2\beta_1 \text{ and } \beta_3 = -\beta_1$$

Let $\beta_1 = 1$, then $\beta_2 = -2$ and $\beta_3 = -1$

$$\beta^{(2)} = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} e^{3t} \\ -2e^{3t} \\ -e^{3t} \end{pmatrix}$$

Let consider $\lambda = \lambda_3 = 5$, then we get:

$$\begin{pmatrix} 7 & -1 & 6 \\ -10 & 4 & -12 \\ -2 & 1 & -1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{5} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \rightarrow \beta_1 = 3, \beta_2 = -6 \text{ and } \beta_3 = -2$$

$$\beta^{(3)} = \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix}$$

$$x = \begin{pmatrix} 3 \\ -6 \\ -2 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} 3e^{5t} \\ -6e^{5t} \\ -2e^{5t} \end{pmatrix}$$

If those three solution are linearly independent, then we have:

$$x = c_1 \begin{pmatrix} e^{2t} \\ -e^{2t} \\ -e^{2t} \end{pmatrix} + c_2 \begin{pmatrix} e^{3t} \\ -2e^{3t} \\ -e^{3t} \end{pmatrix} + c_3 \begin{pmatrix} 3e^{5t} \\ -6e^{5t} \\ -2e^{5t} \end{pmatrix}$$

Where c_1, c_2, c_3 are arbitrary constants.

The general solution is:

$$\begin{cases} x_1 = c_1 e^{2t} - 2c_2 e^{3t} + 3c_3 e^{5t} \\ x_2 = -c_1 e^{2t} - 2c_2 e^{3t} - 6c_3 e^{5t} \\ x_3 = -c_1 e^{2t} - c_2 e^{3t} - 2c_3 e^{5t} \end{cases}$$

Where c_1, c_2, c_3 are arbitrary constants.

Second case: Repeated characteristic values

Example 1.38: Solve the following linear system of differential equation using the matrix method?

$$\begin{cases} x_1' = 3x_1 + x_2 - x_3 \\ x_2' = x_1 + 3x_2 - x_3 \\ x_3' = 3x_1 + 3x_2 - x_3 \end{cases}$$

Answer:

$$\begin{cases} x_1' = 3x_1 + x_2 - x_3 \\ x_2' = x_1 + 3x_2 - x_3 \\ x_3' = 3x_1 + 3x_2 - x_3 \end{cases} \leftrightarrow \begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Assuming a solution of this is of the form: $x = \beta e^{\lambda t}$ that is:

$$x_1 = \beta_1 e^{\lambda_1 t}, x_2 = \beta_2 e^{\lambda_2 t}, x_3 = \beta_3 e^{\lambda_3 t}$$

$$A = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{pmatrix} \rightarrow |A - \lambda I| = \begin{vmatrix} 3 - \lambda & 1 & -1 \\ 1 & 3 - \lambda & -1 \\ 3 & 3 & -1 - \lambda \end{vmatrix} = 0$$

$$\rightarrow \lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 2$$

$$A\beta = \lambda\beta$$

Let consider $\lambda = \lambda_1 = 1$, then we get:

$$\begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{1} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \rightarrow \beta_1 = 1, \beta_2 = 1 \text{ and } \beta_3 = 3$$

Let $\beta_1 = 1$, then $\beta_2 = -1$ and $\beta_3 = -1$

$$\beta^{(1)} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} e^t \leftrightarrow x = \begin{pmatrix} e^t \\ e^t \\ 3e^t \end{pmatrix}$$

Let consider $\lambda = \lambda_2 = \lambda_3 = 2$, then we get:

$$\begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ 3 & 3 & -1 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = \mathbf{2} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \rightarrow \beta_1 = 1, \beta_2 = 0 \text{ and } \beta_3 = 1 \text{ or } \beta_1 = 1, \beta_2 = -1 \text{ and } \beta_3 = 0$$

Let $\beta_1 = 1$, then $\beta_2 = -2$ and $\beta_3 = -1$

$$\beta^{(2)} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} e^{2t} \\ 0 \\ e^{2t} \end{pmatrix}$$

$$\beta^{(2)} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} e^{2t} \\ -e^{2t} \\ 0 \end{pmatrix}$$

If those three solution are linearly independent, then we have:

$$x = c_1 \begin{pmatrix} e^t \\ e^t \\ 3e^t \end{pmatrix} + c_2 \begin{pmatrix} e^{2t} \\ 0 \\ e^{2t} \end{pmatrix} + c_3 \begin{pmatrix} e^{2t} \\ -e^{2t} \\ 0 \end{pmatrix}$$

Where c_1, c_2, c_3 are arbitrary constants.

The general solution is:
$$\begin{cases} x_1 = c_1 e^t + c_2 e^{3t} + c_3 e^{5t} \\ x_2 = c_1 e^{2t} - c_3 e^{5t} \\ x_3 = 3c_1 e^{2t} + c_2 e^{3t} \end{cases}$$

Where c_1, c_2, c_3 are arbitrary constants.

Example 1.39: Solve the following linear system of differential equation using the matrix method?

$$\begin{cases} x_1' = 4x_1 + 3x_2 + x_3 \\ x_2' = -4x_1 - 4x_2 - 2x_3 \\ x_3' = 8x_1 + 12x_2 + 6x_3 \end{cases}$$

Answer:

$$\begin{cases} x_1' = 4x_1 + 3x_2 + x_3 \\ x_2' = -4x_1 - 4x_2 - 2x_3 \\ x_3' = 8x_1 + 12x_2 + 6x_3 \end{cases} \leftrightarrow \begin{pmatrix} x_1' \\ x_2' \\ x_3' \end{pmatrix} = \begin{pmatrix} 4 & 3 & 1 \\ -4 & -4 & -2 \\ 8 & 12 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

Assuming a solution of this is of the form: $x = \beta e^{\lambda t}$ that is:

$$x_1 = \beta_1 e^{\lambda_1 t}, x_2 = \beta_2 e^{\lambda_2 t}, x_3 = \beta_3 e^{\lambda_3 t}$$

$$A = \begin{pmatrix} 4 & 3 & 1 \\ -4 & -4 & -2 \\ 8 & 12 & 6 \end{pmatrix} \rightarrow |A - \lambda I| = \begin{vmatrix} 4 - \lambda & 3 & 1 \\ -4 & -4 - \lambda & -1 \\ 8 & 12 & 6 - \lambda \end{vmatrix} = 0$$

$$\rightarrow \lambda_1 = \lambda_2 = \lambda_3 = 2$$

$$A\beta = \lambda\beta$$

Let consider $\lambda = 2$, then we get:

$$\begin{pmatrix} 4 & 3 & 1 \\ -4 & -4 & -2 \\ 8 & 12 & 6 \end{pmatrix} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} = 2 \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix} \rightarrow \beta_1 = 1, \beta_2 = 0 \text{ and } \beta_3 = -2$$

Let $\beta_1 = 0$, then $\beta_2 = 1$ and $\beta_3 = -3$

$$\beta^{(1)} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} \text{ and } \beta^{(2)} = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix}$$

$$x = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} e^{2t} \\ 0 \\ -2e^{2t} \end{pmatrix}, \text{ and } x = \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} e^{2t} \leftrightarrow x = \begin{pmatrix} 0 \\ e^{2t} \\ -3e^{2t} \end{pmatrix}$$

A third solution corresponds to $\lambda = 2$ is of the form:

$$(\beta t + \gamma) e^{\lambda t}$$

Where β satisfies $(A - \lambda I)\beta = 0$

And γ satisfies $(A - \lambda I)\gamma = \beta$,

$$\text{where } \beta = k_1 \beta^{(1)} + k_2 \beta^{(2)} = \begin{pmatrix} k_1 \\ 0 \\ -2k_1 \end{pmatrix} + \begin{pmatrix} 0 \\ k_2 \\ -3k_2 \end{pmatrix} = \begin{pmatrix} k_1 \\ k_2 \\ -2k_1 - 3k_2 \end{pmatrix}$$

$$\text{Then } (A - \lambda I)\gamma = \beta \rightarrow \begin{pmatrix} 2 & 3 & 1 \\ -4 & -6 & -2 \\ 8 & 12 & 4 \end{pmatrix} \begin{pmatrix} \gamma_1 \\ \gamma_2 \\ \gamma_3 \end{pmatrix} = \begin{pmatrix} k_1 \\ k_2 \\ -2k_1 - 3k_2 \end{pmatrix}$$

$$\rightarrow \begin{cases} 2\gamma_1 + 3\gamma_2 + \gamma_3 = k_1 \\ -4\gamma_1 - 6\gamma_2 - 2\gamma_3 = k_2 \\ 8\gamma_1 + 12\gamma_2 + 4\gamma_3 = -2k_1 - 3k_2 \end{cases} \rightarrow k_2 = -2k_1, \quad \text{if } k_1 = 1, \text{ then } k_2 = -2$$

$$\rightarrow \beta = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix}, \text{ and } \gamma = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$X = (\beta t + \gamma)e^{\lambda t} \leftrightarrow X = \left[\begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} t + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right] e^{2t}$$

$$X = \begin{pmatrix} te^{2t} \\ -2te^{2t} \\ (4t+1)e^{2t} \end{pmatrix}$$

If those three solution are linearly independent, then we have:

$$x = c_1 \begin{pmatrix} e^{2t} \\ 0 \\ -2e^{2t} \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^{2t} \\ -3e^{2t} \end{pmatrix} + c_3 \begin{pmatrix} te^{2t} \\ -2te^{2t} \\ (4t+1)e^{2t} \end{pmatrix}$$

Where c_1, c_2, c_3 are arbitrary constants.

The general solution is:

$$\begin{cases} x_1 = c_1 e^{2t} + c_3 t e^{2t} \\ x_2 = c_2 e^{2t} - 2t c_3 e^{2t} \\ x_3 = -2c_1 e^{2t} - 3c_2 e^{2t} + c_3 (4t+1) e^{2t} \end{cases}$$

Where c_1, c_2, c_3 are arbitrary constants.

EXERCISES

Find the general solution of each of the homogeneous linear systems, using the vector- matrix methods, where in each exercise $x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

$$1. x' = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 3 & -4 \\ 4 & 1 & -4 \end{pmatrix} x \quad 2. x' = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 0 & 3 \\ 2 & 3 & 0 \end{pmatrix} x \quad 3. x' = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} x$$

$$4. x' = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & 1 & -1 \end{pmatrix} x \quad 5. x' = \begin{pmatrix} 1 & -1 & -1 \\ 1 & 3 & 1 \\ -3 & -6 & 6 \end{pmatrix} x \quad 6. x' = \begin{pmatrix} 1 & 0 & 2 \\ 1 & 1 & 0 \\ -2 & -4 & -1 \end{pmatrix} x$$

$$7. x' = \begin{pmatrix} 3 & 7 & -3 \\ 1 & 2 & -2 \\ 1 & 6 & -2 \end{pmatrix} x \quad 8. x' = \begin{pmatrix} -5 & -3 & -3 \\ 8 & 5 & 7 \\ -2 & -1 & -3 \end{pmatrix} x \quad 9. x' = \begin{pmatrix} 1 & -3 & 9 \\ 0 & -5 & 18 \\ 0 & -3 & 10 \end{pmatrix} x$$

$$10. x' = \begin{pmatrix} 11 & 6 & 18 \\ 9 & 8 & 18 \\ -9 & -6 & -16 \end{pmatrix} x \quad 11. x' = \begin{pmatrix} 1 & -3 & 9 \\ 0 & -5 & 18 \\ 0 & -3 & 10 \end{pmatrix} x \quad 12. x' = \begin{pmatrix} 7 & 0 & 4 \\ 8 & 3 & 8 \\ -8 & 0 & -5 \end{pmatrix} x$$

$$13. x' = \begin{pmatrix} 1 & 9 & 9 \\ 0 & 18 & 19 \\ 0 & 9 & 10 \end{pmatrix} x \quad 14. x' = \begin{pmatrix} 3 & -2 & -1 \\ -4 & 2 & 4 \\ 5 & -3 & -3 \end{pmatrix} x \quad 15. x' = \begin{pmatrix} 4 & 6 & -1 \\ -1 & -2 & 1 \\ -2 & -8 & 4 \end{pmatrix} x$$

UNIT II LAPLACE TRANSFORMS

2.1.Introduction

Laplace transform methods have a key role to play in the modern approach to the analysis and design of engineering systems. The stimulus for developing these methods was the pioneering work of the English electrical engineer Oliver Heaviside (1850-1925) in developing a method for the systematic solution of differential equations with constant coefficients.

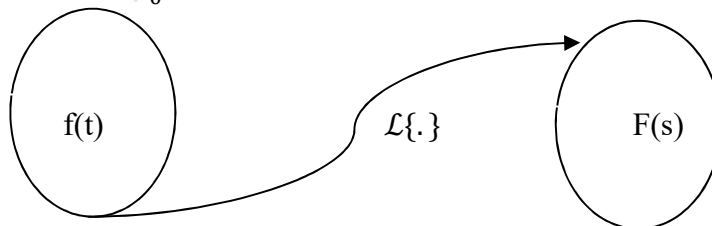
Heaviside was concerned with solving practical problems, and his method was based mainly on intuition; lacking mathematical rigor. Consequently it was frowned upon by the theoreticians at the time. However, Heaviside himself was not concerned with rigorous proves and was satisfied that his method gave the correct results. Using his idea, he was able to solve important practical problems that could not be dealt with using classical methods. This led to many new results in the field such as the propagation of current and voltages along the transmission lines.

Because it worked in practice; Heaviside method was widely accepted by engineers. As its power for problem-solving became more and more apparent, the method attracted the attention of mathematicians who set out to justify it. This provided the stimulus for rapid development in different branches of mathematics including improper integral, asymptotic series, and transform theory. Research on the problem continued for many years before it was eventually recognized that an integral transform developed by the French mathematician Pierre Simon de Laplace (1749-1827), almost a century before, provided a theoretical foundation for Heaviside's work. It was also recognized that the use of this integral provides a more systematic alternative for investigating differential equations than the methods proposed by Heaviside. It is this alternative approach that is the basis of the **Laplace Transform method**.

2.1.1. Definition and notation

We define the Laplace transform of a function $f(t)$ by the expression:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt \quad 2.1$$



t domain (time domain)

s domain (frequency domain)

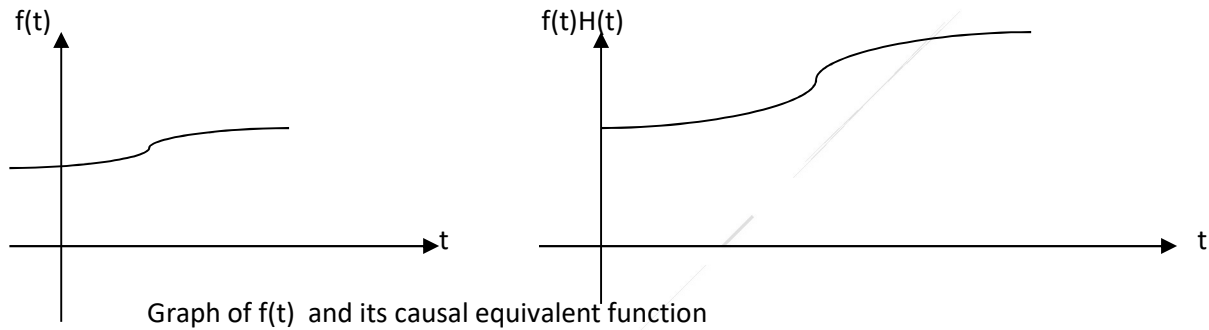
where s is a complex variable and e^{-st} is called the **kernel** of the transformation. It is usual to represent the Laplace transform of a function by the corresponding capital letter. So we write:

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^{\infty} e^{-st} f(t) dt \quad 2.2$$

An alternative notation in common use is to denote $\mathcal{L}\{f(t)\}$ by $\tilde{f}(s)$ or simply $\tilde{f}$. Few observations related to the definition (2.2):

- The symbol $\mathcal{L}$ denotes the Laplace transform operator; when it operates on a function $f(t)$ it transforms it into a function $F(s)$ of the complex variable s .
- As the integral is an improper integral, then $\int_0^{\infty} e^{-st} f(t) dt = \lim_{T \rightarrow \infty} \int_0^T e^{-st} f(t) dt$
- The function, however, unless the domain is clearly specified, a function $f(t)$ is normally interpreted as being defined for all real values, both positive and negative, of t . making use of the Heaviside unit step function $H(t)$, where $H(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases}$ we have

$$f(t)H(t) = \begin{cases} 0 & t < 0 \\ f(t) & t \geq 0 \end{cases}$$



- If the behavior of $f(t)$ for $t < 0$ is of interest, then we need to use the alternative **two-sided** or **bilateral Laplace transform** of the function $f(t)$ defined by :

$$\mathcal{L}_B\{f(t)\} = \int_{-\infty}^{\infty} e^{-st} f(t) dt \quad 2.3$$

The Laplace transform defined by 3.2 with lower limit zero is sometimes referred to as the **one-sided** or **unilateral Laplace transform** of the function $f(t)$. Note that when $f(t)$ is a causal function $\mathcal{L}_B\{f(t)\} = \mathcal{L}\{f(t)\}$.

2.2. Transforms of simple functions

Example 2.1: determine the Laplace transform of the function $f(t) = c$; where c is a constant.

Solution: $\mathcal{L}\{c\} = \int_0^{\infty} e^{-st} c dt = c \lim_{T \rightarrow \infty} \int_0^T e^{-st} dt$

$$= \lim_{T \rightarrow \infty} \left(\frac{-c}{s} e^{-st} \Big|_0^T \right) = \frac{-c}{s} \lim_{T \rightarrow \infty} (e^{-sT} - 1) = \frac{c}{s} \lim_{T \rightarrow \infty} (1 - e^{-sT})$$

Taking $s = \sigma + \omega i$ where $\sigma, \omega \in \mathbb{R}$

$$\lim_{T \rightarrow \infty} e^{-sT} = \lim_{T \rightarrow \infty} e^{-T(\sigma + \omega i)} = \lim_{T \rightarrow \infty} e^{-T\sigma} e^{-i\omega T}$$

A finite limit exists provided that $\sigma = \operatorname{Re}(s) > 0$ when the limit is zero.

$$\mathcal{L}\{c\} = \frac{c}{s}; \operatorname{Re}(s) > 0$$

So

$$\left. \begin{aligned} f(t) &= c \\ F(s) &= \frac{c}{s} \end{aligned} \right\} \operatorname{Re}(s) > 0 \text{ constitutes an example of Laplace transform pair.}$$

Example 2.2: Determine the Laplace transform of the ramp function $f(t) = t$.

$$\text{Solution : } \mathcal{L}\{t\} = \int_0^{\infty} t e^{-st} dt = \lim_{T \rightarrow \infty} \int_0^T t e^{-st} dt$$

$$= \lim_{T \rightarrow \infty} \left(\frac{-t}{s} e^{-st} - \frac{e^{-st}}{s^2} \Big|_0^T \right) = \frac{1}{s^2}$$

$$\left. \begin{aligned} f(t) &= t \\ F(s) &= \frac{1}{s^2} \end{aligned} \right\} \operatorname{Re}(s) > 0$$

Example 2.3: Determine the Laplace transform of the one-sided exponential function $f(t) = e^{kt}$

$$\text{Solution : } \mathcal{L}\{e^{kt}\} = \int_0^{\infty} e^{kt} e^{-st} dt = \int_0^{\infty} e^{t(k-s)} dt = \lim_{T \rightarrow \infty} \int_0^T e^{-(s-k)t} dt$$

$$= -\frac{1}{s-k} \lim_{T \rightarrow \infty} \left(e^{-(s-k)t} \Big|_0^T \right) = \frac{1}{k-s} \left[1 - \lim_{T \rightarrow \infty} e^{-(s-k)T} \right]$$

Writing $s = \sigma + \omega i$ where $\sigma, \omega \in \mathbb{R}$ we have $\lim_{T \rightarrow \infty} e^{-(s-k)T} = \lim_{T \rightarrow \infty} e^{-T(\sigma-k)} e^{-i\omega T}$ if $k \in \mathbb{R} \Rightarrow \operatorname{Re}(s) > k$ if $k \in \mathbb{C} \Rightarrow \operatorname{Re}(s) > \operatorname{Re}(k)$.

$$\mathcal{L}\{e^{kt}\} = \frac{1}{s-k}$$

Giving the following Laplace transform pair

$$\left. \begin{aligned} f(t) &= e^{kt} \\ F(s) &= \frac{1}{s-k} \end{aligned} \right\} \operatorname{Re}(s) > \operatorname{Re}(k)$$

Example 2.4: Determine the Laplace transform of one-sided exponential function $f(t) = e^{-kt}$

Solution: According to the results of the above example we consider $-k = k'$. $f(t) = e^{-kt} = e^{k't}$

$$\left. \begin{aligned} f(t) &= e^{-kt} \\ F(s) &= \frac{1}{s-k'} = \frac{1}{s+k} \end{aligned} \right\} \text{Re}(s) > \text{Re}(k)$$

Example 2.5: Determine the Laplace transform of the sine and cosine functions (i) $\cos(t)$, (ii) $\sin(t)$

Solution:

$$\text{i) } f(t) = \cos(t) = \frac{e^{it} + e^{-i}}{2} = \frac{1}{2}e^{it} + \frac{1}{2}e^{-it}$$

$$F(s) = \frac{1}{2} \left\{ \left[\frac{1}{s-i} \right] + \left[\frac{1}{s+i} \right] \right\} = \frac{1}{2} \left[\frac{s+i+s-i}{s^2+1} \right] = \frac{s}{s^2+1}$$

In general

$$\mathcal{L}\{\cos(at)\} = \frac{s}{s^2 + a^2}, \text{Re}(s) > 0$$

$$\text{(ii) } f(t) = \sin(t) = \frac{e^{it} - e^{-it}}{2i} = \frac{1}{2i}e^{it} - \frac{1}{2i}e^{-it}$$

$$F(s) = \frac{1}{2i} \left\{ \left[\frac{1}{s-i} \right] - \left[\frac{1}{s+i} \right] \right\} = \frac{1}{2i} \left[\frac{s+i-s+i}{s^2+1} \right] = \frac{1}{s^2+1}$$

In general

$$\mathcal{L}\{\sin(at)\} = \frac{a}{s^2 + a^2}, \text{Re}(s) > 0$$

2.3. Basic operational properties

In this section we consider some of the properties of the Laplace transform that will enable us to find further transform pairs $\{f(t), F(s)\}$ without having to compute them directly using the definition.

Property 2.1: The linearity property

A fundamental property of the Laplace transform is its linearity, which may be stated as follows:

If $f(t)$ and $g(t)$ are functions having Laplace transforms and if α and β are any constants, then $\mathcal{L}\{\alpha f(t) \pm \beta g(t)\} = \alpha \mathcal{L}\{f(t)\} \pm \beta \mathcal{L}\{g(t)\}$ 2.4

As a consequence of this property, we say that the Laplace transform U is a **linear operator**.

Proof:

$$\begin{aligned}
\mathcal{L}\{\alpha f(t) \pm \beta g(t)\} &= \int_0^{\infty} e^{-st} [\alpha f(t) \pm \beta g(t)] dt = \int_0^{\infty} [\alpha e^{-st} f(t) \pm \beta e^{-st} g(t)] dt \\
&= \int_0^{\infty} \alpha e^{-st} f(t) dt \pm \int_0^{\infty} \beta e^{-st} g(t) dt = \alpha \int_0^{\infty} e^{-st} f(t) dt \pm \beta \int_0^{\infty} e^{-st} g(t) dt \\
&= \alpha F(s) \pm \beta G(s)
\end{aligned}$$

Example 2.6 : Determine i) $\mathcal{L}\{3t + 2e^{3t}\}$ ii) $\mathcal{L}\{5 - 3t + 4 \sin 2t - 6e^{4t}\}$

Solutions :

$$\text{i) } \mathcal{L}\{3t + 2e^{3t}\} = 3\mathcal{L}\{t\} + 2\mathcal{L}\{e^{3t}\} = 3\frac{1}{s^2} + 2\frac{1}{s-3}, \operatorname{Re}(s) > \max\{0, 3\} \Rightarrow \operatorname{Re}(s) > 3$$

$$\begin{aligned}
\text{ii) } \mathcal{L}\{5 - 3t + 4 \sin 2t - 6e^{4t}\} &= \mathcal{L}\{5\} - 3\mathcal{L}\{t\} + 4\mathcal{L}\{\sin 2t\} - 6\mathcal{L}\{e^{4t}\} \\
&= \frac{5}{s} - \frac{1}{s^2} + \frac{8}{s^2 + 4} - \frac{6}{s-4}, \operatorname{Re}(s) > 4
\end{aligned}$$

Property 2.2 the first shift property

The property is contained in the following theorem, commonly referred to as the first shift theorem or sometimes as the exponential modulation theorem.

Theorem 2.1: The first shift theorem

If $f(t)$ is a function having Laplace transform $F(s)$; with $\operatorname{Re}(s) > \sigma_c$ then the function $e^{at}f(t)$ also has a Laplace transform given by:

$$\mathcal{L}\{e^{at}f(t)\} = F(s-a), \operatorname{Re}(s) > \sigma_c + \operatorname{Re}(a)$$

Example 2.7: Determine i) $\mathcal{L}\{te^{-2t}\}$ ii) $\mathcal{L}\{e^{-3t} \sin 2t\}$

Solutions :

$$\begin{aligned}
\text{i) } \text{Consider } f(t) = t \text{ and } \mathcal{L}\{t\} = F(s) = \frac{1}{s^2}, \operatorname{Re}(s) > 0, \text{ then } \mathcal{L}\{te^{-2t}\} \\
= F(s+2), \operatorname{Re}(s) > 0 - 2
\end{aligned}$$

So, by the first shift theorem we get $\mathcal{L}\{te^{-2t}\} = \frac{1}{(s+2)^2}, \operatorname{Re}(s) > -2$

$$\text{ii) Consider } f(t) = \sin 2t \text{ and } \mathcal{L}\{\sin 2t\} = F(s) = \frac{2}{s^2+4}, \operatorname{Re}(s) > 0, \text{ then } \mathcal{L}\{e^{-3t} \sin 2t\}$$

$$= F(s+3), \operatorname{Re}(s) > 0 - 3$$

So, by the first shift theorem we get $\mathcal{L}\{e^{-3t} \sin 2t\} = \frac{1}{(s+3)^2+4} = \frac{1}{s^2+6s+13}, \operatorname{Re}(s) > -3$

The function $e^{-3t} \sin 2t$ in this example is a member of a general class of functions called **damped sinusoids**. These play an important role in the study of engineering systems, particularly

in the analysis of vibrations. For this reason we add the following two general members of the class to our standard library of Laplace transform pairs:

$$\mathcal{L}\{e^{-kt} \sin at\} = \frac{a}{(s+k)^2 + a^2}, \quad \operatorname{Re}(s) > -k$$

and

$$\mathcal{L}\{e^{-kt} \cos at\} = \frac{s+k}{(s+k)^2 + a^2}, \quad \operatorname{Re}(s) > -k$$

2.3 Transforms of derivatives and Integrals

2.3.1 Transforms of derivatives

Lemma 2.1: Let $F(s) = \mathcal{L}\{f(t)\}$, then

$$\mathcal{L}\{f'(t)\} = sF(s) - f(0) = s\mathcal{L}\{f(t)\} - f(0)$$

Proof:

$$\mathcal{L}\{f'(t)\} = \int_0^{\infty} f'(t)e^{-st} dt = \lim_{T \rightarrow \infty} \int_0^T f'(t)e^{-st} dt$$

Suppose $u = e^{-st} \Rightarrow du = -se^{-st}dt$ and $dv = f'(t)dt \Rightarrow v = f(t)$

$$\lim_{T \rightarrow \infty} \int_0^T f'(t)e^{-st} dt = \lim_{T \rightarrow \infty} \left\{ [f(t)e^{-st}]_0^T + s \int_0^T f(t)e^{-st} dt \right\} = -f(0) + sF(s)$$

Lemma 2.2: $\mathcal{L}\{f''(t)\} = s^2F(s) - sf(0) - f'(0)$

Theorem 2.2:

If $f(t)$ is a function having Laplace transform $F(s)$; with $\operatorname{Re}(s) > \sigma_c$ then the function $t^n f(t)$, $n = 1, 2, 3, \dots$ also has a Laplace transform given by:

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n F(s)}{ds^n}, \quad \operatorname{Re}(s) > \sigma_c + \operatorname{Re}(a)$$

Example 2.8: Determine i) $\mathcal{L}\{t \sin 3t\}$ ii) $\mathcal{L}\{t^2 e^t\}$ iii) $\mathcal{L}\{t^n\}$

Solutions :

i) $\mathcal{L}\{\sin 3t\} = F(s) = \frac{3}{s^2+9}$, $\operatorname{Re}(s) > 0$, so by derivative theorem we obtain

$$\mathcal{L}\{t \sin 3t\} = -\frac{dF}{ds} = \frac{6s}{(s^2+9)^2}, \quad \operatorname{Re}(s) > 0$$

ii) $\mathcal{L}\{e^t\} = F(s) = \frac{1}{s-1}$, $\operatorname{Re}(s) > 1$, so by derivative theorem we obtain

$$\mathcal{L}\{t^2 e^t\} = (-1)^2 \frac{d^2 F}{ds^2} = \frac{d^2}{ds^2} \left(\frac{1}{s-1} \right) = \frac{2}{(s-1)^3}, \quad \operatorname{Re}(s) > 1$$

iii) $\mathcal{L}\{1\} = \frac{1}{s}$, $\operatorname{Re}(s) > 0$, so by derivative theorem we obtain

$$\mathcal{L}\{t^n\} = (-1)^n \frac{d^n F}{ds^n} = \frac{d^n}{ds^n} \left(\frac{1}{s} \right) = \frac{n!}{s^{n+1}}, \quad \operatorname{Re}(s) > 0$$

2.3.2 Transforms of integrals

$$\mathcal{L} \left\{ \int_0^t f(\xi) d\xi \right\} = \frac{1}{s} \mathcal{L}\{f(t)\} = \frac{1}{s} F(s)$$

Example 2.9: Determine $\mathcal{L} \left\{ \int_0^t (\xi^3 + \sin 2\xi) d\xi \right\}$

Solution: In this case $f(t) = t^3 + \sin 2t \Rightarrow F(s) = \mathcal{L}\{f(t)\} = \mathcal{L}\{t^3\} + \mathcal{L}\{\sin 2t\}$

$$= \frac{6}{s^4} + \frac{2}{s^2 + 4} \Rightarrow \mathcal{L} \left\{ \int_0^t (\xi^3 + \sin 2\xi) d\xi \right\} = \frac{1}{s} F(s) = \frac{6}{s^5} + \frac{2}{s(s^2 + 4)}$$

Theorem 2.3: If $F(s) = \mathcal{L}\{f(t)\}$ and $\lim_{t \rightarrow 0^+} \left(\frac{f(t)}{t} \right)$ exists, then $\mathcal{L} \left\{ \frac{f(t)}{t} \right\} = \int_s^\infty F(u) du$

Example 2.10: Determine $\mathcal{L} \left\{ \frac{\sin t}{t} \right\}$

Solutions:

$$\begin{aligned} \text{i) } \quad \lim_{t \rightarrow 0^+} \left(\frac{\sin t}{t} \right) &= 1 \text{ and } \mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1}, \text{ then } \mathcal{L} \left\{ \frac{\sin t}{t} \right\} = \int_s^\infty \frac{1}{u^2 + 1} du \\ &= \lim_{T \rightarrow \infty} \int_s^T \frac{1}{u^2 + 1} du = \lim_{T \rightarrow \infty} [\arctan T - \arctan s] = \frac{\pi}{2} - \arctan s \end{aligned}$$

$$\text{So, } \mathcal{L} \left\{ \frac{\sin t}{t} \right\} = \frac{\pi}{2} - \arctan s$$

2.4 Initial and final values theorems

How can we find the initial and final values of a function $y(t)$ if we know its Laplace transform $Y(s)$? where $t \rightarrow 0^+$ and $t \rightarrow \infty$.

Initial value theorem

$$\lim_{t \rightarrow 0^+} y(t) = y(0^+) = \lim_{s \rightarrow \infty} sY(s)$$

Conditions:

- Laplace transforms of $y(t)$ and $\frac{dy}{dt}$ exist
- $Y(s)$ numerator power (M) is less than the denominator power (N). i.e $M < N$.

Final value theorem:

$$\lim_{t \rightarrow \infty} y(t) = y(\infty) = \lim_{s \rightarrow 0} sY(s)$$

Conditions:

- Laplace transforms of $y(t)$ and $\frac{dy}{dt}$ exist
- $sY(s)$ poles are all on the left plane or at the origin.

Example 2.11:

Find the initial and the final values of $y(t)$ if $Y(s)$ is given by $Y(s) = \frac{10(2s+3)}{s(s^2+2s+5)}$

Solution:

- Initial value: $y(0^+) = \lim_{s \rightarrow \infty} sY(s) = \lim_{s \rightarrow \infty} \frac{10(2s+3)}{s^2+2s+5} = 0$
- Final value: $y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} \frac{10(2s+3)}{s^2+2s+5} = 6$

Table of Laplace transforms

$f(t)$	$\mathcal{L}\{f(t)\}$	Region of convergence
c , constant	$\frac{c}{s}$	$Re(s) > 0$
t	$\frac{1}{s^2}$	$Re(s) > 0$
t^n , n a positive integer	$\frac{n!}{s^{n+1}}$	$Re(s) > 0$
$e^{\pm kt}$, k a constant	$\frac{1}{s \pm k}$	$Re(s) > \pm k$
$\sin at$, $a \in \mathbb{R}$	$\frac{a}{s^2 + a^2}$	$Re(s) > 0$
$\cos at$, $a \in \mathbb{R}$	$\frac{s}{s^2 + a^2}$	$Re(s) > 0$
$e^{\pm kt} \sin at$, $a, k \in \mathbb{R}$	$\frac{a}{(s \pm k)^2 + a^2}$	$Re(s) > \pm k$
$e^{\pm kt} \cos at$, $a, k \in \mathbb{R}$	$\frac{s}{(s \pm k)^2 + a^2}$	$Re(s) > \pm k$

2.5 Inverse transforms

2.5.1 Definition

The symbol $\mathcal{L}^{-1}$ denotes a causal function $f(t)$ whose Laplace transform is $F(s)$; that is, if $\mathcal{L}\{f(t)\} = F(s)$, then $f(t) = \mathcal{L}^{-1}\{F(s)\}$

Example 2.12:

i) Since $\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \Rightarrow \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$

ii) Since $\mathcal{L}\{\sin wt\} = \frac{w}{s^2+w^2} \Rightarrow \mathcal{L}^{-1}\left\{\frac{w}{s^2+w^2}\right\} = \sin wt$

2.5.2 Evaluation of the inverse transforms

The most obvious way of finding the inverse transform of the function $F(s)$ is to make use of a table of transforms such as that given above. Sometimes it is possible to write down the inverse transform directly from the table, but more often than not it is first necessary to carry out some algebraic manipulation on $F(s)$.

Example 2.13: Find the inverse transform $f(t)$ if

a) $F(s) = \frac{1}{s+3}$ b) $F(s) = \frac{1}{(s-2)^2}$ c) $F(s) = \frac{1}{(s+3)(s-2)}$ d) $F(s) = \frac{s+1}{s^2(s^2+9)}$ e) $F(s) = \frac{2}{s^2+6s+13}$
 f) $F(s) = \frac{s+7}{s^2+2s+5}$

Solutions:

a) $F(s) = \frac{1}{s+3} = \frac{1}{s-(-3)} = \mathcal{L}\{e^{-3t}\}$, so $f(t) = e^{-3t}$

b) $F(s) = \frac{1}{(s-2)^2}$ and $\frac{1}{s-2} = \mathcal{L}\{e^{2t}\} \Rightarrow F(s) = \mathcal{L}\{te^{2t}\} \Rightarrow f(t) = te^{2t}$

c) $F(s) = \frac{1}{(s+3)(s-2)} = \frac{A}{(s-3)} + \frac{B}{(s-2)} \Rightarrow F(s) = \frac{-1/5}{(s+3)} + \frac{1/5}{(s-2)}$
 $\Rightarrow f(t) = -1/5 e^{-3t} + 1/5 e^{2t}$

d) $F(s) = \frac{s+1}{s^2(s^2+9)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+9} \Rightarrow s(s^2+9)A + B(s^2+9) + s^2(Cs+D) = s+1$
 $\Leftrightarrow s^3A + 9sA + Bs^2 + 9B + Cs^3 + Ds^2 = S + 1$
 $\Leftrightarrow (A+C)s^3 + (A+D)s^2 + 9As + 9B = S + 1$

$$\Rightarrow \begin{cases} A+C=0 \\ A+D=0 \\ 9A=1 \\ 9B=1 \end{cases} \Rightarrow \begin{cases} C=-1/9 \\ D=-1/9 \\ A=1/9 \\ B=1/9 \end{cases}$$

$$F(s) = \frac{s+1}{s^2(s^2+9)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+9} = 1/9 \left(\frac{1}{s} + \frac{1}{s^2} - \frac{s+1}{s^2+9} \right) = 1/9 \left(\frac{1}{s} + \frac{1}{s^2} - \frac{s}{s^2+9} - \frac{1}{3} \frac{3}{s^2+9} \right)$$

$$\Rightarrow f(t) = 1/9 \left[1 + t - \cos 3t - \frac{1}{3} \sin 3t \right]$$

$$\text{e) } F(s) = \frac{2}{s^2+6s+1} = \frac{2}{(s+3)^2+4} \Rightarrow f(t) = e^{-3t} \sin 2t$$

$$\text{f) } F(s) = \frac{s+7}{s^2+2s+5} = \frac{(s+1)+6}{(s+1)^2+4} = \frac{(s+1)}{(s+1)^2+4} + \frac{6}{(s+1)^2+4} = \frac{(s+1)}{(s+1)^2+4} + 3 \cdot \frac{2}{(s+1)^2+4}$$

$$\Rightarrow f(t) = e^{-t} \cos 2t + 3e^{-t} \sin 2t$$

2.6 Application of Laplace transforms for solving

Linear ordinary differential equations

Having obtained expressions for the Laplace transform of the derivatives, we are now in a position to use Laplace transform methods to solve ordinary linear differential equations with constant coefficients.

2.6.1 The Laplace transforms method for solving first order linear differential equation

Consider the 1st order ODE's of this form $a \frac{dy}{dt} + by(t) = f(t)$, and $Y(s)$ as Laplace transform of $y(t)$ also $F(s)$ as Laplace transform of $f(t)$, then

$$\mathcal{L} \left\{ a \frac{dy}{dt} + by(t) \right\} = \mathcal{L} \{ f(t) \} \Leftrightarrow a \mathcal{L} \left\{ \frac{dy}{dt} \right\} + b \mathcal{L} \{ y(t) \} = \mathcal{L} \{ f(t) \}$$

$$\Leftrightarrow asY(s) - ay(0) + bY(s) = F(s) \Leftrightarrow (as + b)Y(s) = F(s) + ay(0)$$

$$\Rightarrow Y(s) = \frac{F(s) + ay(0)}{as + b}$$

Then $y(t) = \mathcal{L}^{-1} \{ Y(s) \} = \mathcal{L}^{-1} \left\{ \frac{F(s) + ay(0)}{as + b} \right\}$ is a solution of a given 1st order ODE

Example 2.14: Use Laplace transforms for solving the following system of 1st order ODE

$$\begin{cases} \frac{dx}{dt} + \frac{dy}{dt} + 5x + 3y = e^{-t} \\ 2 \frac{dx}{dt} + \frac{dy}{dt} + x + y = 3 \end{cases}$$

Subject to the initial conditions $x = 2$ and $y = 1$ at $t = 0$ means that $x(0) = 2$ and $y(0) = 1$

Solution: Consider $Y(s)$ as Laplace transform of $y(t)$ and $X(s)$ as Laplace transform of $x(t)$, then we get:

$$\begin{aligned}
\begin{cases} \mathcal{L}\left\{\frac{dx}{dt} + \frac{dy}{dt} + 5x + 3y\right\} = \mathcal{L}\{e^{-t}\} \\ \mathcal{L}\left\{2\frac{dx}{dt} + \frac{dy}{dt} + x + y\right\} = \mathcal{L}\{3\} \end{cases} &\Leftrightarrow \begin{cases} \mathcal{L}\left\{\frac{dx}{dt}\right\} + \mathcal{L}\left\{\frac{dy}{dt}\right\} + 5\mathcal{L}\{x\} + 3\mathcal{L}\{y\} = \mathcal{L}\{e^{-t}\} \\ 2\mathcal{L}\left\{\frac{dx}{dt}\right\} + \mathcal{L}\left\{\frac{dy}{dt}\right\} + \mathcal{L}\{x\} + \mathcal{L}\{y\} = \mathcal{L}\{3\} \end{cases} \\
&\Leftrightarrow \begin{cases} sX(s) - x(0) + sY(s) - y(0) + 5X(s) + 3Y(s) = \frac{1}{s+1} \\ 2sX(s) - 2x(0) + sY(s) - y(0) + X(s) + Y(s) = \frac{3}{s} \end{cases} \\
&\Leftrightarrow \begin{cases} (s+5)X(s) - 2 + (s+3)Y(s) - 1 = \frac{1}{s+1} \\ (2s+1)X(s) - 4 + (s+1)Y(s) - 1 = \frac{3}{s} \end{cases} \\
&\Leftrightarrow \begin{cases} (s+5)X(s) + (s+3)Y(s) = \frac{1}{s+1} + 3 = \frac{3s+4}{s+1} \\ (2s+1)X(s) + (s+1)Y(s) = \frac{3}{s} + 5 = \frac{5s+3}{s} \end{cases}
\end{aligned}$$

Using Cramer's method we obtain: $\Delta = \begin{vmatrix} s+5 & s+3 \\ 2s+1 & s+1 \end{vmatrix} = -s^2 - s + 2$, $\Delta_X =$

$$\begin{vmatrix} \frac{3s+4}{s+1} & s+3 \\ \frac{5s+3}{s} & s+1 \end{vmatrix} = \frac{-2s^2-14s-9}{s}, \quad \Delta_Y = \begin{vmatrix} s+5 & \frac{3s+4}{s+1} \\ 2s+1 & \frac{5s+3}{s} \end{vmatrix} = (s+5)\frac{5s+3}{s} - (2s+1)\frac{3s+4}{s+1}$$

$$\text{Then } \begin{cases} X(s) = \frac{\Delta_X}{\Delta} = \frac{\frac{-2s^2-14s-9}{s}}{-s^2-s+2} = \frac{2s^2+14s+9}{s(s^2+s-2)} = \frac{2s^2+14s+9}{s(s+2)(s-1)} = -\frac{9/2}{s} - \frac{11/6}{s+2} + \frac{25/3}{s-1} \\ Y(s) = \frac{\Delta_Y}{\Delta} = \frac{(s+5)\frac{5s+3}{s} - (2s+1)\frac{3s+4}{s+1}}{-s^2-s+2} = \frac{2s^3-22s^2-39s-15}{s(s+2)(s+1)(s-1)} = \frac{15/2}{s} + \frac{1/2}{s+1} + \frac{11/2}{s+2} - \frac{25/2}{s-1} \end{cases}$$

and the solution of a given system of 1st order ODEs is calculated as:

$$\begin{aligned}
\mathcal{L}^{-1}\{X(s)\} &= \mathcal{L}^{-1}\left\{-\frac{9/2}{s} - \frac{11/6}{s+2} + \frac{25/3}{s-1}\right\} = -\frac{9}{2} - \frac{11}{6}e^{-2t} + \frac{25}{3}e^t \\
\mathcal{L}^{-1}\{Y(s)\} &= \mathcal{L}^{-1}\left\{\frac{15/2}{s} + \frac{1/2}{s+1} + \frac{11/2}{s+2} - \frac{25/2}{s-1}\right\} = \frac{15}{2} + \frac{1}{2}e^{-t} + \frac{11}{2}e^{-2t} - \frac{25}{2}e^t
\end{aligned}$$

Thus the solution to the given pair of simultaneous differential equations is

$$\left. \begin{aligned} x(t) &= -\frac{9}{2} - \frac{11}{6}e^{-2t} + \frac{25}{3}e^t \\ y(t) &= \frac{15}{2} + \frac{1}{2}e^{-t} + \frac{11}{2}e^{-2t} - \frac{25}{2}e^t \end{aligned} \right\} \forall t \in \mathbb{R} \geq 0$$

2.6.2 The Laplace transforms method for solving second order linear differential equations

Consider the following 2nd order ODEs of the form:

$a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy(t) = u(t)$, and $Y(s)$ as Laplace transform of $y(t)$ also $U(s)$ as Laplace transform of $u(t)$, then

$$\begin{aligned}\mathcal{L}\left\{a \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + cy(t)\right\} &= \mathcal{L}\{u(t)\} \Leftrightarrow a\mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} + b\mathcal{L}\left\{\frac{dy}{dt}\right\} + c\mathcal{L}\{y(t)\} = \mathcal{L}\{u(t)\} \\ \Leftrightarrow a[s^2 Y(s) - sy(0) - \dot{y}(0)] &+ b[sY(s) - y(0)] + cY(s) = U(s) \\ \Leftrightarrow (as^2 + bs + c)Y(s) &= U(s) + (as + b)y(0) + a\dot{y}(0) \\ \Rightarrow Y(s) &= \frac{U(s) + (as + b)y(0) + a\dot{y}(0)}{as^2 + bs + c}\end{aligned}$$

Then $y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{U(s) + (as + b)y(0) + a\dot{y}(0)}{as^2 + bs + c}\right\}$ is a solution of a given 2nd order ODE

Example 2.15: Solve the following 2nd order ODE

$$\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y = 2e^{-t}, \quad t \geq 0$$

Subject to the initial conditions $y = 1$ and $\dot{y} = 0$ at $t = 0$ means that $y(0) = 1$ and $\dot{y}(0) = 0$

Solution: Consider $Y(s)$ as Laplace transform of $y(t)$, then we get:

$$\begin{aligned}\mathcal{L}\left\{\frac{d^2 y}{dt^2} + 5 \frac{dy}{dt} + 6y\right\} &= \mathcal{L}\{2e^{-t}\} \Leftrightarrow \mathcal{L}\left\{\frac{d^2 y}{dt^2}\right\} + 5\mathcal{L}\left\{\frac{dy}{dt}\right\} + 6\mathcal{L}\{y\} = \mathcal{L}\{2e^{-t}\} \\ \Leftrightarrow (s^2 + 5s + 6)Y(s) &= \frac{2}{s + 1} + s + 5 \Rightarrow Y(s) = \frac{2}{(s + 1)(s + 2)(s + 3)} + \frac{s + 5}{(s + 2)(s + 3)} \\ \Rightarrow Y(s) &= \frac{1}{s + 1} + \frac{1}{s + 2} - \frac{1}{s + 3}\end{aligned}$$

Then $y(t) = \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s + 1} + \frac{1}{s + 2} - \frac{1}{s + 3}\right\} = e^{-t} + e^{-2t} - e^{-3t}, \quad t \geq 0$ is a solution of a given 2nd order ODE

2.7 Engineering applications: electrical circuits

To illustrate the use of Laplace transform, we consider here their application in the analysis of electrical circuits. Since initial conditions are automatically taken into account in the transformation process, but the Laplace transform is particularly attractive for examining the transient behavior of such systems.

Passive electrical circuits are constructed of three basic elements: **resistor** (having resistance R , measured in ohms Ω), **capacitors** (having capacitance C , measured in farad F) and **inductors** (having inductance L , measured in henries H). With the associated variables being

current $i(t)$ (measured in amperes A) and the **voltage** $v(t)$ (measured in volts V). The current flow in the circuit is related to the charge $q(t)$ (measured in coulombs C) by the relationship

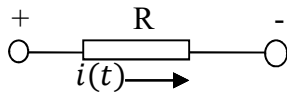
$$i = \frac{dq}{dt}$$

Conventionally, the basis elements are represented symbolically as in the figure bellow. The relationship between the flow of current $i(t)$ and the voltage drops $v(t)$ across these elements at time t are: Voltage drop across resistor = Ri (Ohm's law)

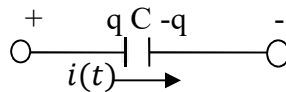
$$\text{Voltage drop across capacitor} = \frac{1}{C} \int i dt = \frac{q}{C}$$

The interaction between the individual elements making up an electrical circuit is determined by **Kirchhoff's law**:

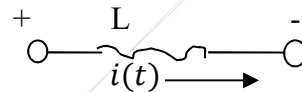
Law 1 The algebraic sum of all the current entering any junction (or node) of a circuit is zero



(a) Resistor



(b) Capacitor



(c) Inductor

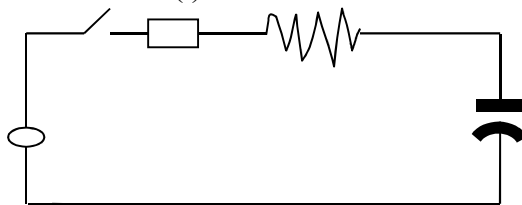
Law 2

The algebraic sum of the voltage drops around any closed group (or path) in a circuit is zero. Use of these laws leads to circuit equations, which may then be analyzed using the Laplace transform techniques.

Example 2.16

The LCR circuit of the figure below consists of a resistor R , a capacitor C and an inductor L connected in series together with a voltage source $e(t)$. Prior to closing the switch at time $t = 0$, both the charge on the capacitor and the resulting current in the circuit are zero. Determine the charge $q(t)$ on the capacitor and the resulting current $i(t)$ in the circuit at time t given that

$$R = 160\Omega, L = 1\text{H}, C = 10^{-4}\text{F} \text{ and } e(t) = 20\text{V}$$



Solution

Applying **Kirchhoff's** second law to the circuit gives

$$Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt = e(t)$$

or, using $\frac{dq}{dt} = i$

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = e(t)$$

Substituting the given values for L; R; C and e(t) gives

$$\frac{d^2q}{dt^2} + 160 \frac{dq}{dt} + 10^4 q = 20$$

Taking Laplace transforms throughout leads to the equation

$$(s^2 + 160s + 10^4)Q(s) = [sq(0) + q'(0)] + \frac{20}{s}$$

Where Q(s) is the transform of q(t). We are given that q(0) = 0

and $q'(0) = i(0) = 0$, so that this reduces to

$$(s^2 + 160s + 104)Q(s) = \frac{20}{s} \Rightarrow Q(s) = \frac{20}{s(s^2 + 160s + 104)}$$

Taking inverse transform, making use of the shift theorem (theorem 2.1), gives

$$q(t) = \frac{1}{500} (1 - e^{-80t} \cos 60t - \frac{4}{3} \sin 60t)$$

The resulting current i(t) in the circuit is then given by

$$i(t) = \frac{dq}{dt} = \frac{1}{3} e^{-80t} \sin 60t$$

EXERCISES

- Use the definition of the Laplace transform to obtain the transforms of $f(t)$ when $f(t)$ is given by: i) $\cosh 2t$ ii) t^2 iii) $3 + t$ iv) te^{-t}
- What are the abscissas of convergence for the following functions?
i) e^{5t} ii) $e^{-5t} + t^2$ iii) $\sinh 3t + \sin 3t$ iv) $3e^{2t} - 2e^{-2t} + \sin 3t$
- Find the Laplace transforms of the following functions and stating the region of convergence: i) $t^2 e^{-2t} + e^{-t} \cos 2t + 3$ ii) $7t^3 e^{-2t} - \sin 3t$
- Find $\mathcal{L}^{-1}\{F(s)\}$ when $F(s)$ is given by:

$$\text{a) } \frac{2s+6}{s^2+4} \quad \text{b) } \frac{s+1}{s^2+2s+5} \quad \text{c) } \frac{1}{(s+1)(s+2)(s^2+2s+1)} \quad \text{d) } \frac{1}{s^2(s^2+16)}$$

5. Using Laplace method, solve for $t \geq 0$ the following differential equations, subject to the specified initial conditions: a) $3 \frac{dx}{dt} - 4x = \sin 2t$ subject to $x(0) = \frac{1}{3}$ b) $\frac{dx}{dt} + 3x = e^{-2t}$ subject

to $x(0) = 2$ c) $\frac{d^2x}{dt^2} - 3 \frac{dx}{dt} + 2x = 4e^{-4t}$ subject to $x(0) = 0$, and $\dot{x}(0) = 1$

d) $\frac{d^2x}{dt^2} + \frac{dx}{dt} - 2x = 5e^{-t} \sin t$ subject to $x(0) = 1$, and $\dot{x}(0) = 0$

6. Using Laplace method, solve for $t \geq 0$ the following simultaneous differential equations, subject to the specified initial conditions:

$$\text{a) } \begin{cases} 5 \frac{d^2x}{dt^2} + 12 \frac{d^2y}{dt^2} + 6x = 0 \\ 5 \frac{d^2x}{dt^2} + 16 \frac{d^2y}{dt^2} + 6y = 0 \end{cases} \text{ subject to } x(0) = \frac{7}{4}, \text{ and } \dot{x}(0) = \dot{y}(0) = 0$$

$$\text{b) } \begin{cases} 2 \frac{d^2x}{dt^2} - \frac{d^2y}{dt^2} - \frac{dx}{dt} - \frac{dy}{dt} = 3y - 9x \\ 2 \frac{d^2x}{dt^2} - \frac{d^2y}{dt^2} + \frac{dx}{dt} + \frac{dy}{dt} = 5y - 7x \end{cases} \text{ subject to } x(0) = \dot{x}(0) = 1, \text{ and } y(0) = \dot{y}(0) = 0$$

UNIT III: MULTIPLE INTEGRALS

3.0. INTRODUCTION

Just as the definite integral of a positive function of one variable represents the **area** of the region between the graph of the function and the x -axis, the **double integral** of a positive function of two variables represents the **volume** of the region between the surface defined by the function (on the three dimensional **Cartesian plane** where $z = f(x, y)$) and the plane which contains its **domain**. (Note that the same volume can be obtained via the **triple integral**—the integral of a function in three variables—of the constant function $f(x, y, z) = 1$ over the above-mentioned region between the surface and the plane.) If there are more variables, a multiple integral will yield **hyper volumes** of multi-dimensional functions.

Multiple integration of a function in n variables: $f(x_1, x_2, \dots, x_n)$ over a domain D is most commonly represented by nested integral signs in the reverse order of execution (the leftmost integral sign is computed last), followed by the function and integrand arguments in proper order (the integral with respect to the rightmost argument is computed last). The domain of integration is either represented symbolically for every argument over each integral sign, or is abbreviated by a variable at the rightmost integral sign:

$$\int \cdots \int_D f(x_1, x_2, \dots, x_n) dx_1 \cdots dx_n$$

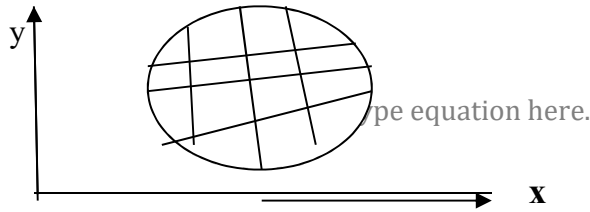
Since the concept of an *ant derivative* is only defined for functions of a single real variable, the usual definition of the *indefinite integral* does not immediately extend to the multiple integral.

The resolution of problems with multiple integrals consists, in most of cases, of finding a way to reduce the multiple integral to an *iterated integral*, a series of integrals of one variable, each being directly solvable. Sometimes, it is possible to obtain the result of the integration by direct examination without any calculations.

3.1. DOUBLE INTEGRATION IN CARTESIAN AND POLAR COORDINATES

3.1.1. Definition

Consider a function $f(x, y)$ defined on domain D bounded by a curve L .



Subdivize D in $s_1, s_2, s_3, \dots, s_n$ of area $\Delta s_1, \Delta s_2, \Delta s_3, \dots, \Delta s_n$ respectively. In each s_i take a point $P_i(\zeta_i, \eta_i)$ and form the sum: $\sum_{i=0}^n f(\zeta_i, \eta_i) \Delta s_i$.

If the limit of that sum when $n \rightarrow \infty$ while the diameter of s_i tends to zero, exists then we say that $f(x, y)$ has a *double integral* in D denoted by $\iint_D f(x, y) dx dy$. So we have:

$$\iint_D f(x, y) dx dy = \lim_{\substack{n \rightarrow \infty \\ \text{diam } s_i \rightarrow 0}} \sum_{i=0}^n f(\zeta_i, \eta_i) \Delta s_i \quad (2.1)$$

Here the limit should be independent of the mode of subdivision

3.1.2. Existence of a double integral

Theorem 3.1: If $f(x, y)$ is continuous on D , then $f(x, y)$ has a double integral on D .

This means the limit of the second member of (2.1) exists and is independent of the mode of subdivision.

3.1.3. Properties

If $f(x, y)$, and $g(x, y)$ are continuous on D , then we have:

- 1) $\iint_D [f(x, y) \pm g(x, y)] dx dy = \iint_D f(x, y) dx dy \pm \iint_D g(x, y) dx dy$
- 2) $\iint_D \lambda f(x, y) dx dy = \lambda \iint_D f(x, y) dx dy, \lambda \in \mathbb{R}$
- 3) If $D = D_1 \cup D_2$ and $D_1^0 \cap D_2^0 = \emptyset$, where D_1^0 , and D_2^0 are interior of D_1 and D_2 respectively, then $\iint_D f(x, y) dx dy = \iint_{D_1} f(x, y) dx dy + \iint_{D_2} f(x, y) dx dy$
- 4) If $D = D_1 \cup D_2 \cup D_3 \cup \dots \cup D_n$ with D_j^0 being two by two disjoint, then:

$$\iint_D f(x, y) dx dy = \iint_{D_1} f(x, y) dx dy + \iint_{D_2} f(x, y) dx dy + \dots + \iint_{D_n} f(x, y) dx dy$$

3.1.4. Evaluation of a double integral

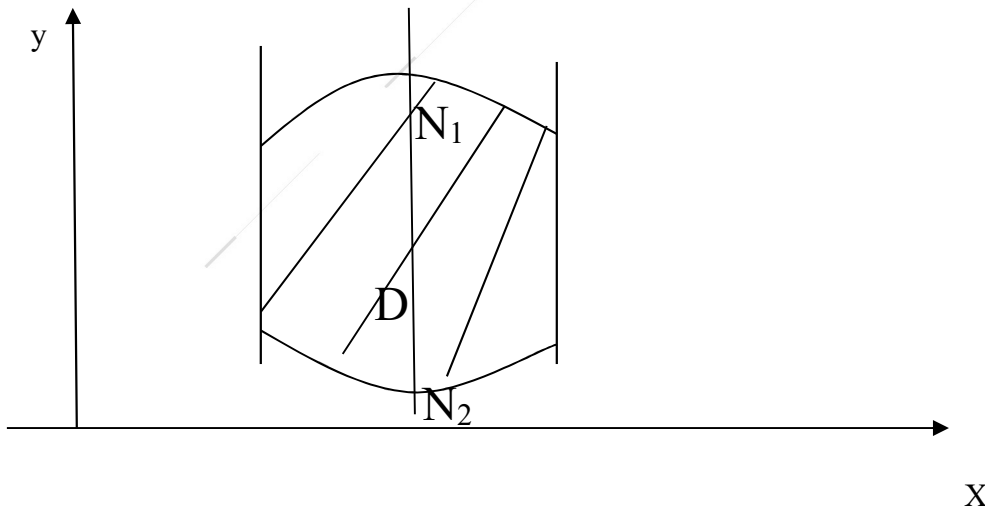
3.1.4.1. Normal (Regular) domains on $\mathbb{R}^2$

This method is applicable to any domain D for which:

- the projection of D onto either the x -axis or the y -axis is bounded by the two values, a and b
- any line perpendicular to this axis that passes between these two values intersects the domain in an interval whose endpoints are given by the graphs of two functions, α and β

a) Regular domain in the direction of y -axis

A domain D is regular in the direction of y -axis if any parallel to oy passing through an interior point cuts the boundary of D in two points N_1 and N_2



If D is normal with respect to the y -axis and $f : D \rightarrow \mathbb{R}$ is a continuous function; then $\alpha(y)$ and $\beta(y)$ (defined on the interval $[a, b]$) are the two functions that determine D . Then:

$$\iint_D f(x, y) \, dx \, dy = \int_a^b dy \int_{\alpha(y)}^{\beta(y)} f(x, y) \, dx.$$

b) Regular domain in the direction of x-axis

If the domain D is normal with respect to the x -axis, and $f : D \rightarrow \mathbb{R}$ is a *continuous function*; then $\alpha(x)$ and $\beta(x)$ (defined on the interval $[a, b]$) are the two functions that determine D . Then

$$\iint_D f(x, y) \, dx \, dy = \int_a^b dx \int_{\alpha(x)}^{\beta(x)} f(x, y) \, dy.$$

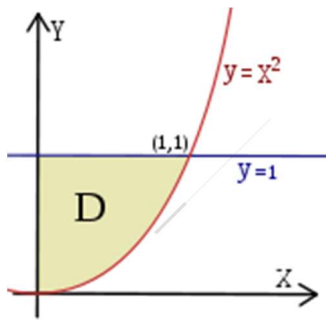
Example 3.1:

Evaluate: a)

$$\iint_D (x + y) \, dx \, dy.$$

Where: $D = \{(x, y) \mid x \geq 0, y \leq 1, y \geq x^2\}$ (please see the graphic in the example).

Answer:



This domain is normal with respect to both the x - and y -axes. To apply the formulae it is required to find the functions that determine D and the intervals over which these are defined.

In this case the two functions are:

$$\alpha(x) = x^2 \text{ and } \beta(x) = 1$$

while the interval is given by the intersections of the functions with $x = 0$, so the interval is $[a, b] = [0, 1]$ (normality has been chosen with respect to the x -axis for a better visual understanding).

It is now possible to apply the formula:

$$\iint_D (x+y) dx dy = \int_0^1 dx \int_{x^2}^1 (x+y) dy = \int_0^1 dx \left[xy + \frac{y^2}{2} \right]_{x^2}^1$$

(at first the second integral is calculated considering x as a constant). The remaining operations consist of applying the basic techniques of integration:

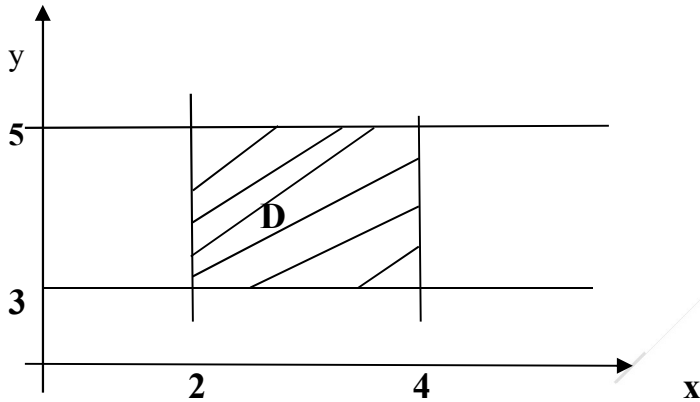
$$\int_0^1 \left[xy + \frac{y^2}{2} \right]_{x^2}^1 dx = \int_0^1 \left(x + \frac{1}{2} - x^3 - \frac{x^4}{2} \right) dx = \dots = \frac{13}{20}.$$

If we choose normality with respect to the y -axis we could calculate

$$\int_0^1 dy \int_0^{\sqrt{y}} (x+y) dx.$$

and obtain the same value.

Evaluate: $\iint_D (x^2 + y^2) dx dy$ where $D = \{(x, y), 2 \leq x \leq 4, 3 \leq y \leq 5\}$



$$\begin{aligned} \iint_D (x^2 + y^2) dx dy &= \int_2^4 \left(\int_3^5 (x^2 + y^2) dy \right) dx = \int_2^4 \left[x^2 y + \frac{y^3}{3} \right]_3^5 dx \\ &= \int_2^4 \left[5x^2 + \frac{5^3}{3} - (3x^2 + \frac{3^3}{3}) \right] dx = \int_2^4 \left[2x^2 + \frac{116}{3} \right] dx = \left[\frac{2}{3} x^3 + \frac{116}{3} x \right]_2^4 \end{aligned}$$

As this domain is also regular in the direction of ox , we may compute this integral as follows:

$$\int_3^5 \left(\int_2^4 (x^2 + y^2) dx \right) dy$$

Example 3.2: Integrating constant functions

When the integrand is a **constant function** c , the integral is equal to the product of c and the measure of the domain of integration. If $c = 1$ and the domain is a sub-region of $\mathbf{R}^2$, the integral gives the area of the region, while if the domain is a sub-region of $\mathbf{R}^3$, the integral gives the volume of the region.

- For example:

$$D = \{(x, y) \in \mathbb{R}^2 : 2 \leq x \leq 4 ; 3 \leq y \leq 6\} \text{ and } f(x, y) = 2$$

in which case

$$\int_3^6 \int_2^4 2 \, dx \, dy = 2 \int_3^6 \int_2^4 1 \, dx \, dy = 2 \cdot \text{area}(D) = (2 \cdot 3) \cdot 2 = 12,$$

since $\int_3^6 \int_2^4 1 \, dx \, dy = \text{area}(D)$ by definition.

3.1.4.2. Change of variables

The limits of integration are often not easily interchangeable (without normality or with complex formulae to integrate). One makes a **change of variables** to rewrite the integral in a more "comfortable" region, which can be described in simpler formulae. To do so, the function must be adapted to the new coordinates.

Example 3.3:

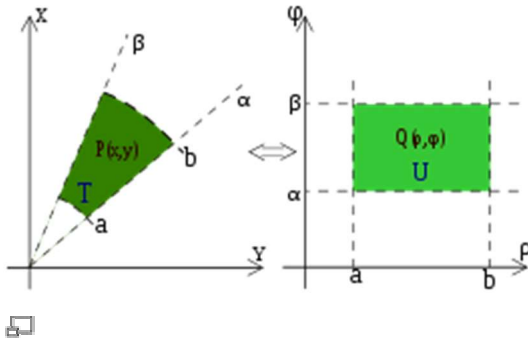
The function is $f(x, y) = (x - 1)^2 + \sqrt{y}$; if one adopts this substitution $x' = x - 1$, $y' = y$ therefore $x = x' + 1$, $y = y'$ one obtains the new function $f_2(x, y) = (x')^2 + \sqrt{y}$.

- Similarly for the domain because it is delimited by the original variables that were transformed before (x and y in example).
- the differentials dx and dy transform via the absolute value of **the determinant of the Jacobian matrix** containing the partial derivatives of the transformations regarding the new variable (consider, as an example, the differential transformation in polar coordinates).

$$\text{Consider } \begin{cases} x = \rho \cos \theta \\ y = \rho \sin \theta \end{cases} \rightarrow J = \frac{\partial(x,y)}{\partial(\rho,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial \rho} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial \rho} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -\rho \sin \theta \\ \sin \theta & \rho \cos \theta \end{vmatrix} = \rho \cos^2 \theta + \rho \sin^2 \theta = \rho$$

There exist three main "kinds" of changes of variable (one in $\mathbb{R}^2$, two in $\mathbb{R}^3$); however, more general substitutions can be made using the same principle.

Polar coordinate system



Transformation from cartesian to polar coordinates.

In $\mathbb{R}^2$ if the domain has a circular symmetry and the function has some particular characteristics you can apply the *transformation to polar coordinates* (see the example in the picture) which means that the generic points $P(x,y)$ in Cartesian coordinates switch to their respective points in polar coordinates. That allows one to change the shape of the domain and simplify the operations.

The fundamental relation to make the transformation is the following:

$$f(x, y) \rightarrow f(\rho \cos \phi, \rho \sin \phi).$$

Example 3.4:

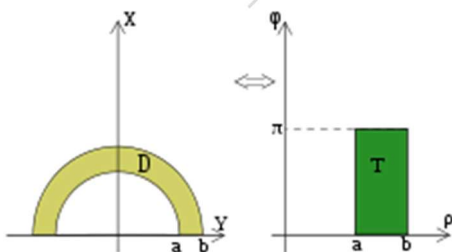
a) The function is $f(x, y) = x + y$ and applying the transformation one obtains

$$f(\rho, \phi) = \rho \cos \phi + \rho \sin \phi = \rho (\cos \phi + \sin \phi).$$

b) The function is $f(x, y) = x^2 + y^2$ In this case one has:

$$f(\rho, \phi) = \rho^2 (\cos^2 \phi + \sin^2 \phi) = \rho^2 \text{ using the } \textit{Pythagorean trigonometric identity} \text{ (very useful to simplify this operation).}$$

The transformation of the domain is made by defining the radius' crown length and the amplitude of the described angle to define the ρ, ϕ intervals starting from x, y .



Example of a domain transformation from cartesian to polar.

c) The domain is $D = x^2 + y^2 \leq 4$, that is a circumference of radius 2; it's evident that the covered angle is the circle angle, so ϕ varies from 0 to 2π , while the crown radius varies from 0 to 2 (the crown with the inside radius null is just a circle).

d) The domain is $D = \{x^2 + y^2 \leq 9, x^2 + y^2 \geq 4, y \geq 0\}$, that is the circular crown in the positive y half-plane (please see the picture in the example); note that ϕ describes a plane angle while ρ varies from 2 to 3. Therefore the transformed domain will be the following rectangle:

$$T = \{2 \leq \rho \leq 3, 0 \leq \phi \leq \pi\}.$$

The **Jacobian determinant** of that transformation is the following:

$$\frac{\partial(x, y)}{\partial(\rho, \phi)} = \begin{vmatrix} \cos \phi & -\rho \sin \phi \\ \sin \phi & \rho \cos \phi \end{vmatrix} = \rho$$

which has been obtained by inserting the partial derivatives of $x = \rho \cos(\phi)$, $y = \rho \sin(\phi)$ in the first column respect to ρ and in the second respect to ϕ , so the $dx dy$ differentials in this transformation becomes $\rho d\rho d\phi$.

Once the function is transformed and the domain evaluated, it is possible to define the formula for the change of variables in polar coordinates:

$$\iint_D f(x, y) dx dy = \iint_T f(\rho \cos \phi, \rho \sin \phi) \rho d\rho d\phi.$$

Please note that ϕ is valid in the $[0, 2\pi]$ interval while ρ , which is a measure of a length, can only have positive values.

e) The function is $f(x, y) = x$ and as the domain the same in 2-d example.

From the previous analysis of D we know the intervals of ρ (from 2 to 3) and of ϕ (from 0 to π).

Now let's change the function:

$$f(x, y) = x \longrightarrow f(\rho, \phi) = \rho \cos \phi \text{ finally let's apply the integration formula:}$$

$$\iint_D x dx dy = \iint_T \rho \cos \phi \rho d\rho d\phi.$$

Once the intervals are known, you have

$$\int_0^\pi \int_2^3 \rho^2 \cos \phi d\rho d\phi = \int_0^\pi \cos \phi d\phi \left[\frac{\rho^3}{3} \right]_2^3 = [\sin \phi]_0^\pi \left(9 - \frac{8}{3} \right) = 0.$$

3.2. CHANGE OF ORDER OF INTEGRATION

Let us assume that we wish to integrate a multivariable function f over a region A .

$$A = \{(x, y) \in \mathbb{R}^2 : 11 \leq x \leq 14 ; 7 \leq y \leq 10\} \text{ and } f(x, y) = x^2 + 4y$$

From this we formulate the double integral

$$\int_7^{10} \int_{11}^{14} (x^2 + 4y) \, dx \, dy$$

The inner integral is performed first, integrating with respect to x and taking y as a constant, as it is not the **variable of integration**. The result of this integral, which is a function depending only on y , is then integrated with respect to y .

$$\begin{aligned} \int_{11}^{14} (x^2 + 4y) \, dx &= \left(\frac{1}{3}x^3 + 4yx \right) \Big|_{x=11}^{x=14} \\ &= \frac{1}{3}(14)^3 + 4y(14) - \frac{1}{3}(11)^3 - 4y(11) \\ &= 471 + 12y \end{aligned}$$

We then integrate the result with respect to y .

$$\begin{aligned} \int_7^{10} (471 + 12y) \, dy &= (471y + 6y^2) \Big|_{y=7}^{y=10} \\ &= 471(10) + 6(10)^2 - 471(7) - 6(7)^2 \\ &= 1719 \end{aligned}$$

Then the multiple integral will give the same result as the iterated integral,

$$\int_{A \times B} f(x, y) \, d(x, y) = \int_A \left(\int_B f(x, y) \, dy \right) dx = \int_B \left(\int_A f(x, y) \, dx \right) dy.$$

In particular this will occur if $|f(x, y)|$ is a **bounded function** and A and B are **bounded sets**.

If the integral is not absolutely convergent, care is needed not to confuse the concepts of *multiple integral* and *iterated integral*, especially since the same notation is often used for either concept. The notation

$$\int_0^1 \int_0^1 f(x, y) \, dy \, dx$$

means, in some cases, an iterated integral rather than a true double integral. In an iterated integral, the outer integral

$$\int_0^1 \cdots dx$$

is the integral with respect to x of the following function of x :

$$g(x) = \int_0^1 f(x, y) dy.$$

A double integral, on the other hand, is defined with respect to area in the xy -plane. If the double integral exists, then it is equal to each of the two iterated integrals (either " $dy dx$ " or " $dx dy$ ") and one often computes it by computing either of the iterated integrals. But sometimes the two iterated integrals exist when the double integral does not, and in some such cases the two iterated integrals are different numbers, i.e., one has

$$\int_0^1 \int_0^1 f(x, y) dy dx \neq \int_0^1 \int_0^1 f(x, y) dx dy.$$

This is an instance of rearrangement of a **conditionally convergent** integral.

The notation

$$\int_{[0,1] \times [0,1]} f(x, y) dx dy$$

may be used if one wishes to be emphatic about intending a double integral rather than an iterated integral distribution.

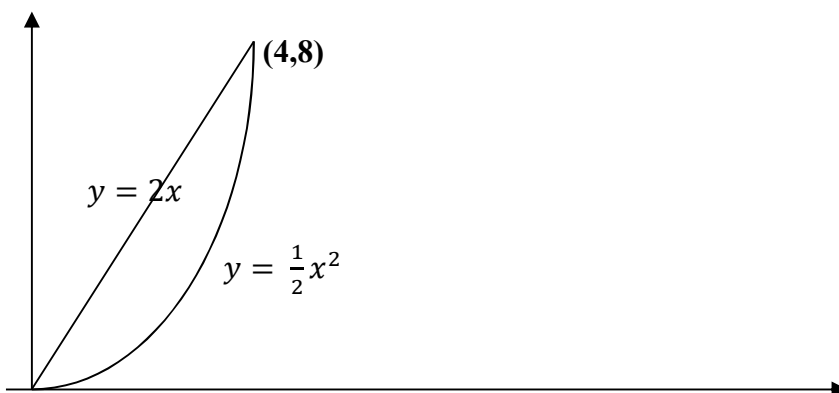
3.3. AREA AS A DOUBLE INTEGRAL

Area of a rectangular domain D is:

$$\text{Area of } D = \iint_D dx dy$$

Example: Use a double integral to find the area of the region D enclosed between the parabola $y = \frac{1}{2}x^2$ and $y = 2x$

Solution:



Area of $D = \iint_D dx dy = \int_0^4 \int_{\frac{1}{2}x^2}^{2x} dy dx = \int_0^4 [y]_{\frac{1}{2}x^2}^{2x} dx = \int_0^4 \left(2x - \frac{1}{2}x^2\right) dx = \left[2x - \frac{1}{2}x^2\right]_0^4 = \frac{16}{3}$

3.4 Triple integrals in Cartesian coordinates

Consider a function $f(x,y,z)$ defined on a domain G of $\mathbb{R}^3$ then:

$$\iiint_G f(x, y, z) dV = \lim_{\substack{x \rightarrow +\infty \\ \text{diam } v_i}} \sum_{i=1}^n f(\xi_i; \eta_i; x_i) \Delta v_i \quad (2)$$

3.4.2 Existence of Triple integral

If $f(x,y,z)$ is continuous, then the limit of the second number of (2) exists and is not dependent of the mode of subdivision. Therefore (2) is called a **Triple integral**.

3.4.3 Properties

1. $\iiint_G [f(x, y, z) \pm g(x, y, z)] dx dy dz = \iiint_G f(x, y, z) dx dy dz \pm \iiint_G g(x, y, z) dx dy dz$
2. $\iiint_G \lambda f(x, y, z) dx dy dz = \lambda \iiint_G f(x, y, z) dx dy dz$
3. $\iiint_G f(x, y, z) dx dy dz = \iiint_{G_1} f(x, y, z) dx dy dz \pm \iiint_{G_2} f(x, y, z) dx dy dz$

3.4.4 Evaluation of triple integrals

Theorem 1: Let V be the rectangular box defined by the inequalities $a \leq x \leq b ; c \leq y \leq d ;$

$k \leq z \leq l$. If f is continuous on the region V , then

$$\iiint_G f(x, y, z) dV = \int_a^b \int_c^d \int_k^l f(x, y, z) dz dy dx \quad (3.1)$$

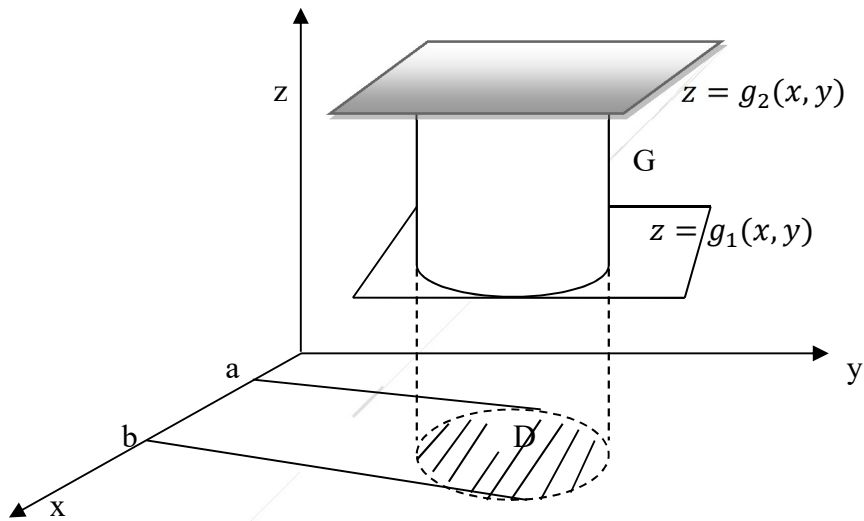
Example: Evaluate the triple integral $\iiint_G 12xy^2z^3 dV$ over the rectangular box G defined by the inequalities $-1 \leq x \leq 2 ; 0 \leq y \leq 3 ; 0 \leq z \leq 2$.

Solution:
$$\iiint_G 12xy^2z^3 dV = \int_{-1}^2 \int_0^3 \int_0^2 12xy^2z^3 dz dy dx = \int_{-1}^2 \int_0^3 \left[3xy^2z^4 \right]_0^2 dy dx = \int_{-1}^2 \int_0^3 48xy^2 dy dx =$$

$$\int_{-1}^2 \left[16xy^3 \right]_0^3 dx = \int_{-1}^2 432x dx = \left[216x^2 \right]_{x=-1}^2 = 648$$

Theorem 2: Let G be a simple solid with upper surface $z = g_2(x, y)$ and lower surface $z = g_1(x, y)$; and D be the projection of G on the xy -plane. If $f(x, y, z)$ is continuous on G , then

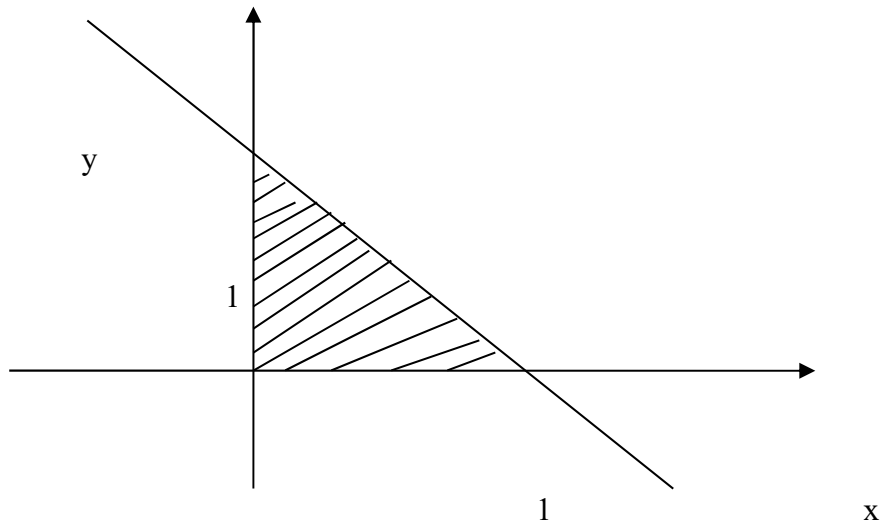
$$\iiint_G f(x, y, z) dV = \iint_D \left[\int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz \right] dA$$



$$\iiint_G f(x, y, z) dV = \int_a^b \left(\int_{y=\varphi_1(x)}^{y=\varphi_2(x)} \left(\int_{z=g_1(x, y)}^{z=g_2(x, y)} f(x, y, z) dz \right) dy \right) dx$$

Example 1: evaluate $\iiint_G xyz dV$ where G is the domain of $\mathbb{R}^3$ limited by: $x=0$, $y=0$, $z=0$, $x+y+z=1$.

Solution: $x + y + z = 1 \Rightarrow z = 1 - x - y$

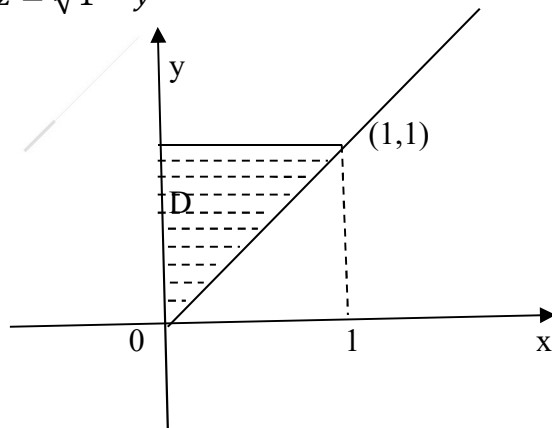


$$\begin{aligned}
 \iiint_G xyz dV &= \int_{x=0}^1 \left(\int_{y=0}^{1-x} \left(\int_{z=0}^{1-x-y} xyz dz \right) dy \right) dx \\
 &= \int_0^1 \left(\int_0^{1-x} \left[\frac{xyz^2}{2} \right]_{z=0}^{1-x-y} dy \right) dx = \int_0^1 \left(\int_0^{1-x} \frac{xy}{2} (1-x-y)^2 dy \right) dx = \frac{1}{720}
 \end{aligned}$$

Example 2 : Let G be the wedge in the first octant cut from the cylinder $y^2 + z^2 = 1$ by the planes $y = x$ and $x = 0$. Evaluate $\iiint_G z dV$.

Solution:

$$y^2 + z^2 = 1 \Rightarrow z = \sqrt{1 - y^2}$$



$$\begin{aligned}
 \iiint_G z dV &= \int_{y=0}^1 \left(\int_{x=0}^y \left(\int_{z=0}^{\sqrt{1-y^2}} z dz \right) dx \right) dy = \int_0^1 \int_0^y \left[\frac{z^2}{2} \right]_0^{\sqrt{1-y^2}} dx dy = \int_0^1 \int_0^y \frac{1}{2} (1 - y^2) dx dy = \frac{1}{2} \int_0^1 [(1 - y^2)x]_0^y dy
 \end{aligned}$$

$$= \frac{1}{2} \int_0^1 (y - y^3) dy = \frac{1}{2} \left[\frac{y^2}{2} - \frac{y^4}{4} \right]_0^1 = \frac{1}{8}$$

3.4.5 Some results on triple integrals

Theorem 3: If $f(x,y,z)$ is continuous on a domain G and if m is the least value and M the greatest value of $f(x,y,z)$ on G , then we have: $m \text{ vol}(V) \leq \iiint_G f(x,y,z) dV \leq M \text{ vol}(V)$

Corollary: Volume $(V) = \iiint_G f(x,y,z) dV$

Example: Use the triple integral to find the volume of the solid enclosed between the cylinder $x^2 + y^2 = 9$ and the planes $z=1$ and $x + z = 5$.

Solution: Volume of $G = \iiint_G dV$

$$\begin{aligned} &= \int_{x=-3}^3 \int_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_1^{5-x} dz dy dx = \int_{x=-3}^3 \int_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} [z]_1^{5-x} dy dx = \int_{x=-3}^3 \int_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x) dy dx \\ &= \int_{-3}^3 (8-2x) \sqrt{9-x^2} dx = 8 \int_{-3}^3 \sqrt{9-x^2} - 2 \int_{-3}^3 x \sqrt{9-x^2} \\ &= 8 * \frac{9}{2} \pi - 0 = 36\pi \end{aligned}$$

Theorem 4: if $f(x,y,z)$ is continuous on a bounded domain G then there exists a point p of coordinates (ξ, η, χ) i.e. $P(\xi, \eta, \chi)$ such that $\iiint_G f(x,y,z) dV = \text{vol}(V) \cdot f(\xi, \eta, \chi)$.

TUTORIAL ON CHAPTER II

TUTORIAL I

1. Evaluate the following double integrals:

(a) $\iint_D x \ln y dx dy$ where $D = \{(x,y) / 0 \leq x \leq 4; 1 \leq y \leq e\}$

(b) $\iint_D (\cos^2 x + \sin^2 y) dx dy$ where $D = \{(x,y) / 0 \leq x \leq \frac{\pi}{4}; 0 \leq y \leq \frac{\pi}{4}\}$

(c) $\iint_D (x-y) dx dy$ where D is a domain limited by the curves of equations $y = 2 - x^2$; $y = 2x - 1$

2. Use the change of variables for evaluating the following double integrals:

- (a) $\iint_D \ln(x^2 + y^2) dA$ where $D = \{(x, y) / x^2 + y^2 = e \text{ and } x^2 + y^2 = 4\}$ (Hint: use polar coordinates)
- (b) $\iint_D (x+y)^3 (x-y)^2 dA$ where $D = \{(x, y) / x+y=1; x-y=1\}$ (Hint: use the transformation $x+y=u$ and $x-y=v$)
- (c) $\iint_D \frac{\sin(\sqrt{x^2 + y^2})}{\sqrt{x^2 + y^2}} dA$ Where $D = \{(x, y) / x^2 + y^2 = \frac{\pi^2}{9} \text{ and } x^2 + y^2 = \pi^2\}$

(Hint: use the polar coordinates)

3.

- (a) Evaluate $\iint_D (x+2y)^{-\frac{1}{2}} dA$ over the region $x-2y \leq 1$ and $x \geq y^2 + 1$
- (b) Evaluate $\iint_D x^2 y dA$ where D is the region $x^2 + y^2 \leq 1$
- (c) Evaluate

I. $\int_0^3 \int_1^2 xy(x+y) dy dx$

II. $\int_{-1}^1 \int_{-2}^2 (2x^2 + y^2) dy dx$

4. Evaluate $\iint_D \sqrt{9-x^2-y^2} dA$ where D is the region in the first quadrant within $x^2 + y^2 = 1$

5. Evaluate the iterated integral:

(a) $\int_{-1}^1 \int_0^2 \int_0^1 (x^2 + y^2 + z^2) dx dy dz$

(b) $\int_{\frac{1}{3}}^{\frac{1}{2}} \int_0^{\pi} \int_0^1 zx \sin xy dz dy dx$

(c) $\int_0^2 \int_{-1}^{y^2} \int_1^z yz dx dy dz$

(d) $\int_1^3 \int_x^2 \int_0^{\ln z^2} x e^z dz dy dx$

6. Evaluate the following triple integral:

(a) $\iiint_G xy \sin yz dV$ where G is the rectangular box defined by the inequalities $0 \leq x \leq \pi, 0 \leq y \leq 1, 0 \leq z \leq \frac{\pi}{6}$

(b) $\iiint_G \cos\left(\frac{z}{y}\right) dV$ where G is the rectangular box defined by the inequalities $\frac{\pi}{6} \leq y \leq \frac{\pi}{2}$, $y \leq x \leq \frac{\pi}{2}$, $0 \leq z \leq xy$

7. Use a triple integral to find the volume of the solid:

(a) The solid bounded by the surface $y = x^2$ and the planes $x + y = 0$ and $z = 0$

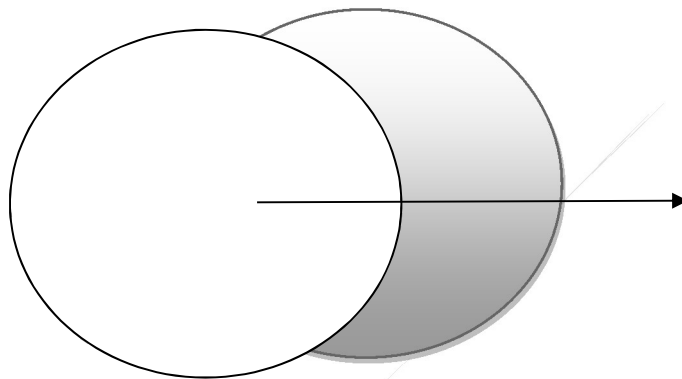
(b) The solid bounded by the surface $z = \sqrt{y}$ and the planes $x + y = 0$ and $z = 0$

8.

(a) Calculate the area of a figure limited by the lines $x = 4y - y^2$; $x + y = 6$

(b) Calculate the area of the figure limited by the circumferences $\rho = 1$; $\rho = \frac{2}{3} \cos \theta$

(Outside of the circumference $\rho = 1$)



9. Calculate $\iiint_G z dV$ where G is the domain defined by the inequalities $0 \leq x \leq \frac{1}{2}$; $x \leq y \leq 2x$;

$$0 \leq z \leq \sqrt{1 - x^2 - y^2}$$

10. Calculate the volume of a solid limited by the surfaces $z = \sqrt{x^2 + y^2}$ and $z = x^2 + y^2$
(Ans: $\frac{\pi}{6}$)

11. Evaluate the triple integrals:

(a) $\int_0^1 dx \int_0^2 dy \int_1^3 x^2 y z dz$

(b) $\int_0^2 \int_1^3 \int_1^4 x y z^2 dz dy dx$

12. Show that $\int_{-1}^1 dz \int_0^2 dx \int_{x-z}^{x+z} (x + y + z) dy = 0$

13. Find the volume of the solid enclosed by the paraboloids $z = 5x^2 + 5y^2$ and $z = 6 - 7x^2 - y^2$

14. Evaluate :

(a) $\iint_D x \sqrt{1 - x^2} dA$; $D = \{(x, y) / 0 \leq x \leq 1; 2 \leq y \leq 3\}$

$$(b) \iint_D \cos(x+y) dA; \quad D = \{(x,y) / -\frac{\pi}{4} \leq x \leq \frac{\pi}{4}; \quad 0 \leq y \leq \frac{\pi}{4}\}$$

15.

- (a) If $u = x + y - 3z$; $v = 3x^2 + x + y$; $w = x + 3y - z$ determine the Jacobian transformation of (x,y,z)
- (b) If $x = \rho \cos \theta$; $y = \rho \sin \theta$; $z = z$ determine the Jacobian transformation of (x,y,z)
- (c) If $x = \rho \sin \varphi \cos \theta$; $y = \rho \sin \varphi \sin \theta$; $z = \rho \cos \varphi$ determine the Jacobian transformation of x,y and z .

UNIT IV. INTRODUCTION TO VECTOR CALCULUS

4.1. GRADIENT, DIVERGENCE, CURL-SOLENOIDAL AND IRRITATION FIELDS

4.1.1. Gradient of a scalar function

Consider a scalar function $\phi(x, y, z)$, then the gradient of $\phi(x, y, z)$ denoted by **grad** $\phi(x, y, z)$ or $\vec{\nabla}\phi(x, y, z)$ is defined as follows:

$$\vec{\nabla}\phi(x, y, z) = \frac{\partial\phi(x,y,z)}{\partial x} \vec{i} + \frac{\partial\phi(x,y,z)}{\partial y} \vec{j} + \frac{\partial\phi(x,y,z)}{\partial z} \vec{k} \quad (4.1)$$

Where $\vec{\nabla} = \frac{\partial}{\partial x} \vec{i} + \frac{\partial}{\partial y} \vec{j} + \frac{\partial}{\partial z} \vec{k}$

Example 4.1: If we have $\phi(x, y, z) = 3x^2yz^3$, then calculate $\text{grad}\phi(x, y, z)$?

Answer:

$$\begin{aligned} \vec{\nabla}\phi(x, y, z) &= \vec{\nabla}[3x^2yz^3] = \frac{\partial(3x^2yz^3)}{\partial x} \vec{i} + \frac{\partial(3x^2yz^3)}{\partial y} \vec{j} + \frac{\partial(3x^2yz^3)}{\partial z} \vec{k} \\ &= 6xyz^3 \vec{i} + 3x^2z^3 \vec{j} + 9x^2yz^2 \vec{k} \end{aligned}$$

4.1.2. Divergence of a vector function

A vector function is like $A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$.

Consider a vector function $\vec{A}(x, y, z) = A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$, then the divergence of $\vec{A}$ denoted by **div** $\vec{A}$ or $\vec{\nabla} \cdot \vec{A}$ is defined as follows:

$$\text{div}\vec{A}(x, y, z) = \frac{\partial A_1(x,y,z)}{\partial x} + \frac{\partial A_2(x,y,z)}{\partial y} + \frac{\partial A_3(x,y,z)}{\partial z} \quad (4.2)$$

Example 4.2: If $\vec{A}(x, y, z) = xyz\vec{i} + (x^2 - y^2 + z^2)\vec{j} + 2xyz^3\vec{k}$, then calculate $\text{div}\vec{A}$?

Answer: $\text{div}\vec{A} = \vec{\nabla} \cdot \vec{A} = \frac{\partial xyz}{\partial x} + \frac{\partial x^2 - y^2 + z^2}{\partial y} + \frac{\partial 2xyz^3}{\partial z} = yz - 2y + 6xyz^2$

4.1.3. Curl of a vector function

Consider a vector function $\vec{A}(x, y, z) = A_1(x, y, z)\vec{i} + A_2(x, y, z)\vec{j} + A_3(x, y, z)\vec{k}$, then the Curl of $\vec{A}$ denoted by $\text{Curl}\vec{A}$ or $\vec{\nabla} \times \vec{A}$ is defined as follows:

$$\text{Curl}\vec{A} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix} = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \vec{i} - \left(\frac{\partial A_3}{\partial x} - \frac{\partial A_1}{\partial z} \right) \vec{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \vec{k} \quad (4.3)$$

Example 4.3: If $\vec{A}(x, y, z) = xyz\vec{i} + (x^2 - y^2 + z^2)\vec{j} + 2xyz^3\vec{k}$, then calculate $\text{curl}\vec{A}$?

$$\begin{aligned} \text{Answer: } \text{Curl}\vec{A} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & x^2 - y^2 + z^2 & 2xyz^3 \end{vmatrix} \\ &= \left(\frac{\partial 2xyz^3}{\partial y} - \frac{\partial (x^2 - y^2 + z^2)}{\partial z} \right) \vec{i} - \left(\frac{\partial (x^2 - y^2 + z^2)}{\partial x} - \frac{\partial xyz}{\partial z} \right) \vec{j} \\ &\quad + \left(\frac{\partial (x^2 - y^2 + z^2)}{\partial x} - \frac{\partial xyz}{\partial y} \right) \vec{k} \\ &= (xz^3 + 2z)\vec{i} - (yz^3 - xy)\vec{j} + (2x - xz)\vec{k} \end{aligned}$$

Note: In the sequence of our course we can use ∇ instead of $\vec{\nabla}$.

4.1.4. Solenoidal and irrotational fields

A **solenoidal vector field** satisfies

$$\nabla \cdot \vec{B} = 0 \quad (4.4)$$

for every vector $\vec{B}$, where $\nabla \cdot \vec{B}$ is the *divergence*. If this condition is satisfied, there exists a vector $\vec{A}$, known as the *vector potential*, such that

$$\vec{B} \equiv \nabla \times \vec{A} \quad (4.5)$$

where $\nabla \times \vec{A}$ is the *curl*. This follows from the vector identity

$$\nabla \cdot \vec{B} = \nabla \cdot (\nabla \times \vec{A})$$

If $\vec{A}$ is an **irrotational field** such that a vector field $\vec{V}$ for which the curl vanishes,

$$\nabla \times \vec{V} = \vec{0} \quad (4.6)$$

4.2. VECTOR IDENTITIES (WITHOUT PROOF)

If the partial derivatives of vector functions $\vec{A}$, $\vec{B}$ and scalar functions ϕ , φ are assumed to exist, then:

1. $\nabla \cdot (\phi + \varphi) = \nabla \phi + \nabla \varphi$ or $grad(\phi + \varphi) = grad\phi + grad\varphi$
2. $\nabla \cdot (\vec{A} + \vec{B}) = \nabla \cdot \vec{A} + \nabla \cdot \vec{B}$ or $div(\vec{A} + \vec{B}) = div\vec{A} + div\vec{B}$
3. $\nabla \times (\vec{A} + \vec{B}) = \nabla \times \vec{A} + \nabla \times \vec{B}$ or $curl(\vec{A} + \vec{B}) = curl\vec{A} + curl\vec{B}$
4. $\nabla \cdot (\varphi \vec{A}) = (\nabla \varphi) \cdot \vec{A} + \varphi (\nabla \cdot \vec{A})$
5. $\nabla \times (\varphi \vec{A}) = (\nabla \varphi) \times \vec{A} + \varphi (\nabla \times \vec{A})$
6. $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$
7. $\nabla \times (\vec{A} \times \vec{B}) = (\vec{B} \cdot \nabla) \vec{A} - \vec{B} (\vec{A} \cdot \nabla) - (\vec{A} \cdot \nabla) \vec{B} + \vec{A} (\nabla \cdot \vec{B})$
8. $\nabla (\vec{A} \cdot \vec{B}) = (\vec{B} \cdot \nabla) \vec{A} - (\vec{A} \cdot \nabla) \vec{B} + \vec{B} \times (\nabla \times \vec{A}) + \vec{A} \times (\nabla \times \vec{B})$
9. $\nabla \cdot (\nabla \varphi) = \nabla^2 \varphi = \frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2}$ is called Laplace of a scalar function φ and $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called the Laplace operator.
10. $\nabla \times (\nabla \varphi) = 0$ The curl of the gradient of a scalar function is always zero.
11. $\nabla \cdot (\nabla \times \vec{A}) = 0$ The gradient of the curl of a vector function is always zero
12. $\nabla \times (\nabla \cdot \vec{A}) = \nabla (\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

4.3. DIRECTIONAL DERIVATIVES

We can calculate the rate of change of $f(x, y, z)$ along any direction we please. If $\hat{u}$ is the unit vector in that direction then $(grad f) \cdot \hat{u} = \nabla f \cdot \hat{u}$ gives the required **directional derivative**, that is, the rate of change of $f(x, y, z)$ in the direction of $\hat{u}$. Remember that $\vec{a} \cdot \vec{b} = |\vec{a}| \cdot |\vec{b}| \cos \theta$, where θ is the angle between the two vectors, it follows that the rate of change of $f(x, y, z)$ is zero along directions perpendicular to $grad f$ and is maximum along the direction parallel to $grad f$.

Example: Find $grad f$ for $f(x, y, z) = 3x^2 + 2y^2 + z^2$ at the point (1,2,3). Hence calculate:

- a) The directional derivative of $f(x, y, z)$ at the point (1,2,3) in the direction of the unit vector $(2,2,1)/3$?

- b) The maximum rate of change of the function at (1,2,3) and its direction?

Solution: $\text{grad } f = 6x\vec{i} + 4y\vec{j} + 2z\vec{k}$, then $\text{grad } f(1,2,3) = 6\vec{i} + 8\vec{j} + 6\vec{k}$

- a) The directional derivative of $f(x, y, z)$ at the point (1,2,3) in the direction of the unit vector $\frac{1}{3}(2,2,1)$ is $(6\vec{i} + 8\vec{j} + 6\vec{k}) \cdot \left(\frac{2}{3}\vec{i} + \frac{2}{3}\vec{j} + \frac{1}{3}\vec{k}\right) = \frac{34}{3}$
- b) The maximum rate of change of the function at (1,2,3) occurs along the direction parallel to $\text{grad } f$ at (1,2,3), that is, parallel to (6,8,6). The unit vector in that direction is $(3,4,3)/\sqrt{34}$ and the maximum rate of change of $f(x, y, z)$ is $2\sqrt{34}$

4.4. LINE, SURFACE AND VOLUME INTEGRALS

4.4.1. LINE INTEGRALS

4.4.1.1. Definition of line integrals

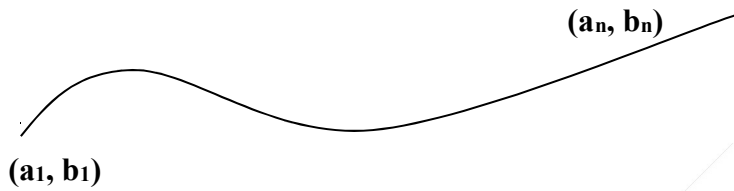


Fig 1. Subdivision of the curve C

Consider a curve C in the XoY plane. Subdivide C as indicated in the fig1. Between (x_{k-1}, y_{k-1}) and (x_k, y_k) take (ζ_k, η_k) and form the following sum:

$\sum_{k=1}^n [P(\zeta_k, \eta_k)\Delta x_k + Q(\zeta_k, \eta_k)\Delta y_k]$ where $P(x, y)$ and $Q(x, y)$ are functions defined on a domain containing C and $\Delta x_k, \Delta y_k$ equal to $|x_k - x_{k-1}|$ and $|y_k - y_{k-1}|$ respectively.

If the limit of this sum exists when $n \rightarrow +\infty$ while Δx_k and Δy_k tend to zero and is independent of the mode of subdivision, then we say that $P(x, y)dx + Q(x, y)dy$ has a line integral with C denoted by:

$$\int_C P(x, y)dx + Q(x, y)dy$$

$$\int_C P(x, y)dx + Q(x, y)dy = \lim_{\substack{\Delta x_k \rightarrow 0 \\ \Delta y_k \rightarrow 0}} \sum_{k=1}^n [P(\zeta_k, \eta_k)\Delta x_k + Q(\zeta_k, \eta_k)\Delta y_k] \quad (4.8)$$

If $P(x, y)$ and $Q(x, y)$ are continuous in a domain D containing the curve C, then the limit in the second hand of (4.8) exists and is independent of the mode of subdivision.

4.4.1.2. Properties of line integrals

$$1. \int_C P(x, y)dx \pm Q(x, y)dy = \int_C P(x, y)dx \pm \int_C Q(x, y)dy$$

Remark: If C:

then

$\int_C P(x, y)dx + Q(x, y)dy$ may be written as:

$$\int_{A_1}^{A_2} P(x, y)dx + Q(x, y)dy \text{ or } \int_{(a_1, b_1)}^{(a_n, b_n)} P(x, y)dx + Q(x, y)dy$$

$$2. \int_{(a_1, b_1)}^{(a_n, b_n)} P(x, y)dx + Q(x, y)dy = - \int_{(a_n, b_n)}^{(a_1, b_1)} P(x, y)dx + Q(x, y)dy$$

3. If the curve C:

Then:

$$\int_C P(x, y)dx + Q(x, y)dy = \int_{(a_1, b_1)}^{(a_3, b_3)} P(x, y)dx + Q(x, y)dy = \int_{(a_1, b_1)}^{(a_2, b_2)} P(x, y)dx + Q(x, y)dy + \int_{(a_2, b_2)}^{(a_3, b_3)} P(x, y)dx + Q(x, y)dy$$

4.4.1.3. Evaluation of line integrals

First case: C is given by $y = f(x)$, $a \leq x \leq b$

$$\int_C P(x, y)dx + Q(x, y)dy = \int_a^b [P(x, f(x)) + Q(x, f(x))f'(x)]dx$$

Second case: C is given by $x = g(y)$, $c \leq y \leq d$

$$\int_C P(x, y)dx + Q(x, y)dy = \int_c^d [P(g(y), y)g'(y) + Q(g(y), y)]dy$$

Third case: C is given in parametric form:

$$C = \begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}, \quad t_1 \leq t \leq t_2$$

$$\int_C P(x, y)dx + Q(x, y)dy = \int_{t_1}^{t_2} [P(\varphi(t), \psi(t))\varphi'(t) + Q(\varphi(t), \psi(t))\psi'(t)]dt$$

Example 4.9: Evaluate $\int_{(0,1)}^{(1,2)} (x^2 - y)dx + (y^2 + x)dy$ along

a) the parabola $x = t$ and $y = t^2 + 1$

b) the straight line (0,1) and (1,2)

Answer:

a) At point (0,1), $t = 0$ At point (1,2), $t = 1$

$$0 \leq t \leq 1$$

If $x = t$, then $dx = dt$, and If $y = t^2 + 1$, then $dy = 2t dt$

$$\int_{(0,1)}^{(1,2)} (x^2 - y)dx + (y^2 + x)dy = \int_0^1 \{[t^2 - (t^2 + 1)] \cdot 1 + [(t^2 + 1)^2 + t] \cdot 2t\} dt$$

$$= \int_0^1 (t^2 - t^2 - 1 + 2t^5 + 4t^3 + 2t + 2t^2) dt$$

$$= \int_0^1 (-1 + 2t^5 + 4t^3 + 2t^2 + 2t) dt$$

$$= \left[-t + \frac{1}{3}t^6 + t^4 + \frac{2}{3}t^3 + t^2 \right]_0^1$$

$$= \frac{1}{3} + 1 + \frac{2}{3} = 2$$

b) The equation of a straight line is $y = x + 1$, and $0 \leq x \leq 1$

If $y = x + 1$, then $dy = dx$

$$\int_{(0,1)}^{(1,2)} (x^2 - y)dx + (y^2 + x)dy = \int_0^1 \{[x^2 - (x + 1)] \cdot 1 + [(x + 1)^2 + x]\} dx$$

$$= \int_0^1 (x^2 - x - 1 + x^2 + 2x + 1 + x) dx$$

$$= \int_0^1 (2x^2 + 2x) dx$$

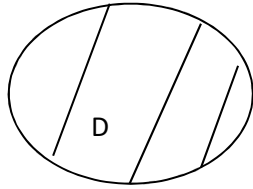
$$= \left[\frac{2}{3}x^3 + x^2 \right]_0^1$$

$$= \frac{2}{3} + 1 = \frac{5}{3}$$

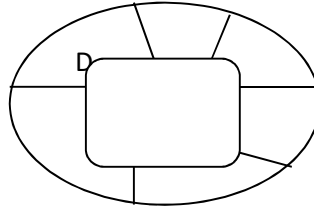
4.4.1.4. Green formula in the plane

1. Definitions:

- A domain D in the plane is said to be a **simply connected**, if when you compress it into one point you find a point belonging the domain.

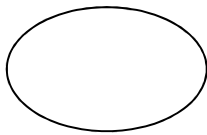


A simply connected

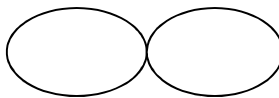


is not simply connected

- A curve C is said to be a simply closed curve if it does not cut itself.



Is simply closed



is not simply closed

2. Theorem:

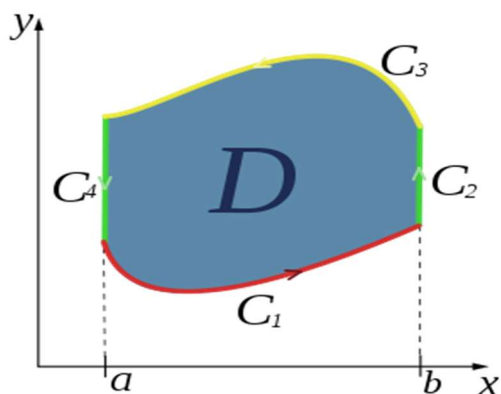
Let C be a positively oriented, piecewise smooth, simple closed curve in the plane $\mathbb{R}^2$, and let D be the region bounded by C . If L and M are functions of (x, y) defined on an **open region** containing D and have continuous partial derivatives $(\frac{\partial M(x,y)}{\partial x}; \frac{\partial N(x,y)}{\partial y})$ there, then

$$\oint_C (L dx + M dy) = \iint_D \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} \right) dx dy.$$

For **positive orientation**, an arrow pointing in the **counterclockwise** direction may be drawn in the small circle in the integral symbol.

In physics, Green's theorem is mostly used to solve two-dimensional flow integrals, stating that the sum of fluid outflows at any point inside a volume is equal to the total outflow summed about an enclosing area. In plane geometry, and in particular, area surveying, Green's theorem can be used to determine the area and centroid of plane figures solely by integrating over the perimeter.

Proof when D is a simple region



If D is a simple region with its boundary consisting of the curves C_1, C_2, C_3, C_4 , Green's theorem can be demonstrated.

The following is a proof of the theorem for the simplified area D , a type I region where C_2 and C_4 are vertical lines. A similar proof exists for when D is a type II region where C_1 and C_3 are straight lines. The general case can be deduced from this special case by approximating the domain D by a union of simple domains.

If it can be shown that

$$\oint_C L \, dx = \iint_D \left(-\frac{\partial L}{\partial y} \right) dA \quad (1)$$

and

$$\oint_C M \, dy = \iint_D \left(\frac{\partial M}{\partial x} \right) dA \quad (2)$$

are true, then Green's theorem is proven in the first case.

Define the type I region D as pictured on the right by

$$D = \{(x, y) | a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$$

where g_1 and g_2 are continuous functions on $[a, b]$. Compute the double integral in (1):

$$\begin{aligned} \iint_D \frac{\partial L}{\partial y} dA &= \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial L}{\partial y}(x, y) dy dx \\ &= \int_a^b \left\{ L(x, g_2(x)) - L(x, g_1(x)) \right\} dx \end{aligned} \quad (3)$$

Now compute the line integral in (1). C can be rewritten as the union of four curves: C_1, C_2, C_3, C_4 .

With C_1 , use the **parametric equations**: $x = x, y = g_1(x), a \leq x \leq b$. Then

$$\int_{C_1} L(x, y) dx = \int_a^b L(x, g_1(x)) dx$$

With C_3 , use the parametric equations: $x = x, y = g_2(x), a \leq x \leq b$. Then

$$\int_{C_3} L(x, y) dx = - \int_{-C_3} L(x, y) dx = - \int_a^b L(x, g_2(x)) dx.$$

The integral over C_3 is negated because it goes in the negative direction from b to a , as C is oriented positively (counterclockwise). On C_2 and C_4 , x remains constant, meaning

$$\int_{C_4} L(x, y) dx = \int_{C_2} L(x, y) dx = 0$$

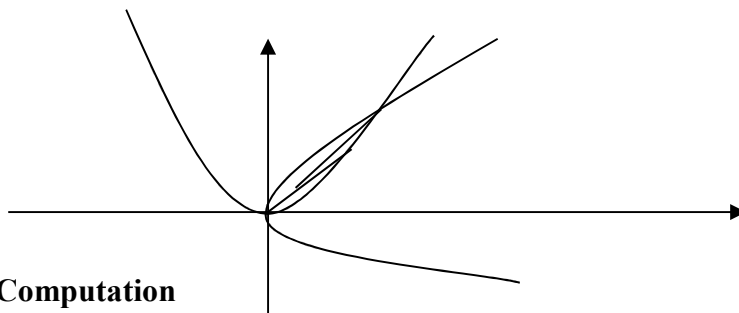
Therefore,

$$\begin{aligned} \int_C L dx &= \int_{C_1} L(x, y) dx + \int_{C_2} L(x, y) dx + \int_{C_3} L(x, y) dx + \int_{C_4} L(x, y) dx \\ &= - \int_a^b L(x, g_2(x)) dx + \int_a^b L(x, g_1(x)) dx \quad (4) \end{aligned}$$

Combining (3) with (4), we get (1). Similar computations give (2).

Example 4.10: Verify the Green formula for $\int_C (2xy - x^2)dx + (x + y^2)dy$. Where C is the curve of the region of the xy plane bounded by $y = x^2$ and $x = y^2$.

Answer:



1^o) Direct Computation

a) Along the curve $y = x^2$

$$\begin{aligned}
 I_a &= \int_0^1 [(2xx^2 - x^2) + (x + x^4)2x] dx = \int_0^1 [(2x^3 - x^2) + (2x^2 + 2x^5)] dx \\
 &= \int_0^1 (2x^3 + x^2 + 2x^5) dx = \left[\frac{x^4}{2} + \frac{x^3}{3} + \frac{x^6}{3} \right]_0^1 = \frac{7}{6}
 \end{aligned}$$

b) Along the curve $x = y^2 \rightarrow dx = 2ydy$

$$\begin{aligned}
 I_b &= - \int_0^1 [(2yy^2 - y^4)2y + (y^2 + y^2)] dy = - \int_0^1 (4y^3 + 2y^2 - 2y^5) dy \\
 &= \left[y^4 + \frac{2y^3}{3} - \frac{y^6}{3} \right]_0^1 = -\frac{17}{15}
 \end{aligned}$$

$$I = I_a + I_b = \frac{7}{6} - \frac{17}{15} = \frac{1}{30}$$

2⁰) Using Green formula

$$M(x, y) = 2xy - x^2; \quad \frac{\partial M}{\partial y} = 2x$$

$$N(x, y) = x + y^2; \quad \frac{\partial N}{\partial x} = 1$$

$$\int_c (2xy - x^2)dx + (x + y^2)dy = \iint_D (1 - 2x) dx dy$$

$$\begin{aligned}
 I &= \int_0^1 \left(\int_{x^2}^{\sqrt{x}} (1 - 2x) dy \right) dx = \int_0^1 [y - 2xy]_{x^2}^{\sqrt{x}} dx = \int_0^1 [(\sqrt{x} - 2x\sqrt{x}) - (x^2 - 2x^3)] dx \\
 &= \frac{1}{30}
 \end{aligned}$$

Or

$$I = \int_0^1 \left(\int_{y^2}^{\sqrt{y}} (1 - 2x) dx \right) dy = \int_0^1 [x - x^2]_{y^2}^{\sqrt{y}} dy = \frac{1}{30}$$

3. Remark:

$Ldx + Mdy$ with L and M satisfying $\frac{\partial L}{\partial y} = \frac{\partial M}{\partial x}$ is called a total differential.

In that case we suppose that $Ldx + Mdy = df$ where L and M satisfying $\frac{\partial L}{\partial y} = \frac{\partial M}{\partial x}$

$$\int_A^B Ldx + Mdy = \int_A^B df = f(B) - f(A)$$

Example 4.11: Evaluate $\int_{(1,2)}^{(3,4)} (6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy$

Answer: $\frac{\partial L}{\partial y} = 12xy - 3y^2 = M$, so there exists a function $f(x, y)$ such that $Ldx + Mdy = df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$.

$$\frac{\partial f}{\partial x} = 6xy^2 - y^3, \frac{\partial f}{\partial y} = 6x^2y - 3xy^2$$

$$f(x, y) = 3x^2y - 3xy^2 + g(y)$$

$$\begin{cases} \frac{\partial f}{\partial y} = 6x^2y - 3xy^2 + g'(y) \\ \frac{\partial f}{\partial y} = 6x^2y - 3xy^2 \end{cases} \rightarrow g'(y) = 0 \rightarrow g(y) = c$$

$$f(x, y) = 3x^2y - 3xy^2 + c$$

$$\begin{aligned} \int_{(1,2)}^{(3,4)} (6xy^2 - y^3)dx + (6x^2y - 3xy^2)dy &= f(x, y)|_{(1,2)}^{(3,4)} = f(3,4) - f(1,2) \text{ where } c=0 \\ &= (3 \cdot 9 \cdot 16 - 4^3 \cdot 3) - (3 \cdot 1 \cdot 2^2 - 2^3 \cdot 1) = 236 \end{aligned}$$

Area Calculation using line integrals

Green's theorem can be used to compute area by line integral. The area of D is given by:

$$A = \iint_D dA.$$

Provided we choose L and M such that:

$$\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y} = 1.$$

Then the area is given by:

$$A = \oint_C (Ldx + Mdy).$$

Possible formulas for the area of D include:

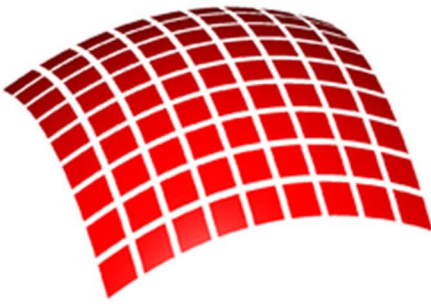
$$A = \oint_C x \, dy = - \oint_C y \, dx = \frac{1}{2} \oint_C (-y \, dx + x \, dy).$$

4.4.2. SURFACE INTEGRALS

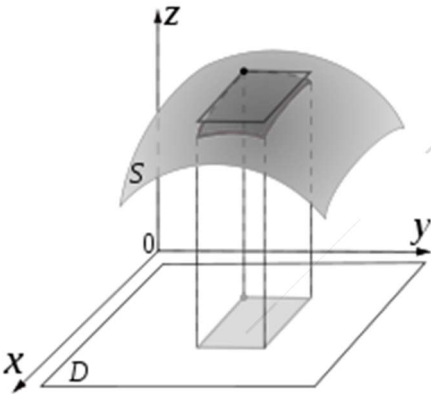
4.4.2.1. Definition

In mathematics, a **surface integral** is a definite integral taken over a surface (which may be a curve set in space); it can be thought of as the double integral analog of the line integral. Given a surface, one may integrate over its scalar fields (that is, functions which return numbers as values), and vector fields (that is, functions which return vectors as values).

Surface integrals have applications in physics, particularly with the classical theory of electromagnetism.



The definition of surface integral relies on splitting the surface into small surface elements.



An illustration of a single surface element. These elements are made infinitesimally small, by the limiting process, so as to approximate the surface.

Consider a two sided surface S having a projection on D on the XoY plane. Let $\phi(x, y, z)$ a function which is continuous at any point of S , then the surface integral of $\phi(x, y, z)$ over S designated by $\iint_S \phi(x, y, z) \, dS$ is defined as follows:

$$\iint_S \phi(x, y, z) dS = \begin{cases} \iint_D \phi(x, y, z) \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy, & \text{if the equation of } S \text{ is } z = f(x, y) \text{ (explicitly)} \\ \iint_D \phi(x, y, z) \frac{\sqrt{\left(\frac{\partial F}{\partial x}\right)^2 + \left(\frac{\partial F}{\partial y}\right)^2 + \left(\frac{\partial F}{\partial z}\right)^2}}{\left|\frac{\partial F}{\partial z}\right|} dx dy, & \text{if the equation of } S \text{ is given by } F(x, y, z) = 0 \text{ (Implicitly)} \end{cases}$$

Example 4.12: Evaluate $\iint_S \mathbf{u}(x, y, z) dS$ where S is the surface of the paraboloid $z = 2 - (x^2 + y^2)$ above the xy -plane and $\mathbf{u}(x, y, z)$ equal to:

- c) 1
d) $x^2 + y^2$

Answer: z is given explicitly $z = f(x, y)$.

$$\frac{\partial z}{\partial x} = -2x \text{ and } \frac{\partial z}{\partial y} = -2y$$

if you project $z = 2 - (x^2 + y^2)$ in the xoy plane then $z = 0$ and $D = \{(x, y) \text{ such that } x^2 + y^2 = 2\}$

a) If $\mathbf{u}(x, y, z) = 1$, then $\iint_S \mathbf{u}(x, y, z) dS = \iint_D \sqrt{1 + 4x^2 + 4y^2} dx dy$

Passing to polar coordinates: $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$ where $0 \leq r \leq \sqrt{2}$ and $0 \leq \theta \leq 2\pi$

$$\iint_S \mathbf{u}(x, y, z) dS = \iint_D \sqrt{1 + 4x^2 + 4y^2} dx dy = \int_0^{2\pi} \left(\int_0^{\sqrt{2}} r \sqrt{1 + 4r^2} dr \right) d\theta = \frac{13\pi}{3}$$

b) If $\mathbf{u}(x, y, z) = x^2 + y^2$

, then $\iint_S \mathbf{u}(x, y, z) dS = \iint_D (x^2 + y^2) \sqrt{1 + 4x^2 + 4y^2} dx dy$

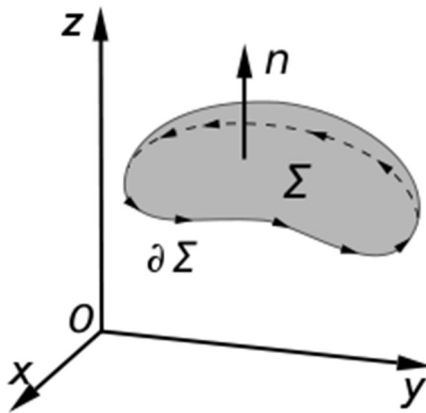
Passing to polar coordinates: $\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases}$ where $0 \leq r \leq \sqrt{2}$ and $0 \leq \theta \leq 2\pi$

$$\begin{aligned} \iint_S \mathbf{u}(x, y, z) dS &= \iint_D (x^2 + y^2) \sqrt{1 + 4x^2 + 4y^2} dx dy \\ &= \int_0^{2\pi} \left(\int_0^{\sqrt{2}} r^3 \sqrt{1 + 4r^2} dr \right) d\theta = \frac{149\pi}{30} \end{aligned}$$

4.4.2.2. Theorems involving surface integrals

Various useful results for surface integrals can be derived using differential geometry and vector calculus, such as the **divergence theorem**, and its generalization, **Stokes' theorem**.

Kelvin–Stokes theorem



An illustration of the Kelvin–Stokes theorem, with surface Σ , its boundary $\partial\Sigma$ and the "normal" vector n .

The classical **Kelvin–Stokes theorem**:

$$\int_{\Sigma} \nabla \times \mathbf{F} \cdot d\mathbf{\Sigma} = \oint_{\partial\Sigma} \mathbf{F} \cdot d\mathbf{r},$$

which relates the surface integral of the curl of a vector field over a surface Σ in Euclidean three-space to the line integral of the vector field over its boundary, is a special case of the general Stokes theorem (with $n = 2$) once we identify a vector field with a 1 form using the metric on Euclidean three-space. The curve of the line integral, $\partial\Sigma$, must have positive orientation, meaning that $d\mathbf{r}$ points counterclockwise when the surface normal, $d\mathbf{\Sigma}$, points toward the viewer, following the right-hand rule.

One consequence of the formula is that the field lines of a vector field with zero curl cannot be closed contours.

The formula can be rewritten as:

$$\begin{aligned} & \iint_{\Sigma} \left\{ \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dy dz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dz dx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy \right\} \\ &= \oint_{\partial\Sigma} \{ P dx + Q dy + R dz \} \end{aligned}$$

where P , Q and R are the components of $\mathbf{F}$.

These variants are frequently used:

$$\int_{\Sigma} (g(\nabla \times \mathbf{F}) + (\nabla g) \times \mathbf{F}) \cdot d\mathbf{\Sigma} = \oint_{\partial\Sigma} g\mathbf{F} \cdot d\mathbf{r},$$

$$\int_{\Sigma} (\mathbf{F}(\nabla \cdot \mathbf{G}) - \mathbf{G}(\nabla \cdot \mathbf{F}) + (\mathbf{G} \cdot \nabla)\mathbf{F} - (\mathbf{F} \cdot \nabla)\mathbf{G}) \cdot d\mathbf{\Sigma}$$

$$= \oint_{\partial\Sigma} (\mathbf{F} \times \mathbf{G}) \cdot d\mathbf{r}.$$

Applications: In electromagnetism

Two of the four **Maxwell equations** involve curls of 3-D vector fields and their differential and integral forms are related by the Kelvin–Stokes theorem. Caution must be taken to avoid cases with moving boundaries: the partial time derivatives are intended to exclude such cases. If moving boundaries are included, interchange of integration and differentiation introduces terms related to boundary motion not included in the results below (see **Differentiation under the integral sign**)

Name	Differential form	Integral form (using Kelvin–Stokes theorem plus relativistic invariance, $\int \frac{\partial}{\partial t} \dots \rightarrow \frac{d}{dt} \int \dots$)
Maxwell-Faraday equation <u>Faraday's law of induction:</u>	$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_C \mathbf{E} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{E} \cdot d\mathbf{A} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{A}$ (with C and S not necessarily stationary)
<u>Ampère's law</u> (with Maxwell's extension):	$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \nabla \times \mathbf{H} \cdot d\mathbf{A}$ $= \int_S \mathbf{J} \cdot d\mathbf{A} + \int_S \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{A}$ (with C and S not necessarily stationary)

4.4.3. VOLUME INTEGRALS

Likewise the **Ostrogradsky-Gauss theorem** (also known as the Divergence theorem or Gauss's theorem)

$$\iiint_V \operatorname{div} \vec{A} \, dV = \iint_S \vec{n} \cdot \vec{A} \, dS$$

is a special case if we identify a vector field with the $n-1$ form obtained by contracting the vector field with the Euclidean volume form.

Example 4.13: Verify the divergence theorem for $\vec{A} = (2x - z)\vec{i} + x^2y\vec{j} - xz^2\vec{k}$ taken over the region bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$

Answer: We have to verify that $\iiint_V \operatorname{div} \vec{A} \, dV = \iint_S \vec{n} \cdot \vec{A} \, dS$

$$\operatorname{div} \vec{A} = 2 + x^2 - 2xz$$

$$\iiint_V \operatorname{div} \vec{A} \, dV = \int_0^1 \left(\int_0^1 \left(\int_0^1 (2 + x^2 - 2xz) dz \right) dy \right) dx = \frac{11}{6}$$

$$\iint_S \vec{n} \cdot \vec{A} \, dS = \iint_{S_1} \vec{n} \cdot \vec{A} \, dS_1 + \iint_{S_2} \vec{n} \cdot \vec{A} \, dS_2 + \dots + \iint_{S_6} \vec{n} \cdot \vec{A} \, dS_6$$

1) Consider Face DEFG as S_1 ; $x = 1$ and $\vec{n} = \vec{i}$

$$\iint_{S_1} \vec{n} \cdot \vec{A} \, dS_1 = \int_0^1 \int_0^1 [(2 - z)\vec{i} + y\vec{j} - z^2\vec{k}] \cdot \vec{i} \, dy \, dz = \int_0^1 \int_0^1 (2 - z) \, dy \, dz = \frac{3}{2}$$

2) Consider Face ABEF as S_3 ; $y = 1$ and $\vec{n} = \vec{j}$

$$\iint_{S_3} \vec{n} \cdot \vec{A} \, dS_3 = \int_0^1 \int_0^1 [(2x - z)\vec{i} + x^2\vec{j} - xz^2\vec{k}] \cdot \vec{j} \, dx \, dz = \int_0^1 \int_0^1 x^2 \, dx \, dz = \frac{1}{3}$$

3) Consider Face ABCO as S_2 ; $X = 0$ and $\vec{n} = -\vec{i}$

$$\iint_{S_2} \vec{n} \cdot \vec{A} \, dS_2 = \int_0^1 \int_0^1 [(2x - z)\vec{i} + x^2y\vec{j} - xz^2\vec{k}] \cdot (-\vec{i}) \, dz \, dy = \int_0^1 \int_0^1 z \, dx \, dz = \frac{1}{2}$$

4) Consider Face BCDE as S_4 , the results is $-\frac{1}{2}$

5) Consider Face AFGO as S_5 , the results is 0

$$\iint_S \vec{n} \cdot \vec{A} \, dS = \frac{3}{2} + \frac{1}{2} + \frac{1}{3} + 0 - \frac{1}{2} + 0 = \frac{11}{6}$$

EXERCISES

1.a) Obtain the Jacobian J of the transformation $s = 2x + y, \quad t = x - 2y$

b) Determine the inverse transformation of the above transformation and obtain its Jacobian J_1 . Confirm that $J_1 = J^{-1}$

2. Find $\text{grad } f$ for $f(x, y, z) = x^2 y z^2$ at the point $(1, 2, 3)$. Hence calculate the directional derivative of $f(x, y, z)$ at $(1, 2, 3)$ in the direction of the vector $(-2, 3, -6)$

b) Find the $\text{grad } f$ where $f(x, y, z)$ is

a) $x^2 + y^2 - z$ b) $z \tan^{-1}(y/x)$ c) $xyz \sin\{\pi(x + y + z)\}$

3. If $\nabla \phi = (2xy + z^2)\vec{i} + (x^2 + z)\vec{j} + (y + 2xz)\vec{k}$

4. Find the divergence of the vector $\vec{v} = (2x - y^2, 3z + x^2, 4y - z^2)$ at the point $(1, 2, 3)$.

5. Find the value of the constant λ such that the vector field defined by $\vec{F} = (2x^2 y^2 + z^2)\vec{i} + (3xy^3 - x^2 z)\vec{j} + (\lambda xy^2 z + xy)\vec{k}$ is solenoidal

6. The vector $\vec{v}$ is defined by $\vec{v} = \vec{r} r^{-1}$, where $\vec{r} = (x, y, z)$ and $r = |\vec{r}|$. Show that $\nabla(\nabla \cdot \vec{v}) \equiv \text{grad div } \vec{v} = -\frac{2}{3}\vec{r}$

7. Find $\vec{u} = \text{curl } \vec{v}$ where $\vec{v} = (3xz^3, -yz, x + 2z)$

8. Maxwell's equation in free space may be written, in Gaussian units, as

$$\text{div } \vec{H} = 0, \quad \text{div } \vec{E} = 0, \quad \nabla \times \vec{H} = \frac{1}{c} \frac{\partial \vec{E}}{\partial t}, \quad \nabla \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{H}}{\partial t}$$

Where c is the velocity of light (assumed constant). Show that these equations are satisfied by

$$\vec{H} = \frac{1}{c} \frac{\partial}{\partial t} \text{grad } \phi \times \vec{k}, \quad \vec{E} = -\vec{k} \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} + \frac{\partial}{\partial z} \text{grad } \phi$$

Where ϕ satisfies $\nabla^2 \phi = \frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2}$ and $\vec{k}$ is a unit vector along the z -axis.

9. a) Evaluate $\int_R (x^2 + y^2) dA$ over the triangle with vertices at $(0, 0)$, $(2, 0)$ and $(1, 1)$

b) Evaluate $\int_R x^2 y dA$, where R is the region $x^2 + y^2 \leq 1$

10. Evaluate $\oint_C [2x(x + y)dx + (x^2 + xy + y^2)dy]$ around the square with vertices at $(0, 0)$, $(1, 0)$, $(1, 1)$ and $(0, 1)$

11. Use Green's theorem in a plane to evaluate the line integral

$\oint_C [(e^x - 3y^2)dx + (e^y + 4x^2)dy]$ where C is the circle $x^2 + y^2 = 4$. (Hint: Use polar coordinates to evaluate the double integral)

12. Evaluate the surface integral $\iint_S (x + y + z)ds$, where S is the portion of the sphere $x^2 + y^2 + z^2 = 1$ that lies in the first quadrant.

13. Verify Stokes' theorem for $\vec{F} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$, where S is the upper half of the sphere $x^2 + y^2 = 1$ and C is its boundary.

14. a) Evaluate $\int_{(1,4)}^{(3,1)} 2xy^3dx + (1 + 3x^2y^2)dy$

b) A particle moves over the semicircle $C: \vec{r}(t) = \cos t \vec{i} + \sin t \vec{j}, 0 \leq t \leq \pi$ while subject to the force $\vec{F}(x, y) = e^y\vec{i} + xe^y\vec{j}$. Find the work done.